

EUROPEAN ECONOMICS ANALYST

UK—The Economic Cost of Brexit, Ten Years On

- The UK economy has materially underperformed since the 2016 EU referendum, shifting from a growth profile closer to the US to one closer to the Euro area. Our updated “Doppelgänger” analysis suggests real GDP is around 6% below its counterfactual path, more than our previous 5% estimate two years ago. While these figures could capture the impact of other factors as well as Brexit, robustness checks point to an underperformance of at least 4%. These estimates are broadly consistent with other studies. Model-based estimates of the cost of leaving the EU average around 5%, while recent empirical research finds a 6–8% impact.
- Analysis of the channels through which Brexit could have affected output also points to a significant drag. Goods trade volumes have underperformed by 10–15% since the referendum. Services trade has held up better, but recent studies nonetheless find a meaningful Brexit-related hit. Lower trade intensity has likely reduced productivity, lowering GDP by 2–4%. Business investment has also been weak since 2016—with business surveys suggesting this is partly due to Brexit—likely lowering output by 2%. The impact on labour supply through migration has been more mixed, with lower EU immigration but higher non-EU inflows. That said, shifting migration patterns have probably contributed to labour shortages in some sectors.
- Taken together, our estimates indicate that Brexit has materially weighed on Britain’s economic performance relative to other advanced economies. While new trade agreements could, in principle, help mitigate some of the costs of Brexit, the benefit from new (non-EU) trade deals is likely to be small.
- That said, we see two reasons why the peak Brexit headwind might now be behind us. First, estimates point to a front-loaded drag, suggesting that the Brexit growth headwind should ease over time. Second, opinion polls now show greater support for a closer EU relationship, and the government is pursuing more EU-UK cooperation. An ambitious reset—such as closer single-market or customs-union alignment—could reverse a meaningful share of the Brexit cost and support stronger trend growth. Consistent with this, we expect average growth in the UK to pick up from 1.2% since 2016 to 1.5% over the coming years.

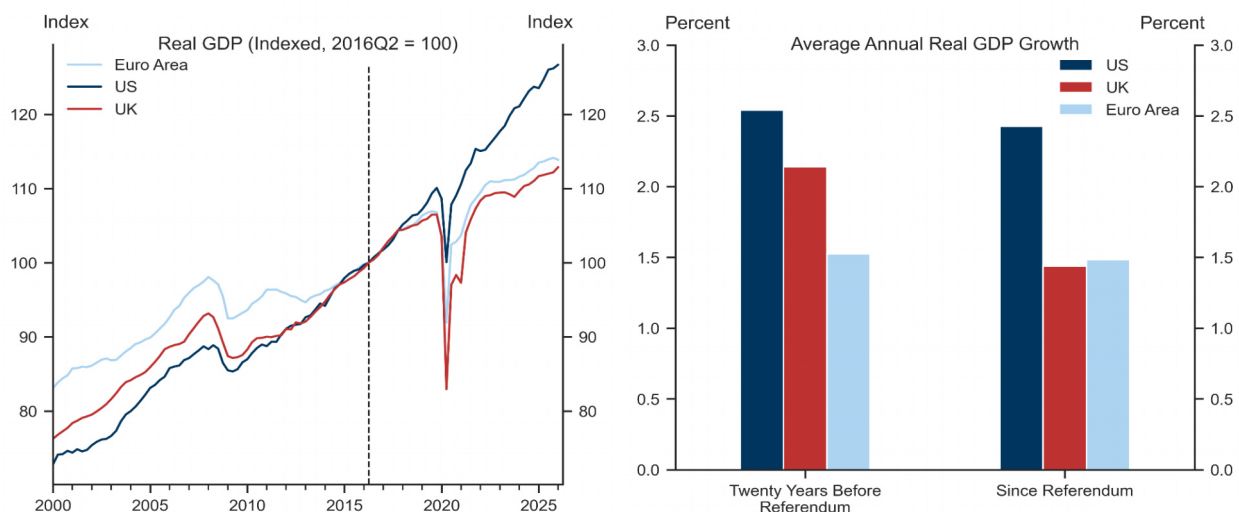
James Moberly
+44(20)7774-9444 |
james.r.moberly@gs.com
Goldman Sachs International

Sven Jari Stehn
+44(20)7774-8061 | jari.stehn@gs.com
Goldman Sachs International

UK—The Economic Cost of Brexit, Ten Years On

The UK voted to leave the EU ten years ago this week and has underperformed other advanced economies significantly since then (Exhibit 1). UK growth looked more similar to the US in the twenty years leading up to the referendum, but more similar to the Euro area in the subsequent ten years.

Exhibit 1: UK Growth Was Closer to the US Before Brexit, More Like the Euro Area Since Then



Source: Goldman Sachs Global Investment Research, Haver Analytics

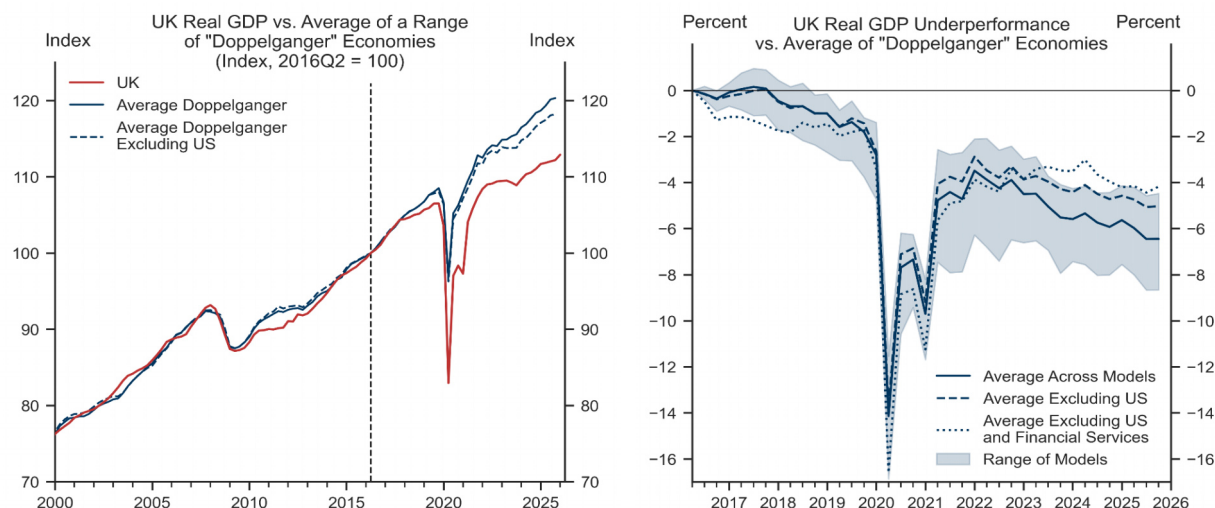
Given the ten-year anniversary of the referendum, we take stock of the macroeconomic costs of the Brexit vote and update our 2024 estimates.

Counting the Cost

We start by examining empirically how much UK real GDP has underperformed other advanced economies since 2016 using a “Doppelgänger” analysis.¹ This method uses statistical techniques to find the best combination of other countries that match the path of UK real GDP before the referendum and uses this same combination to project what might have happened thereafter.

Our estimates suggest that the UK has fallen short of its Doppelgänger by 6% since the referendum, compared with the 5% we estimated two years ago (Exhibit 2, left). Varying the group of countries in the analysis—including the G7, G10, OECD, OECD Europe—points to a range of 4–8% for the output cost (Exhibit 2, right).

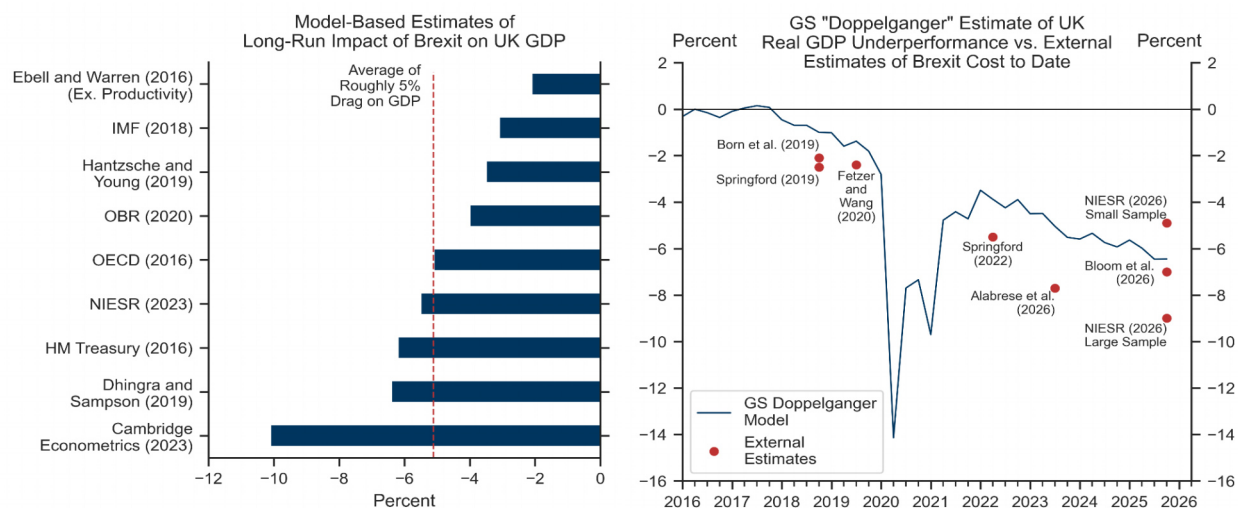
¹ This synthetic control method was applied to Brexit by Benjamin Born and co-authors in “The Costs of Economic Nationalism: Evidence from the Brexit Experiment,” May 2019.

Exhibit 2: Our Doppelgänger Model Points to Significant Underperformance Since the Referendum

Source: Goldman Sachs Global Investment Research, Haver Analytics

A shortcoming of the Doppelgänger approach is that not all of the post-2016 underperformance can be attributed to Brexit, as other structural changes have affected Britain's relative economic performance. In particular, the inclusion of the US could skew the estimates, as the US has benefited from a much larger technology boom than other advanced economies during this time. Excluding the US from the Doppelgänger analysis points to a smaller output cost at 5% (dashed line in Exhibit 2, right). It is also possible that UK growth was temporarily boosted by a booming financial sector before the referendum relative to other advanced economies. Re-running the analysis on UK GDP without the financial sector and again excluding the US points to 4% underperformance, which is likely to be a lower-bound estimate (dotted line in Exhibit 2, right).

Our Doppelgänger estimates are consistent with prior studies for the economic cost of Brexit. Model-based studies—which simulate the economic repercussions of greater trade frictions outside of the EU—point to a total output cost of Brexit of around 5% (Exhibit 3, left). The latest empirical studies—often using Doppelgänger models—point to a hit of 6-8%, which has grown notably in recent years (Exhibit 3, right). We provide a more detailed summary of these studies in the appendix.

Exhibit 3: Studies Point to a Large (and Rising) Cost of Brexit

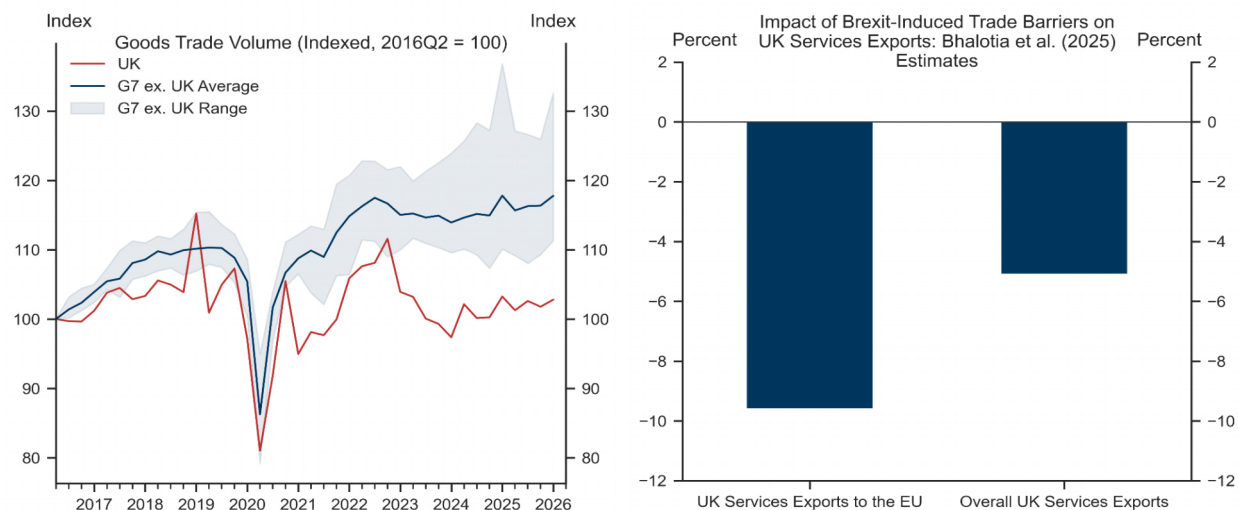
Ebell and Warren (2016) note that incorporating productivity losses would substantially increase their estimates.

Source: Goldman Sachs Global Investment Research, Various academic studies

Assessing the Channels

Given uncertainty around the top-down estimates, we consider the mechanisms of Brexit to help gauge how much of the UK's underperformance can be attributed to Brexit. Studies generally consider three mechanisms through which Brexit could have lowered long-term output: (1) international trade, (2) business investment and (3) immigration.

First, the UK has materially underperformed other advanced economies in terms of trade since 2016. We estimate that goods trade volumes would be 10-15% higher in a counterfactual in which volumes had grown in line with the average of other G7 economies, reflecting higher trade frictions outside of the EU (Exhibit 4, left). Services trade has held up better. Recent studies, however, also point to a significant hit to services trade as a result of leaving the EU (Exhibit 4, right).

Exhibit 4: UK Trade Has Been Hit

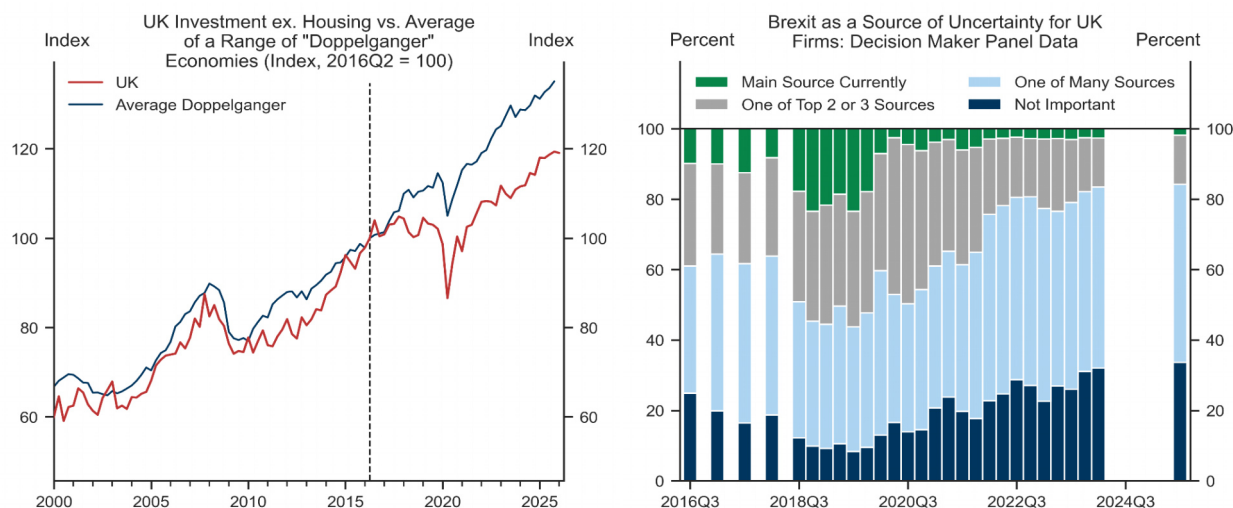
Source: Goldman Sachs Global Investment Research, Haver Analytics, Bhalotia et al. (2025)

Overall, the trade estimates point to a roughly 5-10% decline in trade volumes due to Brexit. Previous studies suggest that a decline in trade intensity of this magnitude could have reduced GDP by 2-4% through reduced productivity.²

Second, UK investment growth stalled after the referendum, falling significantly below its pre-2016 trend (Exhibit 5, left). Our Doppelgänger estimates for investment point to a shortfall of around 10% relative to other economies. At least some of this underperformance likely reflected Brexit uncertainty after the referendum, with surveys clearly flagging Brexit as a source of uncertainty (Exhibit 5, right). Our estimates on the business investment channel imply an output hit of just over 2% by end-2025.³

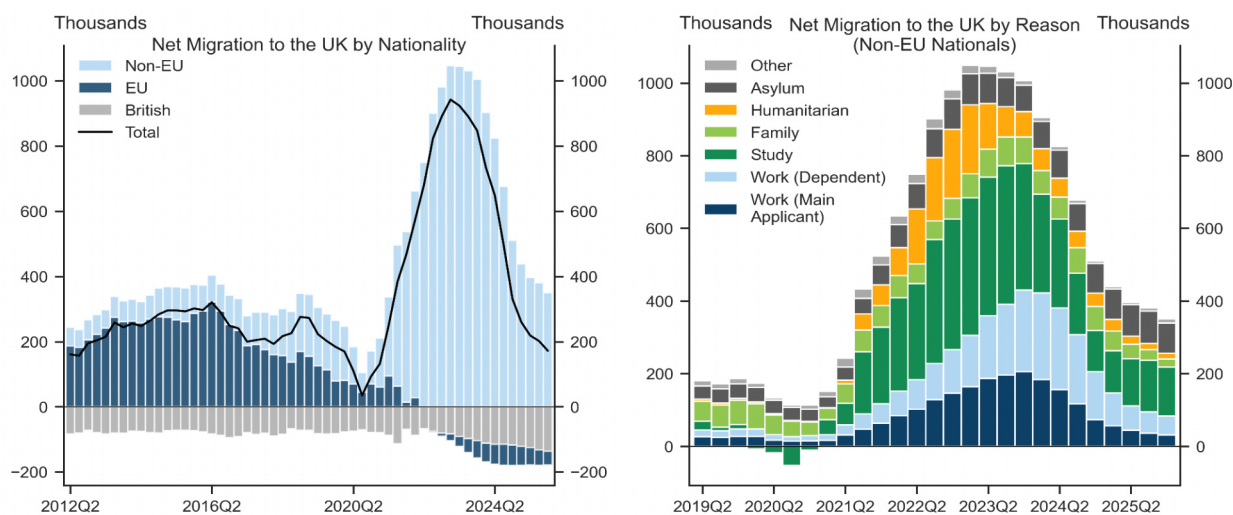
² Two studies by Feyrer using different methodologies have estimated that each 1% reduction in trade lowers GDP by around $\frac{1}{4}\%$ or $\frac{1}{2}\%$.

³ Haskel and Martin (2023) estimated that weakness in business investment after the referendum had lowered output by 1.3% by 2022.

Exhibit 5: Investment Has Underperformed

Source: Goldman Sachs Global Investment Research, Haver Analytics, Bank of England

Third, Brexit has affected immigration flows into the UK. Net migration from the EU has fallen sharply from a peak of over 300k in the year to 2016Q2 to -42k in 2025 (Exhibit 6, left). At the same time, net migration from outside the EU rose significantly to over 1m per year after the pandemic. Although much of the pickup in non-EU immigration reflects reasons other than work (Exhibit 6, right), recent [research](#) suggests that the new immigration system has so far led to a modest net increase in the number of foreign-born people in work. That said, net migration has dropped sharply recently, partly reflecting new government policies.

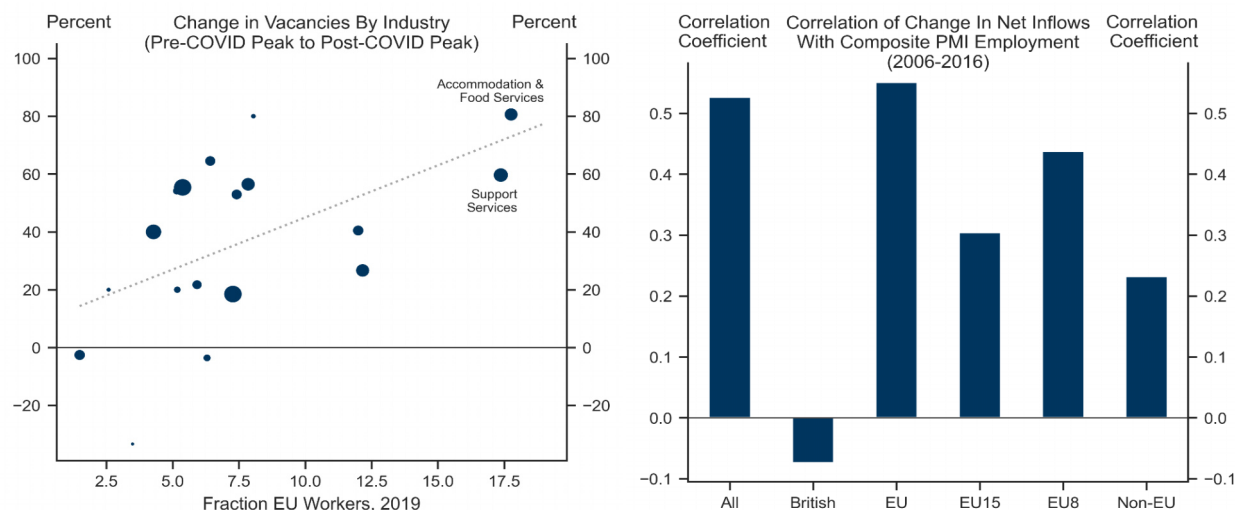
Exhibit 6: Net Migration from the EU Has Fallen but Overall Inflows Rose Post-Pandemic

Source: ONS

Despite the rise in net migration after Brexit, our previous [work](#) suggests that changes in the immigration system have probably contributed to labour shortages. EU nationals have historically been concentrated in lower-skilled occupations and are harder to replace under the new skill-selective migration regime. The sectors that are most reliant on EU workers have experienced the largest increases in vacancy rates since the Covid

pandemic (Exhibit 7, left). Going forward, reduced cyclical volatility of migration flows compared with the pre-Brexit period (Exhibit 7, right) could lead to greater volatility in labour market tightness and domestic inflationary pressures.

Exhibit 7: Lower Immigration from the EU Helps Explain Labour Shortages



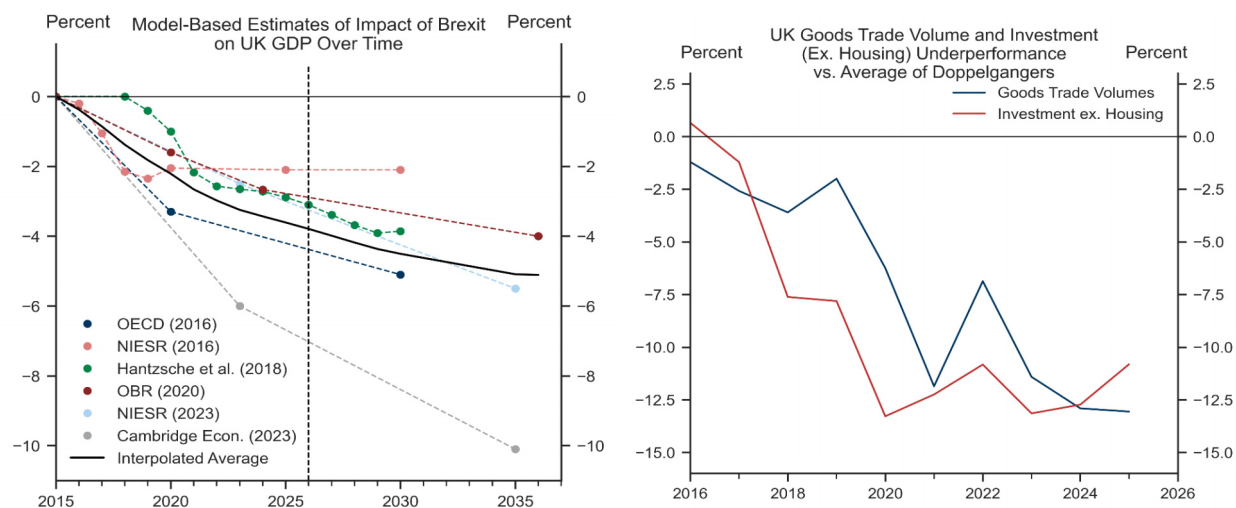
Source: Goldman Sachs Global Investment Research, Haver Analytics, ONS

As Bad as It Gets?

Taken together, our estimates show that Brexit has materially weighed on Britain's economic performance relative to other advanced economies. While new trade agreements could, in principle, help mitigate some of the costs of Brexit, the benefit from new (non-EU) trade deals is likely to be small.

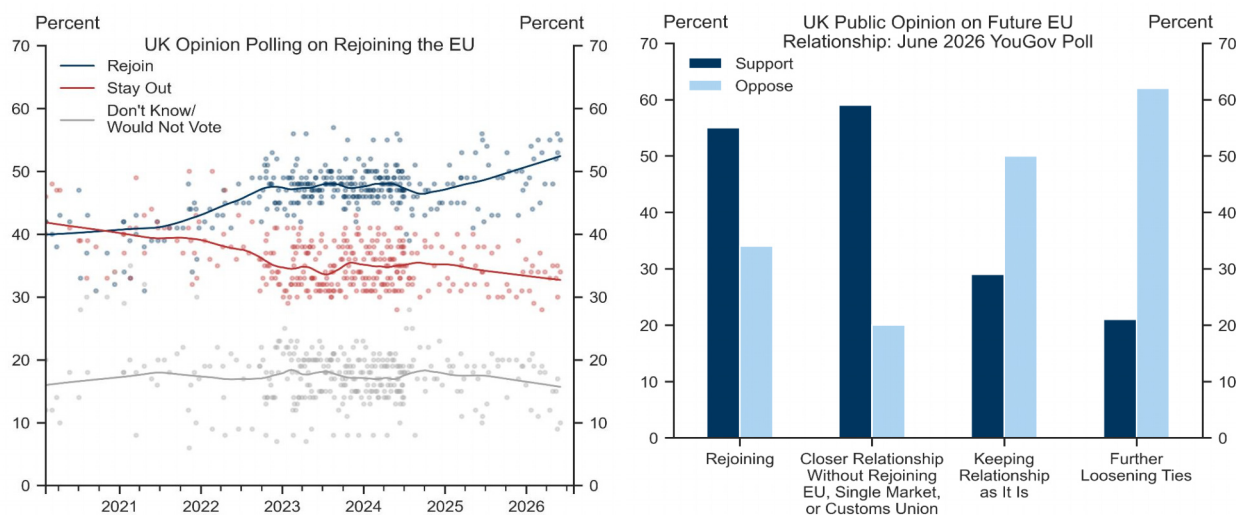
That said, we see two reasons to think that the worst of the Brexit headwind might now be behind us.

First, although the full impact of leaving the EU is likely to take time to materialise, several studies point to a front-loaded growth drag (Exhibit 8, left). Decision Maker Panel data and our own analysis indicate that much of the impact on investment had already come through before the UK left the EU at the end of 2020 as businesses anticipated future disruption to trade barriers. Although there is evidence that trade flows can take 7-10 years to adjust fully to new trade barriers, the shortfall in goods trade volumes relative to other economies shows some signs of levelling off in the last two years (Exhibit 8, right). It is true that the UK's underperformance on GDP has continued to build, perhaps reflecting lags from lower investment and trade into productivity. But these factors suggest that the growth hit is likely to gradually ease over time.

Exhibit 8: Past the Peak Brexit Growth Headwind

Source: Goldman Sachs Global Investment Research, Haver Analytics, Various academic studies

Second, recent opinion polls and political shifts point to building support for a closer EU-UK relationship. Public opinion polls show that a majority of the UK population now support rejoining the EU (52%), up notably since Covid and now comfortably ahead of the 33% share supporting Brexit (Exhibit 9, left). Public support is highest for a closer relationship with Europe without rejoining the EU, Single Market or Customs Union (Exhibit 9, right). As a result, the government has worked towards a “reset” in EU-UK relations and Andy Burnham (who is likely to succeed Prime Minister Starmer) has historically supported closer integration with Europe.

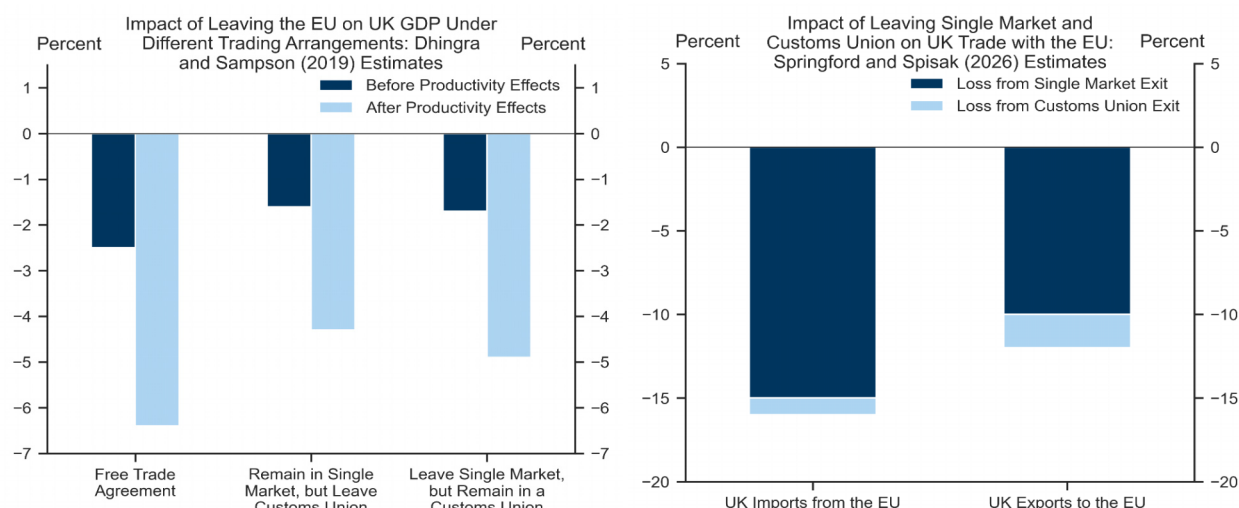
Exhibit 9: Public Opinion Has Shifted in Favour of a Closer Relationship

Source: Goldman Sachs Global Investment Research, Various polling companies, YouGov

The government could start by finalising the veterinary agreement, negotiating mutual recognition of professional qualifications, and aligning with EU rules where it is of clear economic benefit (e.g., for chemicals or the automotive sector). A more ambitious reset further down the line—such as rejoining the single market or negotiating a customs union—could raise GDP by 1-2% and thereby undo a meaningful share of the economic

cost incurred by Brexit (Exhibit 10).

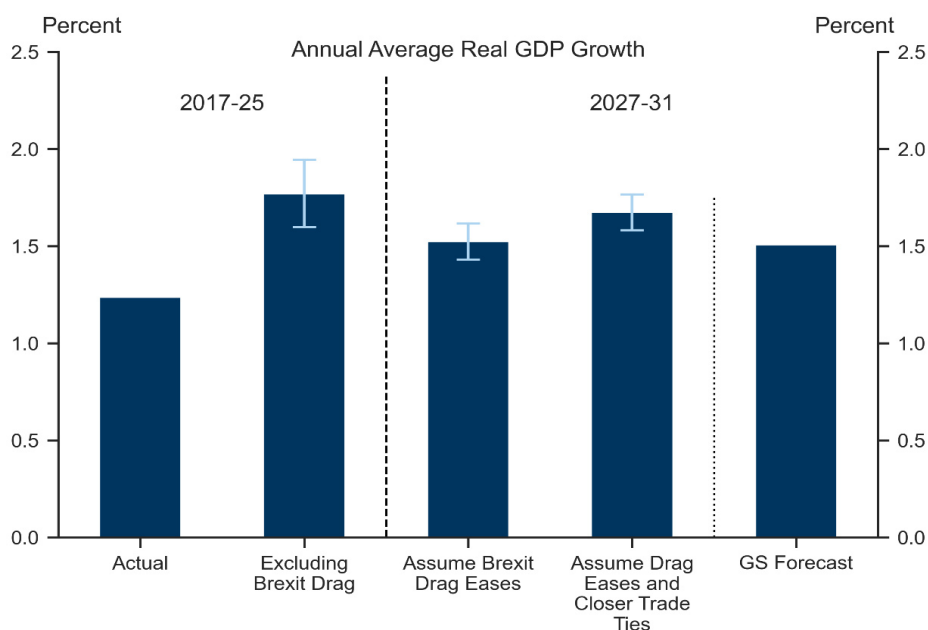
Exhibit 10: A Closer Economic Relationship Could Mitigate Some of the Brexit Costs



Source: Dhingra and Sampson (2019), Springford and Spisak (2026)

The combination of a receding Brexit headwind and a closer economic relationship with the EU could improve the UK's growth performance significantly in coming years (Exhibit 11). Simple calculations suggest that—assuming an unchanged underlying growth pace—UK trend growth could pick up to 1.5% in coming years with a diminishing Brexit drag. That figure could rise to 1.7% in the event of a significant reset of EU-UK relations.

Exhibit 11: An Easing Brexit Drag Could See Trend Growth Improve



“Excluding Brexit drag” uses the average of the GS doppelganger estimate and the growth headwind implied by the model-based estimates shown in Exhibit 8 (left). The light blue bracket shows the range from those two approaches. “Assuming Brexit Drag Eases” assumes that the drag fades in line with the time profile from the model-based estimates. “Closer Trade Ties” assumes a boost of 1-2% from a more ambitious reset over 10 years.

Source: Goldman Sachs Global Investment Research, Haver Analytics

While there is considerable uncertainty around the exact timing and magnitude of that improvement, the likely scale of the drag on growth over the past decade and the potential upside from a closer EU-UK relationship suggests that the upside is material. Consistent with this, we expect average growth in the UK to pick up from 1.2% since 2016 to 1.5% over the coming years.

James Moberly

Sven Jari Stehn

Appendix

Exhibit 12: Studies Estimating the Impact of Brexit on UK GDP

Model-Based Estimate of Long-Run Impact of Brexit			Tracking Estimates of the Impact of Brexit on GDP to Date		
HM Treasury	2016	6.2% under a bilateral agreement over 15 years	Born et al.	2019	1.7-2.5% by the end of 2018
OECD	2016	3.3% by 2020, 5.1% by 2030	Springford (CER)	2019	2.5% by the end of 2018
NIESR	2016	2.1% under FTA ex. productivity; productivity effects could increase cost significantly further	Fetzer and Wang	2020	2.3-2.5% aggregate impact by 2019, 8.5% in worst affected regions
Dhingra et al. (LSE/CEP)	2017	1.3-2.7% ex. productivity; 6.3-9.4% inc. productivity	Springford (CER)	2022	5.5% by 2022Q2
IMF	2018	2.6-3.9% under FTA, average of 3.1%	Alabrese et al.	2026	7.7% by 2023
Hantzsche et al. (NIESR)	2018	3.9% under May deal	Bloom et al.	2026	6-8% by end of 2025
Dhingra and Sampson (LSE/CEP)	2019	2.5% ex. productivity, 6.4% inc. productivity under Johnson deal	NIESR	2026	4.9-9.0% by end 2025, depending on comparison group of countries
Hantzsche and Young (NIESR)	2019	3.5% under Johnson deal			
OBR	2020	4% in the long-run under an FTA			
NIESR	2023	2-3% by 2023, 5-6% by 2035			
Cambridge Econometrics	2023	6.0% by 2023, 10.1% by 2035			

Source: Data compiled by Goldman Sachs Global Investment Research, Various academic studies

European Economics Team

Sven Jari Stehn

+44(20)7774-8061
jari.stehn@gs.com
Goldman Sachs International

James Moberly

+44(20)7774-9444
james.r.moberly@gs.com
Goldman Sachs International

Giovanni Pierdomenico

+44(20)7051-6807
giovanni.pierdomenico@gs.com
Goldman Sachs International

Filippo Taddei

+44(20)7774-5458
filippo.taddei@gs.com
Goldman Sachs International

Niklas Garnadt

+49(69)7532-1537
niklas.garnadt@gs.com
Goldman Sachs Bank Europe SE

Alexandre Stott

+33(1)4212-1108
alexandre.stott@gs.com
Goldman Sachs Bank Europe SE - Paris Branch

Katya Vashkinskaya

+44(20)7774-4833
katya.vashkinskaya@gs.com
Goldman Sachs International

Disclosure Appendix

Reg AC

We, James Moberly and Sven Jari Stehn, hereby certify that all of the views expressed in this report accurately reflect our personal views, which have not been influenced by considerations of the firm's business or client relationships.

Unless otherwise stated, the individuals listed on the cover page of this report are analysts in Goldman Sachs' Global Investment Research division.

Contributing Authors: James Moberly Goldman Sachs International, Sven Jari Stehn Goldman Sachs International.

Unless otherwise stated, the individuals listed in the Contributing Authors disclosure of this report are analysts in Goldman Sachs' Global Investment Research division.

Disclosures

Regulatory disclosures

Disclosures required by United States laws and regulations

See company-specific regulatory disclosures above for any of the following disclosures required as to companies referred to in this report: manager or co-manager in a pending transaction; 1% or other ownership; compensation for certain services; types of client relationships; managed/co-managed public offerings in prior periods; directorships; for equity securities, market making and/or specialist role. Goldman Sachs trades or may trade as a principal in debt securities (or in related derivatives) of issuers discussed in this report.

The following are additional required disclosures: **Ownership and material conflicts of interest:** Goldman Sachs policy prohibits its analysts, professionals reporting to analysts and members of their households from owning securities of any company in the analyst's area of coverage. **Analyst compensation:** Analysts are paid in part based on the profitability of Goldman Sachs, which includes investment banking revenues. **Analyst as officer or director:** Goldman Sachs policy generally prohibits its analysts, persons reporting to analysts or members of their households from serving as an officer, director or advisor of any company in the analyst's area of coverage. **Non-U.S. Analysts:** Non-U.S. analysts may not be associated persons of Goldman Sachs & Co. LLC and therefore may not be subject to FINRA Rule 2241 or FINRA Rule 2242 restrictions on communications with a subject company, public appearances and trading in securities covered by the analysts.

Additional disclosures required under the laws and regulations of jurisdictions other than the United States

The following disclosures are those required by the jurisdiction indicated, except to the extent already made above pursuant to United States laws and regulations. **Australia:** Goldman Sachs Australia Pty Ltd and its affiliates are not authorised deposit-taking institutions (as that term is defined in the Banking Act 1959 (Cth)) in Australia and do not provide banking services, nor carry on a banking business, in Australia. This research, and any access to it, is intended only for "wholesale clients" within the meaning of the Australian Corporations Act, unless otherwise agreed by Goldman Sachs. In producing research reports, members of Global Investment Research of Goldman Sachs Australia may attend site visits and other meetings hosted by the companies and other entities which are the subject of its research reports. In some instances the costs of such site visits or meetings may be met in part or in whole by the issuers concerned if Goldman Sachs Australia considers it is appropriate and reasonable in the specific circumstances relating to the site visit or meeting. To the extent that the contents of this document contains any financial product advice, it is general advice only and has been prepared by Goldman Sachs without taking into account a client's objectives, financial situation or needs. A client should, before acting on any such advice, consider the appropriateness of the advice having regard to the client's own objectives, financial situation and needs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests and a copy of Goldman Sachs' Australian Sell-Side Research Independence Policy Statement are available at: <https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html>. **Brazil:** Disclosure information in relation to CVM Resolution n. 20 is available at <https://www.gs.com/worldwide/brazil/area/gir/index.html>. Where applicable, the Brazil-registered analyst primarily responsible for the content of this research report, as defined in Article 20 of CVM Resolution n. 20, is the first author named at the beginning of this report, unless indicated otherwise at the end of the text. **Canada:** This information is being provided to you for information purposes only and is not, and under no circumstances should be construed as, an advertisement, offering or solicitation by Goldman Sachs & Co. LLC for purchasers of securities in Canada to trade in any Canadian security. Goldman Sachs & Co. LLC is not registered as a dealer in any jurisdiction in Canada under applicable Canadian securities laws and generally is not permitted to trade in Canadian securities and may be prohibited from selling certain securities and products in certain jurisdictions in Canada. If you wish to trade in any Canadian securities or other products in Canada please contact Goldman Sachs Canada Inc., an affiliate of The Goldman Sachs Group Inc., or another registered Canadian dealer. **Hong Kong:** Further information on the securities of covered companies referred to in this research may be obtained on request from Goldman Sachs (Asia) L.L.C. **India:** Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (India) Securities Private Limited, Research Analyst - SEBI Registration Number INH000001493, 10th Floor, Ascent-Worli, Sudam Kalu Ahire Marg, Worli, Mumbai-400 025, India, Corporate Identity Number U74140MH2006FTC160634, Phone +91 22 6616 9000, Fax +91 22 6616 9001. Goldman Sachs may beneficially own 1% or more of the securities (as such term is defined in clause 2 (h) the Indian Securities Contracts (Regulation) Act, 1956) of the subject company or companies referred to in this research report. Investment in securities market are subject to market risks. Read all the related documents carefully before investing. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Goldman Sachs (India) Securities Private Limited compliance officer and investor grievance contact details can be found at: <https://www.goldmansachs.com/worldwide/india/documents/Grievance-Redressal-and-Escalation-Matrix.pdf>, and a copy of the annual audit compliance report can be found at this link: <https://publishing.gs.com/content/site/india-annual-compliance-report.html>. **Japan:** See below. **Korea:** This research, and any access to it, is intended only for "professional investors" within the meaning of the Financial Services and Capital Markets Act, unless otherwise agreed by Goldman Sachs. Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (Asia) L.L.C., Seoul Branch. **New Zealand:** Goldman Sachs New Zealand Limited and its affiliates are neither "registered banks" nor "deposit takers" (as defined in the Reserve Bank of New Zealand Act 1989) in New Zealand. This research, and any access to it, is intended for "wholesale clients" (as defined in the Financial Advisers Act 2008) unless otherwise agreed by Goldman Sachs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests is available at: <https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html>. **Russia:** Research reports distributed in the Russian Federation are not advertising as defined in the Russian legislation, but are information and analysis not having product promotion as their main purpose and do not provide appraisal within the meaning of the Russian legislation on appraisal activity. Research reports do not constitute a personalized investment recommendation as defined in Russian laws and regulations, are not addressed to a specific client, and are prepared without analyzing the financial circumstances, investment profiles or risk profiles of clients. Goldman Sachs assumes no responsibility for any investment decisions that may be taken by a client or any other person based on this research report. **Singapore:** Goldman Sachs (Singapore) Pte. (Company Number: 198602165W), which is regulated by the Monetary Authority of Singapore, accepts legal responsibility for this research, and should be contacted with respect to any matters arising from, or in connection with, this research. **Taiwan:** This material is for reference only and must not be reprinted without permission. Investors should carefully consider their own investment risk. Investment results are the responsibility of the individual investor. **United Kingdom:** Persons who would be categorized as retail clients in the United Kingdom, as such term is defined in the rules of the Financial Conduct Authority, should read this research in conjunction with prior Goldman Sachs research on the covered

companies referred to herein and should refer to the risk warnings that have been sent to them by Goldman Sachs International. A copy of these risks warnings, and a glossary of certain financial terms used in this report, are available from Goldman Sachs International on request.

European Union and United Kingdom: Disclosure information in relation to Article 6 (2) of the European Commission Delegated Regulation (EU) (2016/958) supplementing Regulation (EU) No 596/2014 of the European Parliament and of the Council (including as that Delegated Regulation is implemented into United Kingdom domestic law and regulation following the United Kingdom's departure from the European Union and the European Economic Area) with regard to regulatory technical standards for the technical arrangements for objective presentation of investment recommendations or other information recommending or suggesting an investment strategy and for disclosure of particular interests or indications of conflicts of interest is available at <https://www.gs.com/disclosures/europeanpolicy.html> which states the European Policy for Managing Conflicts of Interest in Connection with Investment Research.

Japan: Goldman Sachs Japan Co., Ltd. is a Financial Instrument Dealer registered with the Kanto Financial Bureau under registration number Kinsho 69, and a member of Japan Securities Dealers Association, Financial Futures Association of Japan Type II Financial Instruments Firms Association, and Investment Management Association of Japan. Sales and purchase of equities are subject to commission pre-determined with clients plus consumption tax. See company-specific disclosures as to any applicable disclosures required by Japanese stock exchanges, the Japanese Securities Dealers Association or the Japanese Securities Finance Company.

Global product; distributing entities

Goldman Sachs Global Investment Research produces and distributes research products for clients of Goldman Sachs on a global basis. Analysts based in Goldman Sachs offices around the world produce research on industries and companies, and research on macroeconomics, currencies, commodities and portfolio strategy. This research is disseminated in Australia by Goldman Sachs Australia Pty Ltd (ABN 21 006 797 897); in Brazil by Goldman Sachs do Brasil Corretora de Títulos e Valores Mobiliários S.A.; Public Communication Channel Goldman Sachs Brazil: 0800 727 5764 and / or contatogoldmanbrasil@gs.com. Available Weekdays (except holidays), from 9am to 6pm. Canal de Comunicação com o Público Goldman Sachs Brasil: 0800 727 5764 e/ou contatogoldmanbrasil@gs.com. Horário de funcionamento: segunda-feira à sexta-feira (exceto feriados), das 9h às 18h; in Canada by Goldman Sachs & Co. LLC; in Hong Kong by Goldman Sachs (Asia) L.L.C.; in India by Goldman Sachs (India) Securities Private Ltd.; in Japan by Goldman Sachs Japan Co., Ltd.; in the Republic of Korea by Goldman Sachs (Asia) L.L.C., Seoul Branch; in New Zealand by Goldman Sachs New Zealand Limited; in Russia by OOO Goldman Sachs; in Singapore by Goldman Sachs (Singapore) Pte. (Company Number: 198602165W); and in the United States of America by Goldman Sachs & Co. LLC. Goldman Sachs International has approved this research in connection with its distribution in the United Kingdom.

Goldman Sachs International ("GSI"), authorised by the Prudential Regulation Authority ("PRA") and regulated by the Financial Conduct Authority ("FCA") and the PRA, has approved this research in connection with its distribution in the United Kingdom.

European Economic Area: Goldman Sachs Bank Europe SE ("GSBE") is a credit institution incorporated in Germany and, within the Single Supervisory Mechanism, subject to direct prudential supervision by the European Central Bank and in other respects supervised by German Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin) and Deutsche Bundesbank and disseminates research within the European Economic Area.

General disclosures

This research is for our clients only. Other than disclosures relating to Goldman Sachs, this research is based on current public information that we consider reliable, but we do not represent it is accurate or complete, and it should not be relied on as such. The information, opinions, estimates and forecasts contained herein are as of the date hereof and are subject to change without prior notification. We seek to update our research as appropriate, but various regulations may prevent us from doing so. Other than certain industry reports published on a periodic basis, the large majority of reports are published at irregular intervals as appropriate in the analyst's judgment.

Goldman Sachs conducts a global full-service, integrated investment banking, investment management, and brokerage business. We have investment banking and other business relationships with a substantial percentage of the companies covered by Global Investment Research. Goldman Sachs & Co. LLC, the United States broker dealer, is a member of SIPC (<https://www.sipc.org>).

Our salespeople, traders, and other professionals may provide oral or written market commentary or trading strategies to our clients and principal trading desks that reflect opinions that are contrary to the opinions expressed in this research. Our asset management area, principal trading desks and investing businesses may make investment decisions that are inconsistent with the recommendations or views expressed in this research.

We and our affiliates, officers, directors, and employees will from time to time have long or short positions in, act as principal in, and buy or sell, the securities or derivatives, if any, referred to in this research, unless otherwise prohibited by regulation or Goldman Sachs policy.

The views attributed to third party presenters at Goldman Sachs arranged conferences, including individuals from other parts of Goldman Sachs, do not necessarily reflect those of Global Investment Research and are not an official view of Goldman Sachs.

Any third party referenced herein, including any salespeople, traders and other professionals or members of their household, may have positions in the products mentioned that are inconsistent with the views expressed by analysts named in this report.

This research is focused on investment themes across markets, industries and sectors. It does not attempt to distinguish between the prospects or performance of, or provide analysis of, individual companies within any industry or sector we describe.

Any trading recommendation in this research relating to an equity or credit security or securities within an industry or sector is reflective of the investment theme being discussed and is not a recommendation of any such security in isolation.

This research is not an offer to sell or the solicitation of an offer to buy any security in any jurisdiction where such an offer or solicitation would be illegal. It does not constitute a personal recommendation or take into account the particular investment objectives, financial situations, or needs of individual clients. Clients should consider whether any advice or recommendation in this research is suitable for their particular circumstances and, if appropriate, seek professional advice, including tax advice. The price and value of investments referred to in this research and the income from them may fluctuate. Past performance is not a guide to future performance, future returns are not guaranteed, and a loss of original capital may occur. Fluctuations in exchange rates could have adverse effects on the value or price of, or income derived from, certain investments.

Certain transactions, including those involving futures, options, and other derivatives, give rise to substantial risk and are not suitable for all investors. Investors should review current options and futures disclosure documents which are available from Goldman Sachs sales representatives or at <https://www.theocc.com/about/publications/character-risks.jsp> and https://www.goldmansachs.com/disclosures/cftc_fcm_disclosures. Transaction costs may be significant in option strategies calling for multiple purchase and sales of options such as spreads. Supporting documentation will be supplied upon request.

Differing Levels of Service provided by Global Investment Research: The level and types of services provided to you by Goldman Sachs Global Investment Research may vary as compared to that provided to internal and other external clients of GS, depending on various factors including your individual preferences as to the frequency and manner of receiving communication, your risk profile and investment focus and perspective (e.g.,

marketwide, sector specific, long term, short term), the size and scope of your overall client relationship with GS, and legal and regulatory constraints. As an example, certain clients may request to receive notifications when research on specific securities is published, and certain clients may request that specific data underlying analysts' fundamental analysis available on our internal client websites be delivered to them electronically through data feeds or otherwise. No change to an analyst's fundamental research views (e.g., ratings, price targets, or material changes to earnings estimates for equity securities), will be communicated to any client prior to inclusion of such information in a research report broadly disseminated through electronic publication to our internal client websites or through other means, as necessary, to all clients who are entitled to receive such reports.

All research reports are disseminated and available to all clients simultaneously through electronic publication to our internal client websites. Not all research content is redistributed to our clients or available to third-party aggregators, nor is Goldman Sachs responsible for the redistribution of our research by third party aggregators. For research, models or other data related to one or more securities, markets or asset classes (including related services) that may be available to you, please contact your GS representative or go to <https://research.gs.com>.

Disclosure information is also available at <https://www.gs.com/research/hedge.html> or from Research Compliance, 200 West Street, New York, NY 10282.

© 2026 Goldman Sachs.

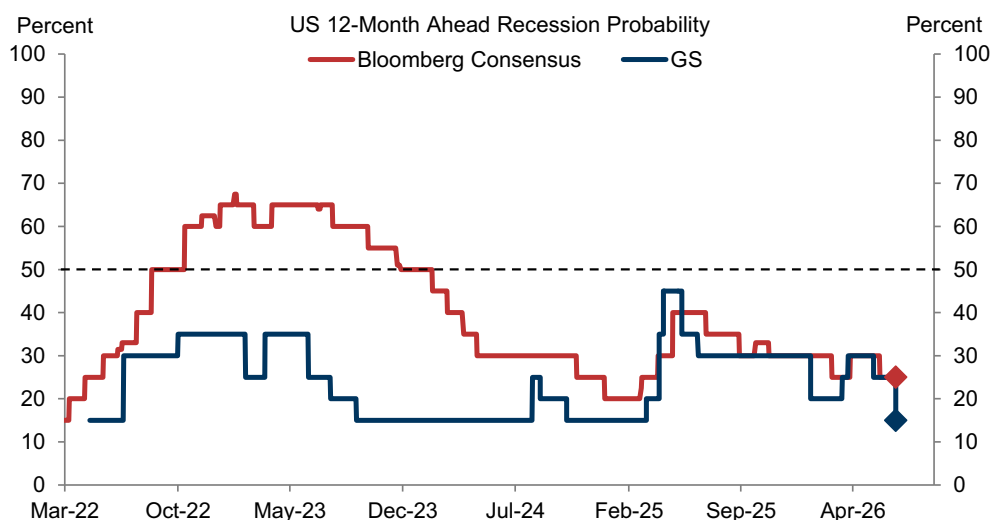
You are permitted to store, display, analyze, modify, reformat, and print the information made available to you via this service only for your own use. You may not resell or reverse engineer this information to calculate or develop any index for disclosure and/or marketing or create any other derivative works or commercial product(s), data or offering(s) without the express written consent of Goldman Sachs. You are not permitted to publish, transmit, or otherwise reproduce this information, in whole or in part, in any format to any third party without the express written consent of Goldman Sachs. This foregoing restriction includes, without limitation, using, extracting, downloading or retrieving this information, in whole or in part, to train or finetune a machine learning or artificial intelligence system, or to provide or reproduce this information, in whole or in part, as a prompt or input to any such system.

Global Views: More Crude, Less Concern

1. The agreement between the US and Iran has reduced the downside risks to our economic outlook. Our commodities strategists now see Brent crude at \$80 per barrel by the end of 2026, with two-sided risks. On the upside, Iran's announcement on Saturday that the Strait was closed again served as a reminder that oil flows might only recover slowly. On the downside, a near-term glut could develop as oil is released quickly into a market that was already oversupplied before the war. The agreement has led us to cut our 12-month US recession risk estimate further from 25% to the long-term norm of 15%. (This is below our 20% estimate on the eve of the war because the labor market improvement since then indicates greater underlying resilience.)

Jan Hatzius
+1(212)902-0394 | jan.hatzius@gs.com
Goldman Sachs & Co. LLC

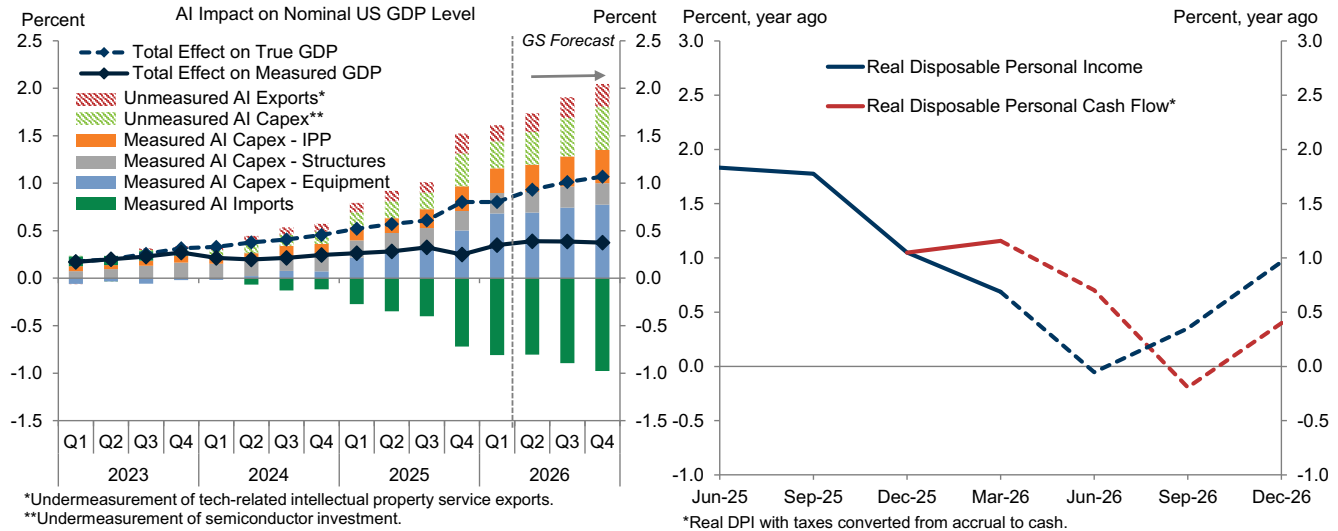
Exhibit 1: US Recession Risk Back to the Long-Term Norm of 15%



Source: Bloomberg, Goldman Sachs Global Investment Research

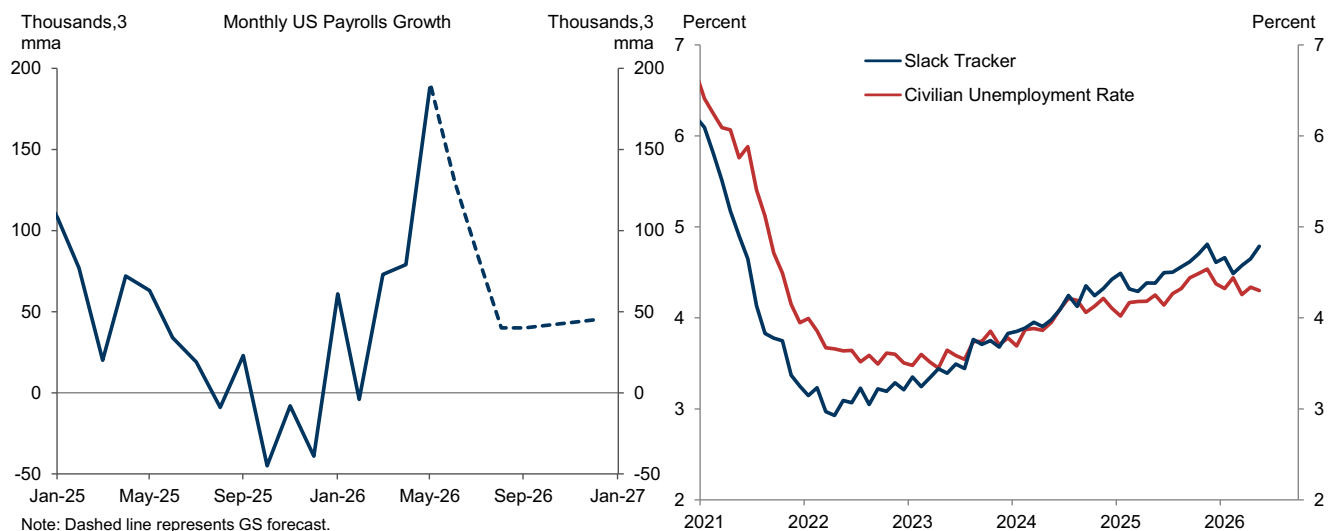
2. Alongside the cut to our recession risk estimate, we have nudged up our sequential H2 GDP growth forecast to 2%. The slightly stronger path reflects a positive sequential impulse to real income from lower gas prices, at a time when the economy continues to benefit from the AI boom via higher equity wealth as well as strong capex. But we still expect growth to remain moderate. First, AI capex largely consists of goods that are imported from Asia and, in the case of semiconductors, don't even show up in GDP. Second, even with the drop in gas prices, real income defined on a cash flow basis is likely to slow in H2 as the impact of higher tax refunds and lower final tax payments in Q2 peters out. This is likely to keep real consumer spending growth at just 1½%.

Investors should consider this report as only a single factor in making their investment decision. For Reg AC certification and other important disclosures, see the Disclosure Appendix, or go to www.gs.com/research/hedge.html.

Exhibit 2: AI Investment Surge Adds Little to US GDP and the US Consumer Remains Under Pressure

Source: Bureau of Economic Analysis, Haver Analytics, Goldman Sachs Global Investment Research

3. With GDP still growing modestly below potential, we expect a slowdown in payroll gains from the outsized 188k rate over the past three months to just below our 60k breakeven estimate. Beyond the growth forecast, we would note the failure of other labor market indicators to confirm a sharp recent improvement. Household employment has grown much more slowly than payrolls, the employment/population ratio has trended sideways to lower, and our updated slack tracker—which summarizes ten key indicators of labor market utilization—has recently resumed its gradual loosening of the prior three years. Consistent with this, we expect wage and unit labor cost growth to remain muted despite the earlier increase in headline inflation.

Exhibit 3: The Payroll Surge Is an Outlier Relative to Other Labor Market Indicators

Source: US Bureau of Labor Statistics, Haver Analytics, Goldman Sachs Global Investment Research

4. The plunge in gasoline prices so far in June should result in an outright decline in seasonally adjusted consumer prices. We also expect core CPI inflation to average just 0.17% in the next three months as airfares decline on the back of lower jet fuel prices, hotel rates—which are measured at the time of booking—drop from their World Cup

highs, and rent inflation reverses what looks like a high-side outlier in the Northeast in May. Core PCE inflation is likely to remain more stubborn, in part because of ongoing upward pressure on imputed financial services prices from the equity market rally and price increases in software/accessories whose weight in core PCE is 30 times larger than in core CPI. More statistically based measures such as trimmed-mean PCE should continue to look more benign, however.

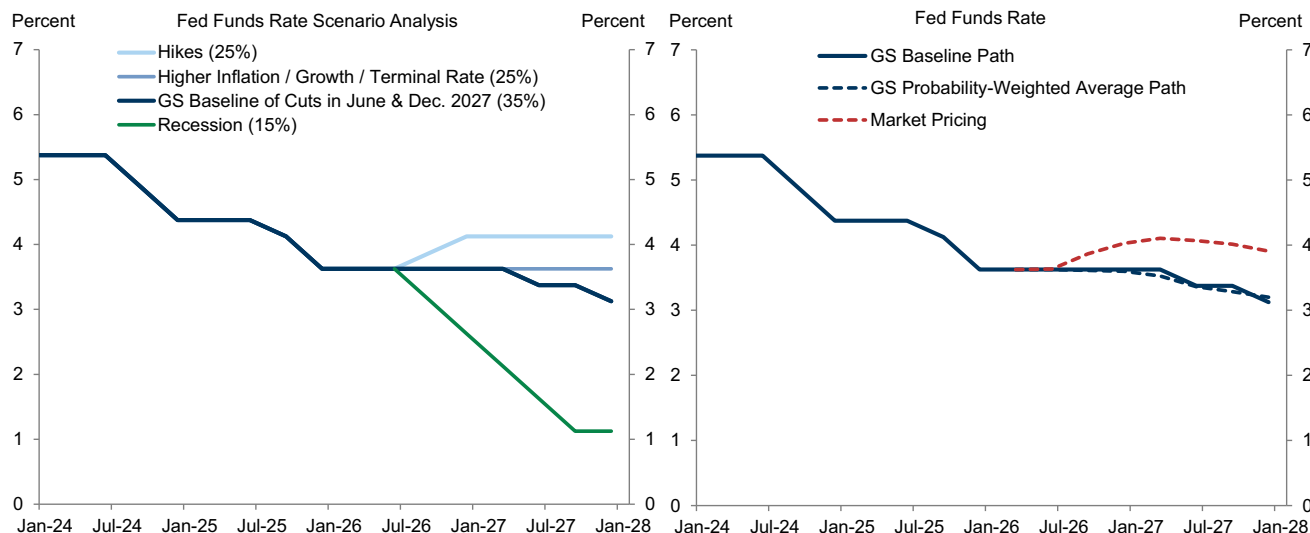
Exhibit 4: Trimmed-Mean PCE Still Looks More Benign Than Core PCE



Note: May 2026 values are GS forecasts based on inflation tracking estimates.

Source: Haver Analytics, Goldman Sachs Global Investment Research

5. The first FOMC meeting under Chairman Warsh was more hawkish than expected. Half of the 18 participants that submitted interest rate projections penciled in one or more rate hikes in the remainder of 2026, and the median projection for core PCE inflation in 2027Q4 rose to 2.5%. While this has increased the upside risks, our baseline forecast remains for no hikes. First, we suspect that about half of the 2026 hikers are nonvoting Federal Reserve Bank presidents, which would leave a majority of voters projecting unchanged or lower rates under appropriate policy. Second, the projections might be stale given the rapid improvement in the news from the Middle East and the plunge in energy prices over the past week. If inflation comes down in coming months—rapidly for headline and gradually for core—and growth remains muted, we expect most voters to continue supporting unchanged rates.

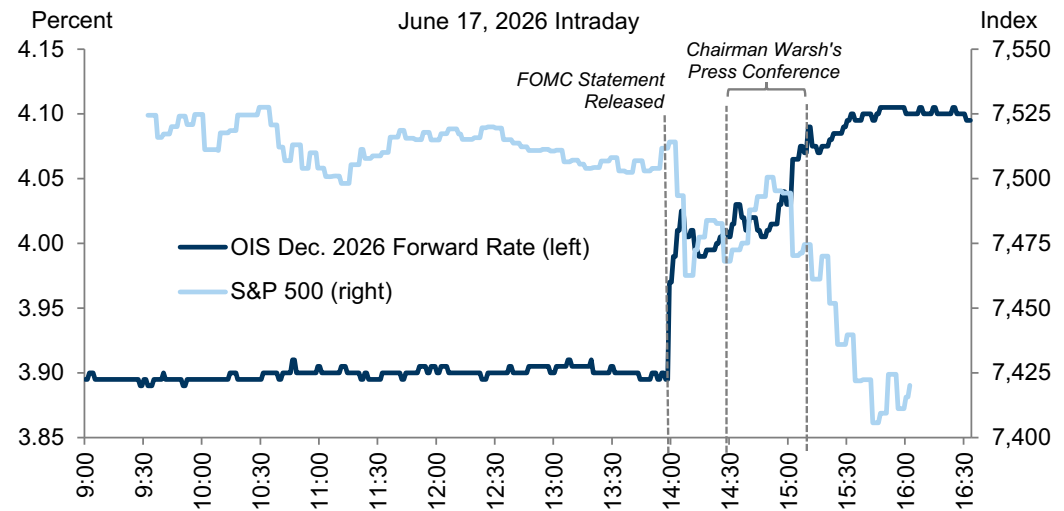
Exhibit 5: While Hikes Have Become More Likely, We Think They Are Less Likely Than Markets Are Pricing

Source: Bloomberg, Goldman Sachs Global Investment Research

6. We had expected Warsh's refusal to provide either a "dot" or verbal guidance about upcoming rate decisions but his broader comments on Fed communications with financial markets went further:

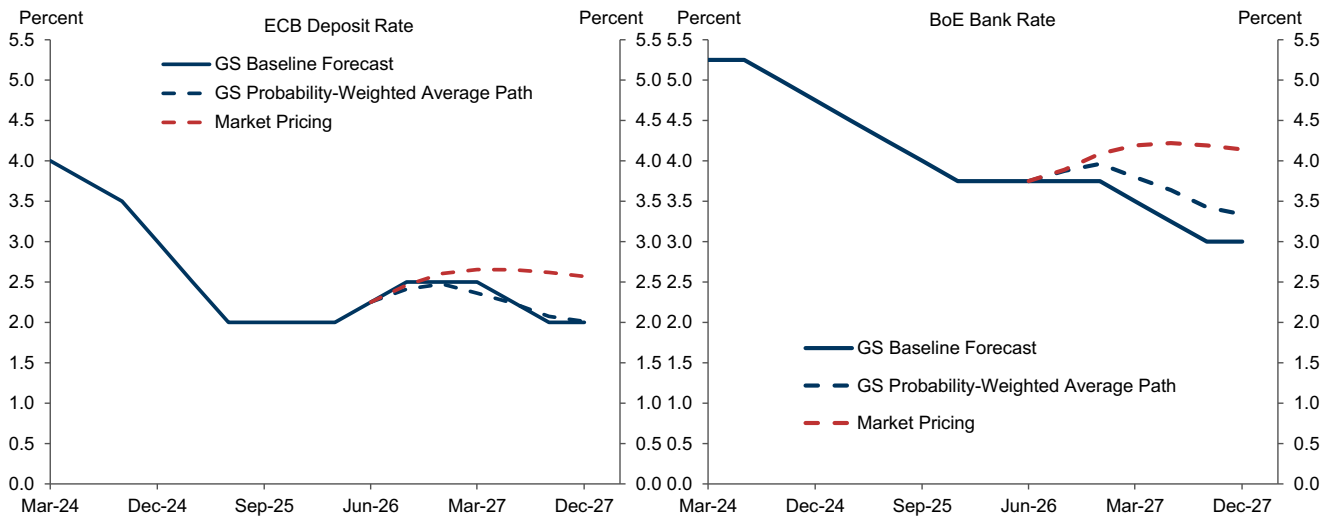
"So I think financial markets perform best when they react to incoming data. I think the financial markets work less efficiently when they ask [the] question 'how will the Federal Reserve react to that incoming information?' The more that markets are paying attention to what's happening in the real economy, deciding what's good data and what's less good data, the more financial markets can price what they believe is the most likely and what are the tail risks."

Since the Fed essentially controls short-term interest rates and the interest rate path matters for asset pricing, it is inevitable that markets ask how the Fed will react to incoming information. Admittedly, this can turn into an unproductive obsession with computing the Fed's reaction function, which is often harder to summarize and subject to greater change than many market participants think. Still, a significant reduction in Fed transparency could make financial conditions—and thus economic outcomes—more volatile as market assumptions about the Fed's thinking become subject to fewer reality checks.

Exhibit 6: FOMC Meeting Illustrates the Importance of Fed Communications for Financial Conditions


Source: Bloomberg, Goldman Sachs Global Investment Research

7. The ECB hike to 2.25% was widely expected, but the drop in energy prices has shifted the balance of risks around our forecast of one more 25bp move in September to the downside. Either way, we continue to expect a return to a 2% deposit rate in 2027 and thus remain dovish relative to market pricing. In the UK, Sir Keir Starmer has stepped down as prime minister in favor of Andy Burnham, who inherits an economy with more potential than widely appreciated. On monetary policy, the improvement in wage and price inflation even prior to the recent oil price drop has strengthened our view that the MPC won't hike in coming months and will ultimately cut Bank Rate to 3% in 2027, well below market pricing.

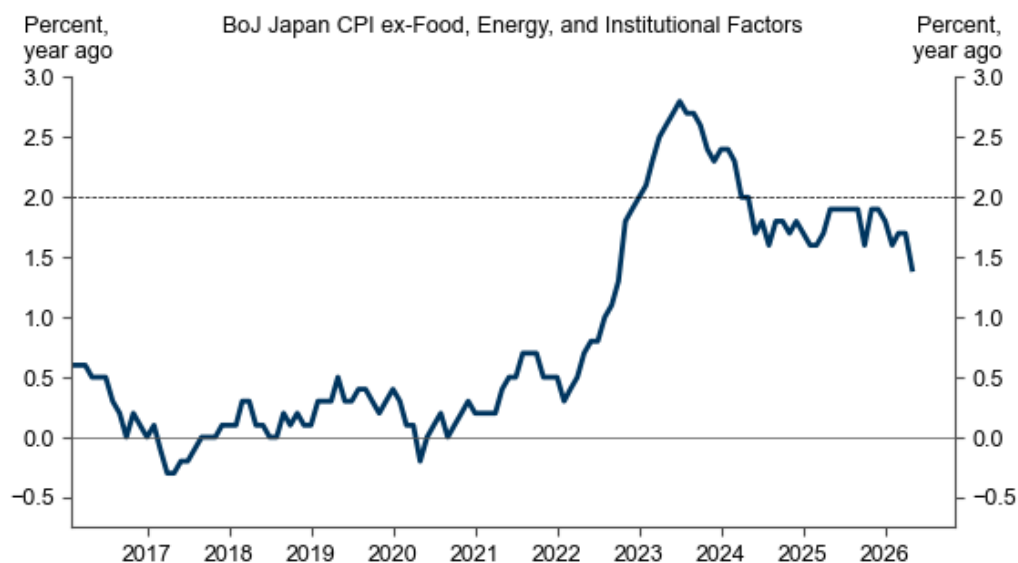
Exhibit 7: Our ECB and BoE Rate Views Remain Below Market Pricing


Source: Bloomberg, Goldman Sachs Global Investment Research

8. The Bank of Japan hiked by 25bp to 1%, the highest level in 30 years. We expect two more 25bp hikes every six months or so until the policy rate hits 1½% next summer. The BoJ can take its time because inflation—using its estimate of CPI excluding all food and

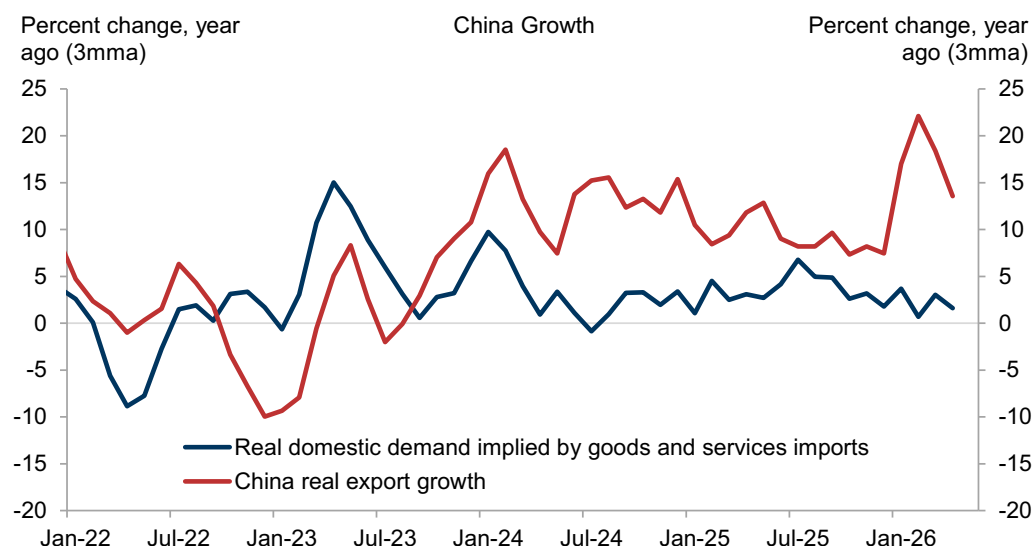
energy as well as institutional factors such as changes in school tuition rates—is running at only about 1½% year-on-year and GDP is likely to grow just 0.5% in 2026. The biggest challenge for the Japanese economy remains fiscal sustainability, as the decline in the debt/GDP ratio over the past decade could reverse if Prime Minister Takaichi implements her expansionary plans and long-term interest rates climb further toward the levels seen in other G10 economies.

Exhibit 8: Underlying Inflation in Japan Remains Benign



Source: BoJ, Haver Analytics, Goldman Sachs Global Investment Research

9. We attribute most of the downside growth surprises in China in April and May to a combination of the oil shock, payback for the consumer goods trade-in program, and the government's decision to slow fiscal spending after the strong Q1 GDP print. June is likely to remain sluggish because of heavy rainfall, but we expect a rebound in Q3 as the temporary drags abate, the oil shock reverses, and policymakers respond to the recent weakness with more fiscal spending. Underneath the short-term ups and downs, the gap between domestic demand and exports continues to grow, which drives up the current account surplus with negative effects on many of China's trading partners.

Exhibit 9: China's Domestic Demand Is Growing Far More Slowly Than Its Exports

Source: Haver Analytics, Goldman Sachs Global Investment Research

10. US financial markets have digested the hawkish FOMC shock well, with declines in breakeven inflation and 30-year Treasury yields alongside a rebound in broad equity indices. If the near-term economic outlook settles into calmer waters, the focus on the sustainability of the AI trade is likely to grow even further. The work of our equity and cross-asset strategists suggests that ongoing upside surprises in AI capex, strength in S&P earnings, and the relative absence of private sector financial imbalances should all help equity markets make further headway in coming months. However, valuations of AI-related stocks are now high even relative to our optimistic views regarding the economic value that is likely to be created by AI over the next decade, so it is getting harder to justify further large gains.

Jan Hatzius

Disclosure Appendix

Reg AC

I, Jan Hatzius, hereby certify that all of the views expressed in this report accurately reflect my personal views, which have not been influenced by considerations of the firm's business or client relationships.

Unless otherwise stated, the individuals listed on the cover page of this report are analysts in Goldman Sachs' Global Investment Research division.

Contributing Authors: Jan Hatzius Goldman Sachs & Co. LLC.

Unless otherwise stated, the individuals listed in the Contributing Authors disclosure of this report are analysts in Goldman Sachs' Global Investment Research division.

Disclosures

Regulatory disclosures

Disclosures required by United States laws and regulations

See company-specific regulatory disclosures above for any of the following disclosures required as to companies referred to in this report: manager or co-manager in a pending transaction; 1% or other ownership; compensation for certain services; types of client relationships; managed/co-managed public offerings in prior periods; directorships; for equity securities, market making and/or specialist role. Goldman Sachs trades or may trade as a principal in debt securities (or in related derivatives) of issuers discussed in this report.

The following are additional required disclosures: **Ownership and material conflicts of interest:** Goldman Sachs policy prohibits its analysts, professionals reporting to analysts and members of their households from owning securities of any company in the analyst's area of coverage. **Analyst compensation:** Analysts are paid in part based on the profitability of Goldman Sachs, which includes investment banking revenues. **Analyst as officer or director:** Goldman Sachs policy generally prohibits its analysts, persons reporting to analysts or members of their households from serving as an officer, director or advisor of any company in the analyst's area of coverage. **Non-U.S. Analysts:** Non-U.S. analysts may not be associated persons of Goldman Sachs & Co. LLC and therefore may not be subject to FINRA Rule 2241 or FINRA Rule 2242 restrictions on communications with a subject company, public appearances and trading in securities covered by the analysts.

Additional disclosures required under the laws and regulations of jurisdictions other than the United States

The following disclosures are those required by the jurisdiction indicated, except to the extent already made above pursuant to United States laws and regulations. **Australia:** Goldman Sachs Australia Pty Ltd and its affiliates are not authorised deposit-taking institutions (as that term is defined in the Banking Act 1959 (Cth)) in Australia and do not provide banking services, nor carry on a banking business, in Australia. This research, and any access to it, is intended only for "wholesale clients" within the meaning of the Australian Corporations Act, unless otherwise agreed by Goldman Sachs. In producing research reports, members of Global Investment Research of Goldman Sachs Australia may attend site visits and other meetings hosted by the companies and other entities which are the subject of its research reports. In some instances the costs of such site visits or meetings may be met in part or in whole by the issuers concerned if Goldman Sachs Australia considers it is appropriate and reasonable in the specific circumstances relating to the site visit or meeting. To the extent that the contents of this document contains any financial product advice, it is general advice only and has been prepared by Goldman Sachs without taking into account a client's objectives, financial situation or needs. A client should, before acting on any such advice, consider the appropriateness of the advice having regard to the client's own objectives, financial situation and needs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests and a copy of Goldman Sachs' Australian Sell-Side Research Independence Policy Statement are available at: <https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html>. **Brazil:** Disclosure information in relation to CVM Resolution n. 20 is available at <https://www.gs.com/worldwide/brazil/area/gir/index.html>. Where applicable, the Brazil-registered analyst primarily responsible for the content of this research report, as defined in Article 20 of CVM Resolution n. 20, is the first author named at the beginning of this report, unless indicated otherwise at the end of the text. **Canada:** This information is being provided to you for information purposes only and is not, and under no circumstances should be construed as, an advertisement, offering or solicitation by Goldman Sachs & Co. LLC for purchasers of securities in Canada to trade in any Canadian security. Goldman Sachs & Co. LLC is not registered as a dealer in any jurisdiction in Canada under applicable Canadian securities laws and generally is not permitted to trade in Canadian securities and may be prohibited from selling certain securities and products in certain jurisdictions in Canada. If you wish to trade in any Canadian securities or other products in Canada please contact Goldman Sachs Canada Inc., an affiliate of The Goldman Sachs Group Inc., or another registered Canadian dealer. **Hong Kong:** Further information on the securities of covered companies referred to in this research may be obtained on request from Goldman Sachs (Asia) L.L.C. **India:** Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (India) Securities Private Limited, Research Analyst - SEBI Registration Number INH000001493, 10th Floor, Ascent-Worli, Sudam Kalu Ahire Marg, Worli, Mumbai-400 025, India, Corporate Identity Number U74140MH2006FTC160634, Phone +91 22 6616 9000, Fax +91 22 6616 9001. Goldman Sachs may beneficially own 1% or more of the securities (as such term is defined in clause 2 (h) the Indian Securities Contracts (Regulation) Act, 1956) of the subject company or companies referred to in this research report. Investment in securities market are subject to market risks. Read all the related documents carefully before investing. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Goldman Sachs (India) Securities Private Limited compliance officer and investor grievance contact details can be found at: <https://www.goldmansachs.com/worldwide/india/documents/Grievance-Redressal-and-Escalation-Matrix.pdf>, and a copy of the annual audit compliance report can be found at this link: <https://publishing.gs.com/content/site/india-annual-compliance-report.html>. **Japan:** See below. **Korea:** This research, and any access to it, is intended only for "professional investors" within the meaning of the Financial Services and Capital Markets Act, unless otherwise agreed by Goldman Sachs. Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (Asia) L.L.C., Seoul Branch. **New Zealand:** Goldman Sachs New Zealand Limited and its affiliates are neither "registered banks" nor "deposit takers" (as defined in the Reserve Bank of New Zealand Act 1989) in New Zealand. This research, and any access to it, is intended for "wholesale clients" (as defined in the Financial Advisers Act 2008) unless otherwise agreed by Goldman Sachs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests is available at: <https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html>. **Russia:** Research reports distributed in the Russian Federation are not advertising as defined in the Russian legislation, but are information and analysis not having product promotion as their main purpose and do not provide appraisal within the meaning of the Russian legislation on appraisal activity. Research reports do not constitute a personalized investment recommendation as defined in Russian laws and regulations, are not addressed to a specific client, and are prepared without analyzing the financial circumstances, investment profiles or risk profiles of clients. Goldman Sachs assumes no responsibility for any investment decisions that may be taken by a client or any other person based on this research report. **Singapore:** Goldman Sachs (Singapore) Pte. (Company Number: 198602165W), which is regulated by the Monetary Authority of Singapore, accepts legal responsibility for this research, and should be contacted with respect to any matters arising from, or in connection with, this research. **Taiwan:** This material is for reference only and must not be reprinted without permission. Investors should carefully consider their own investment risk. Investment results are the responsibility of the individual investor. **United Kingdom:** Persons who would be categorized as retail clients in the United Kingdom, as such term is defined in the rules of the Financial Conduct Authority, should read this research in conjunction with prior Goldman Sachs research on the covered

companies referred to herein and should refer to the risk warnings that have been sent to them by Goldman Sachs International. A copy of these risks warnings, and a glossary of certain financial terms used in this report, are available from Goldman Sachs International on request.

European Union and United Kingdom: Disclosure information in relation to Article 6 (2) of the European Commission Delegated Regulation (EU) (2016/958) supplementing Regulation (EU) No 596/2014 of the European Parliament and of the Council (including as that Delegated Regulation is implemented into United Kingdom domestic law and regulation following the United Kingdom's departure from the European Union and the European Economic Area) with regard to regulatory technical standards for the technical arrangements for objective presentation of investment recommendations or other information recommending or suggesting an investment strategy and for disclosure of particular interests or indications of conflicts of interest is available at <https://www.gs.com/disclosures/europeanpolicy.html> which states the European Policy for Managing Conflicts of Interest in Connection with Investment Research.

Japan: Goldman Sachs Japan Co., Ltd. is a Financial Instrument Dealer registered with the Kanto Financial Bureau under registration number Kinsho 69, and a member of Japan Securities Dealers Association, Financial Futures Association of Japan Type II Financial Instruments Firms Association, and Investment Management Association of Japan. Sales and purchase of equities are subject to commission pre-determined with clients plus consumption tax. See company-specific disclosures as to any applicable disclosures required by Japanese stock exchanges, the Japanese Securities Dealers Association or the Japanese Securities Finance Company.

Global product; distributing entities

Goldman Sachs Global Investment Research produces and distributes research products for clients of Goldman Sachs on a global basis. Analysts based in Goldman Sachs offices around the world produce research on industries and companies, and research on macroeconomics, currencies, commodities and portfolio strategy. This research is disseminated in Australia by Goldman Sachs Australia Pty Ltd (ABN 21 006 797 897); in Brazil by Goldman Sachs do Brasil Corretora de Títulos e Valores Mobiliários S.A.; Public Communication Channel Goldman Sachs Brazil: 0800 727 5764 and / or contatogoldmanbrasil@gs.com. Available Weekdays (except holidays), from 9am to 6pm. Canal de Comunicação com o Público Goldman Sachs Brasil: 0800 727 5764 e/ou contatogoldmanbrasil@gs.com. Horário de funcionamento: segunda-feira à sexta-feira (exceto feriados), das 9h às 18h; in Canada by Goldman Sachs & Co. LLC; in Hong Kong by Goldman Sachs (Asia) L.L.C.; in India by Goldman Sachs (India) Securities Private Ltd.; in Japan by Goldman Sachs Japan Co., Ltd.; in the Republic of Korea by Goldman Sachs (Asia) L.L.C., Seoul Branch; in New Zealand by Goldman Sachs New Zealand Limited; in Russia by OOO Goldman Sachs; in Singapore by Goldman Sachs (Singapore) Pte. (Company Number: 198602165W); and in the United States of America by Goldman Sachs & Co. LLC. Goldman Sachs International has approved this research in connection with its distribution in the United Kingdom.

Goldman Sachs International ("GSI"), authorised by the Prudential Regulation Authority ("PRA") and regulated by the Financial Conduct Authority ("FCA") and the PRA, has approved this research in connection with its distribution in the United Kingdom.

European Economic Area: Goldman Sachs Bank Europe SE ("GSBE") is a credit institution incorporated in Germany and, within the Single Supervisory Mechanism, subject to direct prudential supervision by the European Central Bank and in other respects supervised by German Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin) and Deutsche Bundesbank and disseminates research within the European Economic Area.

General disclosures

This research is for our clients only. Other than disclosures relating to Goldman Sachs, this research is based on current public information that we consider reliable, but we do not represent it is accurate or complete, and it should not be relied on as such. The information, opinions, estimates and forecasts contained herein are as of the date hereof and are subject to change without prior notification. We seek to update our research as appropriate, but various regulations may prevent us from doing so. Other than certain industry reports published on a periodic basis, the large majority of reports are published at irregular intervals as appropriate in the analyst's judgment.

Goldman Sachs conducts a global full-service, integrated investment banking, investment management, and brokerage business. We have investment banking and other business relationships with a substantial percentage of the companies covered by Global Investment Research. Goldman Sachs & Co. LLC, the United States broker dealer, is a member of SIPC (<https://www.sipc.org>).

Our salespeople, traders, and other professionals may provide oral or written market commentary or trading strategies to our clients and principal trading desks that reflect opinions that are contrary to the opinions expressed in this research. Our asset management area, principal trading desks and investing businesses may make investment decisions that are inconsistent with the recommendations or views expressed in this research.

We and our affiliates, officers, directors, and employees will from time to time have long or short positions in, act as principal in, and buy or sell, the securities or derivatives, if any, referred to in this research, unless otherwise prohibited by regulation or Goldman Sachs policy.

The views attributed to third party presenters at Goldman Sachs arranged conferences, including individuals from other parts of Goldman Sachs, do not necessarily reflect those of Global Investment Research and are not an official view of Goldman Sachs.

Any third party referenced herein, including any salespeople, traders and other professionals or members of their household, may have positions in the products mentioned that are inconsistent with the views expressed by analysts named in this report.

This research is focused on investment themes across markets, industries and sectors. It does not attempt to distinguish between the prospects or performance of, or provide analysis of, individual companies within any industry or sector we describe.

Any trading recommendation in this research relating to an equity or credit security or securities within an industry or sector is reflective of the investment theme being discussed and is not a recommendation of any such security in isolation.

This research is not an offer to sell or the solicitation of an offer to buy any security in any jurisdiction where such an offer or solicitation would be illegal. It does not constitute a personal recommendation or take into account the particular investment objectives, financial situations, or needs of individual clients. Clients should consider whether any advice or recommendation in this research is suitable for their particular circumstances and, if appropriate, seek professional advice, including tax advice. The price and value of investments referred to in this research and the income from them may fluctuate. Past performance is not a guide to future performance, future returns are not guaranteed, and a loss of original capital may occur. Fluctuations in exchange rates could have adverse effects on the value or price of, or income derived from, certain investments.

Certain transactions, including those involving futures, options, and other derivatives, give rise to substantial risk and are not suitable for all investors. Investors should review current options and futures disclosure documents which are available from Goldman Sachs sales representatives or at <https://www.theocc.com/about/publications/character-risks.jsp> and https://www.goldmansachs.com/disclosures/cftc_fcm_disclosures. Transaction costs may be significant in option strategies calling for multiple purchase and sales of options such as spreads. Supporting documentation will be supplied upon request.

Differing Levels of Service provided by Global Investment Research: The level and types of services provided to you by Goldman Sachs Global Investment Research may vary as compared to that provided to internal and other external clients of GS, depending on various factors including your individual preferences as to the frequency and manner of receiving communication, your risk profile and investment focus and perspective (e.g.,

marketwide, sector specific, long term, short term), the size and scope of your overall client relationship with GS, and legal and regulatory constraints. As an example, certain clients may request to receive notifications when research on specific securities is published, and certain clients may request that specific data underlying analysts' fundamental analysis available on our internal client websites be delivered to them electronically through data feeds or otherwise. No change to an analyst's fundamental research views (e.g., ratings, price targets, or material changes to earnings estimates for equity securities), will be communicated to any client prior to inclusion of such information in a research report broadly disseminated through electronic publication to our internal client websites or through other means, as necessary, to all clients who are entitled to receive such reports.

All research reports are disseminated and available to all clients simultaneously through electronic publication to our internal client websites. Not all research content is redistributed to our clients or available to third-party aggregators, nor is Goldman Sachs responsible for the redistribution of our research by third party aggregators. For research, models or other data related to one or more securities, markets or asset classes (including related services) that may be available to you, please contact your GS representative or go to <https://research.gs.com>.

Disclosure information is also available at <https://www.gs.com/research/hedge.html> or from Research Compliance, 200 West Street, New York, NY 10282.

© 2026 Goldman Sachs.

You are permitted to store, display, analyze, modify, reformat, and print the information made available to you via this service only for your own use. You may not resell or reverse engineer this information to calculate or develop any index for disclosure and/or marketing or create any other derivative works or commercial product(s), data or offering(s) without the express written consent of Goldman Sachs. You are not permitted to publish, transmit, or otherwise reproduce this information, in whole or in part, in any format to any third party without the express written consent of Goldman Sachs. This foregoing restriction includes, without limitation, using, extracting, downloading or retrieving this information, in whole or in part, to train or finetune a machine learning or artificial intelligence system, or to provide or reproduce this information, in whole or in part, as a prompt or input to any such system.

US Daily: June FOMC Recap: Hikes Look More Likely but Are Still Not Our Base Case

- The June FOMC meeting delivered a hawkish surprise, with nine participants projecting a hike in 2026 (vs. our expectation of three) and the median projection for 2027 Q4/Q4 core PCE inflation rising to 2.5% (vs. our expectation of 2.3%). Market pricing for the fed funds rate at the end of 2026 rose by 20bp in response to the projections and Chairman Kevin Warsh's press conference.
- Today's meeting raises the risk of interest rate hikes later this year, but our base case is still that the FOMC will leave the policy rate unchanged this year. A majority could support a hike, perhaps for a range of different reasons, if upcoming inflation prints are uncomfortable and job growth remains solid. But we suspect that most of the voters still lean toward an unchanged policy rate and that participants treated the news about a deal with Iran and the reopening of the Strait of Hormuz cautiously for now. If so, today's projections could prove stale if trade resumes, which should resolve the most serious source of upside risk to inflation fairly quickly.
- Warsh also announced that he is creating five task forces to reconsider Fed communications, the Fed's balance sheet, the Fed's use of and reliance on existing data sources, productivity and jobs in "an era of transformation," and the Fed's inflation frameworks. He appeared particularly focused on rethinking Fed communications. Fed officials have already discussed some of these topics recently, and it is hard to know if the task forces will reach different conclusions.

David Mericle
+1(212)357-2619 |
david.mericle@gs.com
Goldman Sachs & Co. LLC

June FOMC Recap: Hikes Look More Likely but Are Still Not Our Base Case

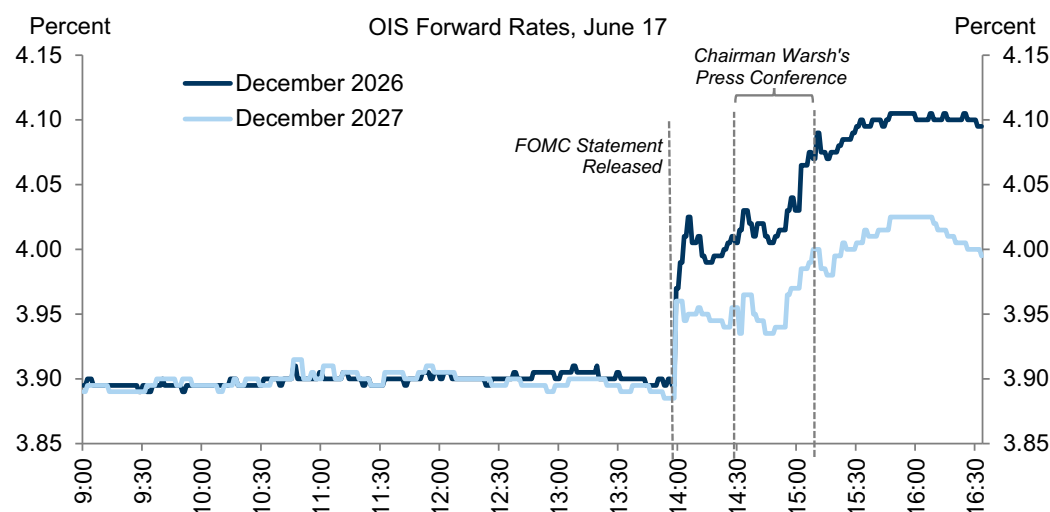
As widely expected, the FOMC left the funds rate unchanged at 3.5-3.75% today and removed the previous forward guidance suggesting cuts from its statement at the June FOMC meeting today. The FOMC also substantially shortened and simplified its post-meeting statement.

But the meeting also delivered a significant hawkish surprise, with nine participants projecting a rate hike in 2026, against our expectation that just three would. Because the new Chairman Kevin Warsh chose not to submit projections, this resulted in a nine to nine tie in the dot plot between those projecting at least one hike and those projecting an unchanged policy rate or, in one case, a rate cut.

Consistent with the hawkish message from the dots, the key surprise in the economic projections was that the median forecast for 2027 Q4/Q4 core PCE inflation rose 0.3pp to 2.5%, above our expectation that it would rise more modestly to 2.3%. This suggests that participants were more concerned that the recent firmer inflation news, caused in part by the energy cost shock from the war with Iran, has implications for inflation further down the road.

It was not clear where Warsh stood on the likelihood of a hike being appropriate from the press conference. He took a balanced view on whether the current stance of monetary policy is restrictive, noting that it is likely restraining housing activity but it is hard to say the same of the impact on financial markets. But he repeatedly emphasized the Fed's commitment to price stability, which might make him sympathetic to a hike if upcoming inflation reports are discouraging. And he said that "financial market prices are probably the most important source of information to guide central bankers," which might imply that he sees market pricing of a higher funds rate later this year as supporting the case for hiking.

In response to both the projections and Warsh's press conference, market pricing for the fed funds rate at the end of 2026 rose by 20bp and pricing for the end of 2027 rose by 11bp (Exhibit 1).

Exhibit 1: The Bond Market Took Both the FOMC's Projections at 2pm and Chairman Warsh's Press Conference Starting at 2:30 as Substantial Hawkish Surprises


Source: Goldman Sachs Global Investment Research

Today's meeting raises the risk of interest rate hikes later this year, but our base case is still that the FOMC will leave the policy rate unchanged this year. A majority could support a hike, perhaps for a range of different reasons, if upcoming inflation prints are uncomfortable and job growth remains solid. Some might be more concerned about recent inflation trends, while others might be less convinced that a hike is necessary but see little harm from a starting point where job growth has been running strong. A great deal will depend on upcoming inflation reports, and we do expect that May core PCE inflation will be firm at 31bp, though 12bp of that would come from financial services.

But we suspect that a majority of the twelve voters, as opposed to the nineteen participants, still lean toward an unchanged policy rate. We also suspect that participants treated the news about a deal with Iran and the reopening of the Strait of Hormuz cautiously for now. If so, today's projections could prove stale if trade resumes, which should resolve the most serious source of upside risk to inflation fairly quickly. Warsh noted that "my colleagues around the table, when they submitted their dots, understand the world is changing quite quickly" and will not feel bound by their projections.

Warsh also announced that he is creating five task forces to investigate Fed communications, the Fed's balance sheet, the Fed's use of and reliance on existing data sources, productivity and jobs in "an era of transformation," and the Fed's inflation frameworks. The task forces will include Fed staff economists as well as people from outside the Fed, including non-economists.

Warsh appeared particularly focused on reconsidering Fed communications and especially on making changes to the Summary of Economic Projections. He also said that "press conferences are useful, but when you have one you want to make sure you have something important to say," which could hint at a break from the current pattern of press conferences after each of the eight FOMC meetings.

Fed officials have already discussed some of the topics assigned to the task forces

recently. As Warsh noted, the FOMC discussed its communications practices as part of its 2025 framework review. It also discussed balance sheet policy as part of its 2020 framework review, and the Fed staff have written a number of research notes on the balance sheet recently. At this early stage, it is hard to know if the new task forces will reach different conclusions and ultimately produce meaningful policy changes.

David Mericle

The US Economics Team

Jan Hatzius

+1(212)902-0394
jan.hatzius@gs.com
Goldman Sachs & Co. LLC

Alec Phillips

+1(202)637-3746
alec.phillips@gs.com
Goldman Sachs & Co. LLC

David Mericle

+1(212)357-2619
david.mericle@gs.com
Goldman Sachs & Co. LLC

Ronnie Walker

+1(917)343-4543
ronnie.walker@gs.com
Goldman Sachs & Co. LLC

Manuel Abecasis

+1(212)902-8357
manuel.abecasis@gs.com
Goldman Sachs & Co. LLC

Elsie Peng

+1(212)357-3137
elsie.peng@gs.com
Goldman Sachs & Co. LLC

Pierfrancesco Mei

+1(212)902-8809
pierfrancesco.mei@gs.com
Goldman Sachs & Co. LLC

Jessica Rindels

+1(972)368-1516
jessica.rindels@gs.com
Goldman Sachs & Co. LLC

Disclosure Appendix

Reg AC

I, David Mericle, hereby certify that all of the views expressed in this report accurately reflect my personal views, which have not been influenced by considerations of the firm's business or client relationships.

Unless otherwise stated, the individuals listed on the cover page of this report are analysts in Goldman Sachs' Global Investment Research division.

Contributing Authors: David Mericle Goldman Sachs & Co. LLC.

Unless otherwise stated, the individuals listed in the Contributing Authors disclosure of this report are analysts in Goldman Sachs' Global Investment Research division.

Disclosures

Regulatory disclosures

Disclosures required by United States laws and regulations

See company-specific regulatory disclosures above for any of the following disclosures required as to companies referred to in this report: manager or co-manager in a pending transaction; 1% or other ownership; compensation for certain services; types of client relationships; managed/co-managed public offerings in prior periods; directorships; for equity securities, market making and/or specialist role. Goldman Sachs trades or may trade as a principal in debt securities (or in related derivatives) of issuers discussed in this report.

The following are additional required disclosures: **Ownership and material conflicts of interest:** Goldman Sachs policy prohibits its analysts, professionals reporting to analysts and members of their households from owning securities of any company in the analyst's area of coverage. **Analyst compensation:** Analysts are paid in part based on the profitability of Goldman Sachs, which includes investment banking revenues. **Analyst as officer or director:** Goldman Sachs policy generally prohibits its analysts, persons reporting to analysts or members of their households from serving as an officer, director or advisor of any company in the analyst's area of coverage. **Non-U.S. Analysts:** Non-U.S. analysts may not be associated persons of Goldman Sachs & Co. LLC and therefore may not be subject to FINRA Rule 2241 or FINRA Rule 2242 restrictions on communications with a subject company, public appearances and trading in securities covered by the analysts.

Additional disclosures required under the laws and regulations of jurisdictions other than the United States

The following disclosures are those required by the jurisdiction indicated, except to the extent already made above pursuant to United States laws and regulations. **Australia:** Goldman Sachs Australia Pty Ltd and its affiliates are not authorised deposit-taking institutions (as that term is defined in the Banking Act 1959 (Cth)) in Australia and do not provide banking services, nor carry on a banking business, in Australia. This research, and any access to it, is intended only for "wholesale clients" within the meaning of the Australian Corporations Act, unless otherwise agreed by Goldman Sachs. In producing research reports, members of Global Investment Research of Goldman Sachs Australia may attend site visits and other meetings hosted by the companies and other entities which are the subject of its research reports. In some instances the costs of such site visits or meetings may be met in part or in whole by the issuers concerned if Goldman Sachs Australia considers it is appropriate and reasonable in the specific circumstances relating to the site visit or meeting. To the extent that the contents of this document contains any financial product advice, it is general advice only and has been prepared by Goldman Sachs without taking into account a client's objectives, financial situation or needs. A client should, before acting on any such advice, consider the appropriateness of the advice having regard to the client's own objectives, financial situation and needs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests and a copy of Goldman Sachs' Australian Sell-Side Research Independence Policy Statement are available at: <https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html>. **Brazil:** Disclosure information in relation to CVM Resolution n. 20 is available at <https://www.gs.com/worldwide/brazil/area/gir/index.html>. Where applicable, the Brazil-registered analyst primarily responsible for the content of this research report, as defined in Article 20 of CVM Resolution n. 20, is the first author named at the beginning of this report, unless indicated otherwise at the end of the text. **Canada:** This information is being provided to you for information purposes only and is not, and under no circumstances should be construed as, an advertisement, offering or solicitation by Goldman Sachs & Co. LLC for purchasers of securities in Canada to trade in any Canadian security. Goldman Sachs & Co. LLC is not registered as a dealer in any jurisdiction in Canada under applicable Canadian securities laws and generally is not permitted to trade in Canadian securities and may be prohibited from selling certain securities and products in certain jurisdictions in Canada. If you wish to trade in any Canadian securities or other products in Canada please contact Goldman Sachs Canada Inc., an affiliate of The Goldman Sachs Group Inc., or another registered Canadian dealer. **Hong Kong:** Further information on the securities of covered companies referred to in this research may be obtained on request from Goldman Sachs (Asia) L.L.C. **India:** Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (India) Securities Private Limited, Research Analyst - SEBI Registration Number INH000001493, 10th Floor, Ascent-Worli, Sudam Kalu Ahire Marg, Worli, Mumbai-400 025, India, Corporate Identity Number U74140MH2006FTC160634, Phone +91 22 6616 9000, Fax +91 22 6616 9001. Goldman Sachs may beneficially own 1% or more of the securities (as such term is defined in clause 2 (h) the Indian Securities Contracts (Regulation) Act, 1956) of the subject company or companies referred to in this research report. Investment in securities market are subject to market risks. Read all the related documents carefully before investing. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Goldman Sachs (India) Securities Private Limited compliance officer and investor grievance contact details can be found at: <https://www.goldmansachs.com/worldwide/india/documents/Grievance-Redressal-and-Escalation-Matrix.pdf>, and a copy of the annual audit compliance report can be found at this link: <https://publishing.gs.com/content/site/india-annual-compliance-report.html>. **Japan:** See below. **Korea:** This research, and any access to it, is intended only for "professional investors" within the meaning of the Financial Services and Capital Markets Act, unless otherwise agreed by Goldman Sachs. Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (Asia) L.L.C., Seoul Branch. **New Zealand:** Goldman Sachs New Zealand Limited and its affiliates are neither "registered banks" nor "deposit takers" (as defined in the Reserve Bank of New Zealand Act 1989) in New Zealand. This research, and any access to it, is intended for "wholesale clients" (as defined in the Financial Advisers Act 2008) unless otherwise agreed by Goldman Sachs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests is available at: <https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html>. **Russia:** Research reports distributed in the Russian Federation are not advertising as defined in the Russian legislation, but are information and analysis not having product promotion as their main purpose and do not provide appraisal within the meaning of the Russian legislation on appraisal activity. Research reports do not constitute a personalized investment recommendation as defined in Russian laws and regulations, are not addressed to a specific client, and are prepared without analyzing the financial circumstances, investment profiles or risk profiles of clients. Goldman Sachs assumes no responsibility for any investment decisions that may be taken by a client or any other person based on this research report. **Singapore:** Goldman Sachs (Singapore) Pte. (Company Number: 198602165W), which is regulated by the Monetary Authority of Singapore, accepts legal responsibility for this research, and should be contacted with respect to any matters arising from, or in connection with, this research. **Taiwan:** This material is for reference only and must not be reprinted without permission. Investors should carefully consider their own investment risk. Investment results are the responsibility of the individual investor. **United Kingdom:** Persons who would be categorized as retail clients in the United Kingdom, as such term is defined in the rules of the Financial Conduct Authority, should read this research in conjunction with prior Goldman Sachs research on the covered

companies referred to herein and should refer to the risk warnings that have been sent to them by Goldman Sachs International. A copy of these risks warnings, and a glossary of certain financial terms used in this report, are available from Goldman Sachs International on request.

European Union and United Kingdom: Disclosure information in relation to Article 6 (2) of the European Commission Delegated Regulation (EU) (2016/958) supplementing Regulation (EU) No 596/2014 of the European Parliament and of the Council (including as that Delegated Regulation is implemented into United Kingdom domestic law and regulation following the United Kingdom's departure from the European Union and the European Economic Area) with regard to regulatory technical standards for the technical arrangements for objective presentation of investment recommendations or other information recommending or suggesting an investment strategy and for disclosure of particular interests or indications of conflicts of interest is available at <https://www.gs.com/disclosures/europeanpolicy.html> which states the European Policy for Managing Conflicts of Interest in Connection with Investment Research.

Japan: Goldman Sachs Japan Co., Ltd. is a Financial Instrument Dealer registered with the Kanto Financial Bureau under registration number Kinsho 69, and a member of Japan Securities Dealers Association, Financial Futures Association of Japan Type II Financial Instruments Firms Association, and Investment Management Association of Japan. Sales and purchase of equities are subject to commission pre-determined with clients plus consumption tax. See company-specific disclosures as to any applicable disclosures required by Japanese stock exchanges, the Japanese Securities Dealers Association or the Japanese Securities Finance Company.

Global product; distributing entities

Goldman Sachs Global Investment Research produces and distributes research products for clients of Goldman Sachs on a global basis. Analysts based in Goldman Sachs offices around the world produce research on industries and companies, and research on macroeconomics, currencies, commodities and portfolio strategy. This research is disseminated in Australia by Goldman Sachs Australia Pty Ltd (ABN 21 006 797 897); in Brazil by Goldman Sachs do Brasil Corretora de Títulos e Valores Mobiliários S.A.; Public Communication Channel Goldman Sachs Brazil: 0800 727 5764 and / or contatogoldmanbrasil@gs.com. Available Weekdays (except holidays), from 9am to 6pm. Canal de Comunicação com o Público Goldman Sachs Brasil: 0800 727 5764 e/ou contatogoldmanbrasil@gs.com. Horário de funcionamento: segunda-feira à sexta-feira (exceto feriados), das 9h às 18h; in Canada by Goldman Sachs & Co. LLC; in Hong Kong by Goldman Sachs (Asia) L.L.C.; in India by Goldman Sachs (India) Securities Private Ltd.; in Japan by Goldman Sachs Japan Co., Ltd.; in the Republic of Korea by Goldman Sachs (Asia) L.L.C., Seoul Branch; in New Zealand by Goldman Sachs New Zealand Limited; in Russia by OOO Goldman Sachs; in Singapore by Goldman Sachs (Singapore) Pte. (Company Number: 198602165W); and in the United States of America by Goldman Sachs & Co. LLC. Goldman Sachs International has approved this research in connection with its distribution in the United Kingdom.

Goldman Sachs International ("GSI"), authorised by the Prudential Regulation Authority ("PRA") and regulated by the Financial Conduct Authority ("FCA") and the PRA, has approved this research in connection with its distribution in the United Kingdom.

European Economic Area: Goldman Sachs Bank Europe SE ("GSBE") is a credit institution incorporated in Germany and, within the Single Supervisory Mechanism, subject to direct prudential supervision by the European Central Bank and in other respects supervised by German Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin) and Deutsche Bundesbank and disseminates research within the European Economic Area.

General disclosures

This research is for our clients only. Other than disclosures relating to Goldman Sachs, this research is based on current public information that we consider reliable, but we do not represent it is accurate or complete, and it should not be relied on as such. The information, opinions, estimates and forecasts contained herein are as of the date hereof and are subject to change without prior notification. We seek to update our research as appropriate, but various regulations may prevent us from doing so. Other than certain industry reports published on a periodic basis, the large majority of reports are published at irregular intervals as appropriate in the analyst's judgment.

Goldman Sachs conducts a global full-service, integrated investment banking, investment management, and brokerage business. We have investment banking and other business relationships with a substantial percentage of the companies covered by Global Investment Research. Goldman Sachs & Co. LLC, the United States broker dealer, is a member of SIPC (<https://www.sipc.org>).

Our salespeople, traders, and other professionals may provide oral or written market commentary or trading strategies to our clients and principal trading desks that reflect opinions that are contrary to the opinions expressed in this research. Our asset management area, principal trading desks and investing businesses may make investment decisions that are inconsistent with the recommendations or views expressed in this research.

We and our affiliates, officers, directors, and employees will from time to time have long or short positions in, act as principal in, and buy or sell, the securities or derivatives, if any, referred to in this research, unless otherwise prohibited by regulation or Goldman Sachs policy.

The views attributed to third party presenters at Goldman Sachs arranged conferences, including individuals from other parts of Goldman Sachs, do not necessarily reflect those of Global Investment Research and are not an official view of Goldman Sachs.

Any third party referenced herein, including any salespeople, traders and other professionals or members of their household, may have positions in the products mentioned that are inconsistent with the views expressed by analysts named in this report.

This research is focused on investment themes across markets, industries and sectors. It does not attempt to distinguish between the prospects or performance of, or provide analysis of, individual companies within any industry or sector we describe.

Any trading recommendation in this research relating to an equity or credit security or securities within an industry or sector is reflective of the investment theme being discussed and is not a recommendation of any such security in isolation.

This research is not an offer to sell or the solicitation of an offer to buy any security in any jurisdiction where such an offer or solicitation would be illegal. It does not constitute a personal recommendation or take into account the particular investment objectives, financial situations, or needs of individual clients. Clients should consider whether any advice or recommendation in this research is suitable for their particular circumstances and, if appropriate, seek professional advice, including tax advice. The price and value of investments referred to in this research and the income from them may fluctuate. Past performance is not a guide to future performance, future returns are not guaranteed, and a loss of original capital may occur. Fluctuations in exchange rates could have adverse effects on the value or price of, or income derived from, certain investments.

Certain transactions, including those involving futures, options, and other derivatives, give rise to substantial risk and are not suitable for all investors. Investors should review current options and futures disclosure documents which are available from Goldman Sachs sales representatives or at <https://www.theocc.com/about/publications/character-risks.jsp> and https://www.goldmansachs.com/disclosures/cftc_fcm_disclosures. Transaction costs may be significant in option strategies calling for multiple purchase and sales of options such as spreads. Supporting documentation will be supplied upon request.

Differing Levels of Service provided by Global Investment Research: The level and types of services provided to you by Goldman Sachs Global Investment Research may vary as compared to that provided to internal and other external clients of GS, depending on various factors including your individual preferences as to the frequency and manner of receiving communication, your risk profile and investment focus and perspective (e.g.,

marketwide, sector specific, long term, short term), the size and scope of your overall client relationship with GS, and legal and regulatory constraints. As an example, certain clients may request to receive notifications when research on specific securities is published, and certain clients may request that specific data underlying analysts' fundamental analysis available on our internal client websites be delivered to them electronically through data feeds or otherwise. No change to an analyst's fundamental research views (e.g., ratings, price targets, or material changes to earnings estimates for equity securities), will be communicated to any client prior to inclusion of such information in a research report broadly disseminated through electronic publication to our internal client websites or through other means, as necessary, to all clients who are entitled to receive such reports.

All research reports are disseminated and available to all clients simultaneously through electronic publication to our internal client websites. Not all research content is redistributed to our clients or available to third-party aggregators, nor is Goldman Sachs responsible for the redistribution of our research by third party aggregators. For research, models or other data related to one or more securities, markets or asset classes (including related services) that may be available to you, please contact your GS representative or go to <https://research.gs.com>.

Disclosure information is also available at <https://www.gs.com/research/hedge.html> or from Research Compliance, 200 West Street, New York, NY 10282.

© 2026 Goldman Sachs.

You are permitted to store, display, analyze, modify, reformat, and print the information made available to you via this service only for your own use. You may not resell or reverse engineer this information to calculate or develop any index for disclosure and/or marketing or create any other derivative works or commercial product(s), data or offering(s) without the express written consent of Goldman Sachs. You are not permitted to publish, transmit, or otherwise reproduce this information, in whole or in part, in any format to any third party without the express written consent of Goldman Sachs. This foregoing restriction includes, without limitation, using, extracting, downloading or retrieving this information, in whole or in part, to train or finetune a machine learning or artificial intelligence system, or to provide or reproduce this information, in whole or in part, as a prompt or input to any such system.

US WEEKLY KICKSTART

The impact of the AI capex boom on S&P 500 return on equity

- **Following a volatile week, the S&P 500 has returned 9% YTD but trades at the same 21x P/E multiple as it did a year ago.** The S&P 500 YTD return reflects a 17% rise in consensus forward 12-month earnings estimates alongside a 7% decline in the P/E multiple, from 22x to 21x. The current 21x P/E ranks in the 87th percentile since 1980. Going forward, we expect earnings growth will continue to drive the S&P 500. Within the market, recent Momentum factor volatility has tracked with the historical precedent following sharp rallies.
- **Record profitability has been one of the factors supporting high S&P 500 valuations.** S&P 500 trailing 4-quarter ROE registers a record high 22%. Whether S&P 500 valuations revert toward their long-term average or prove structurally higher going forward will depend in part on the trajectory of corporate profitability. In our macro model, every 1 percentage point change in S&P 500 ROE is associated with about a 1x turn in S&P 500 P/E. Differences in profitability also help explain the variation in valuations across equity indices.
- **S&P 500 ROE has surged by 150 bp during the past four quarters, driven by a boost from the mega-cap technology stocks to index profit margins.** Among DuPont components, rising S&P 500 profit margins have also been the primary driver of the index's 800 bp ROE expansion since 1985. In recent quarters, ROE for the aggregate S&P 500 has benefited from the growing weight of the mega-cap tech stocks while ROE for the median S&P 500 company has declined. The seven largest tech stocks currently have a collective ROE of 44%, which has expanded by 900 bp during the past three years.
- **The AI capex boom has lifted semiconductor profitability but will be an increasing headwind to mega-cap tech ROE going forward.** The hyperscalers have grown in asset intensity and experienced declining sales/asset turnover since the start of the AI boom. During the next few years, rising hyperscaler depreciation expenses will reverse some of the S&P 500 ROE benefit that AI capex has generated in the form of elevated semiconductor profit margins. Equity raises to fund investment spending will also pressure ROE. Consensus estimates imply that ROE for the seven largest technology stocks will decline by an average of 700 bp next year.
- **AI also creates a key long-term upside case for ROE, both for the mega-cap tech stocks and for the broad equity market.** For the hyperscalers, the net impact of the AI capex boom on profitability will depend on the magnitude of

Ben Snider

+1(212)357-1744 | ben.snider@gs.com
Goldman Sachs & Co. LLC

Ryan Hammond

+1(212)902-5625 |
ryan.hammond@gs.com
Goldman Sachs & Co. LLC

Jenny Ma

+1(212)357-5775 | jenny.ma@gs.com
Goldman Sachs & Co. LLC

Daniel Chavez

+1(212)357-7657 |
daniel.chavez@gs.com
Goldman Sachs & Co. LLC

Kartik Jayachandran

+1(212)855-7744 |
kartik.jayachandran@gs.com
Goldman Sachs & Co. LLC

Christophe Sung

+1(212)902-3841 |
christophe.sung@gs.com
Goldman Sachs & Co. LLC

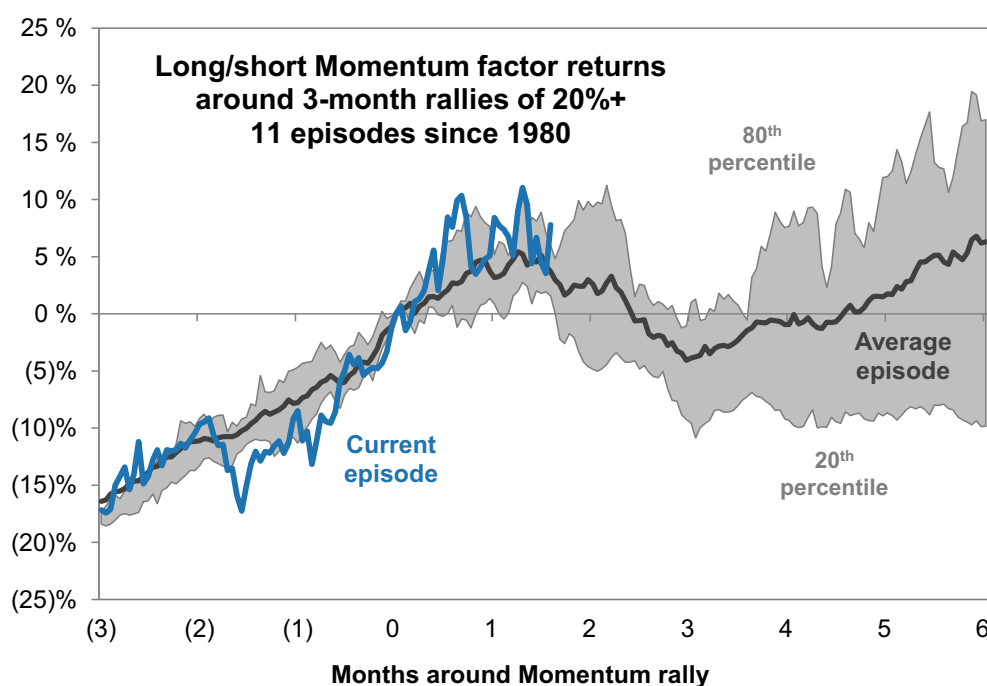
returns on their investments. For the rest of the market, AI-related productivity gains should eventually lift ROE via higher asset turnover and potentially higher profit margins. However, the share of AI-related economic gains that accrue to companies will depend on a number of factors, including government tax policy. Falling effective tax rates have helped lift S&P 500 ROE during the past few decades, but the corporate profit share of US GDP is now at the highest level since 1950.

- **We rebalance our ROE Growth basket (GSTHGROE) in this report.** The basket includes the 50 stocks with the highest consensus ROE growth during the next 12 months.

The outlook for S&P 500 ROE

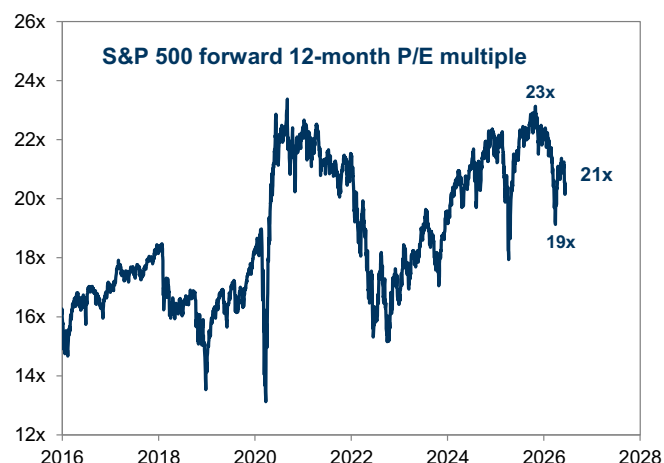
Volatile market rotations during the past week continued to follow the pattern of the sharpest Momentum rallies in recent decades. Before the sell-off that started last week, a narrow-breadth Momentum rally had driven the sharpest two-month S&P 500 return since 1971, when scaled relative to volatility. Following similarly sharp rallies in the past, the Momentum factor has usually struggled during the subsequent few months. In addition to the historical precedent, the current market characteristics of elevated trading leverage, still-narrow market breadth ([Exhibit 36](#)), and uncertainty about both the macro outlook and the AI build-out suggest volatility will persist.

Exhibit 1: Momentum has been volatile following its sharp rally earlier this year

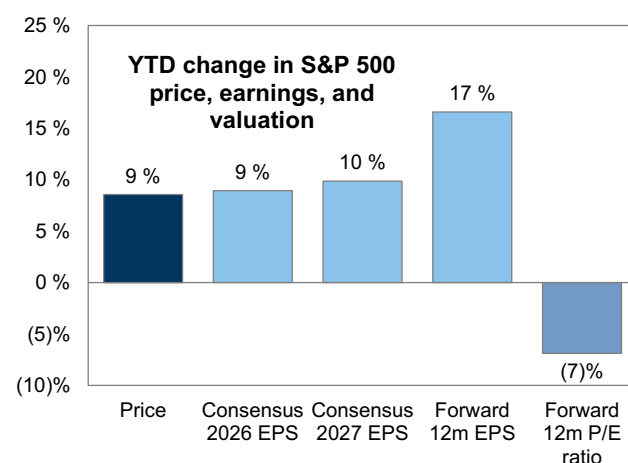


Source: Goldman Sachs Global Investment Research

The S&P 500 P/E multiple has declined during the past few months but remains elevated versus history. The S&P 500 trades at 21x forward earnings, in the 87th percentile since 1980 but 11% below the peak of 23x in late 2025. The median S&P 500 stock trades at 18x, also high by historical standards but in line with the last several years.

Exhibit 2: S&P 500 NTM P/E is 11% below late 2025 peak

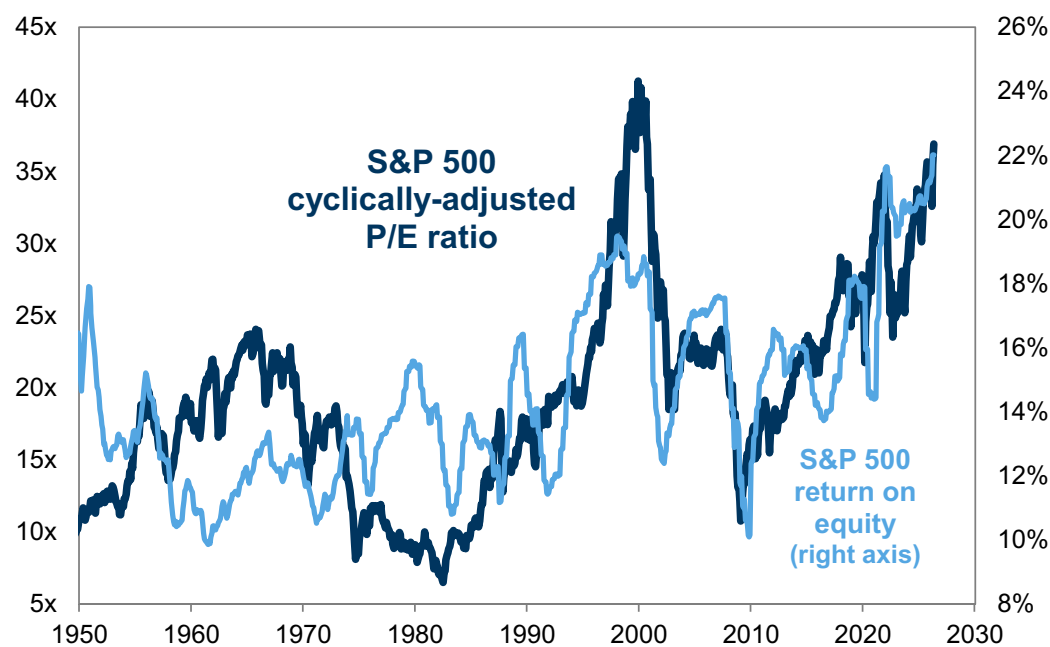
Source: FactSet, Goldman Sachs Global Investment Research

Exhibit 3: S&P 500 forward P/E has contracted YTD

Source: FactSet, Goldman Sachs Global Investment Research

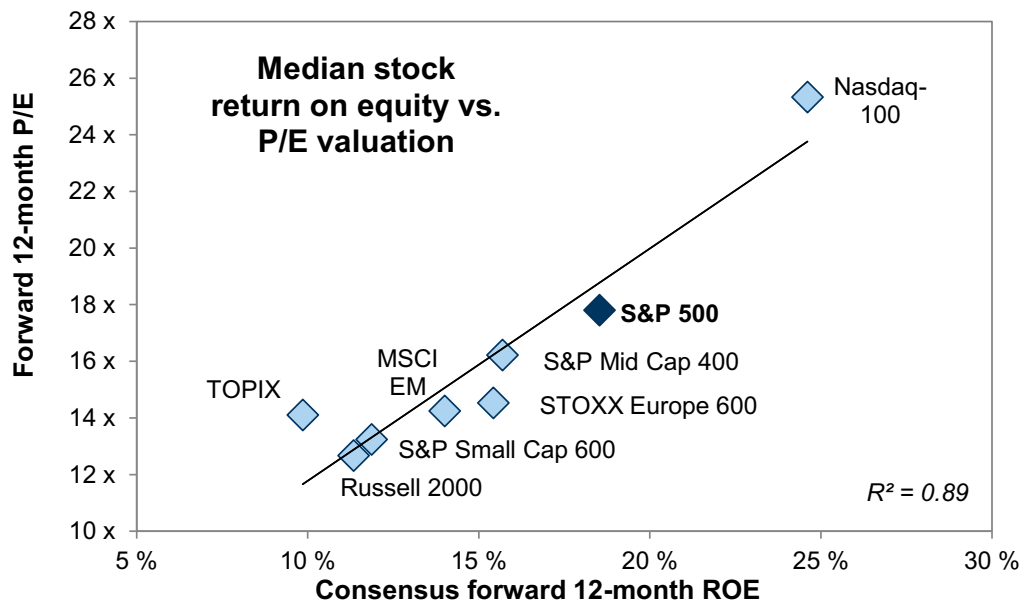
Record profitability has been one of the factors supporting high S&P 500 valuations.

The current S&P 500 trailing 4-quarter return on equity ("ROE") is 22%, the highest on record. Although the P/E is elevated relative to history, it is roughly in line with the fair value indicated by our macro models. Just as investors typically assign higher valuation multiples to stocks with higher profitability, history suggests that the profitability of the aggregate equity market helps determine its P/E multiple, along with macro factors like interest rates and inflation.

Exhibit 4: Higher ROE has supported higher P/E multiples

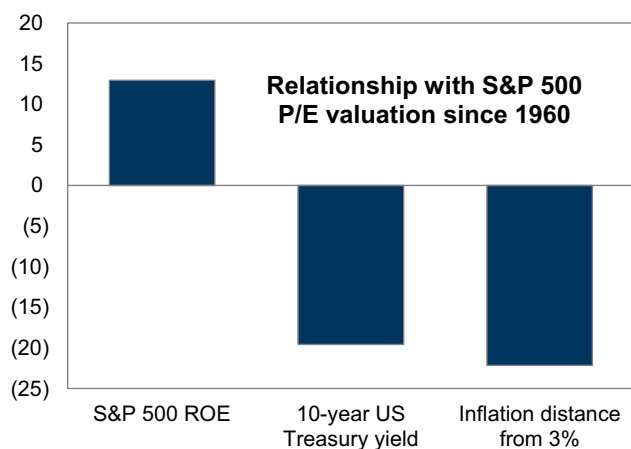
Source: Robert Shiller, Compustat, Goldman Sachs Global Investment Research

Variation in profitability also helps explain the variation in valuations across equity indices. The relationship between ROE and P/E valuation multiples is evident both among US equity indices and across a selection of major global indices.

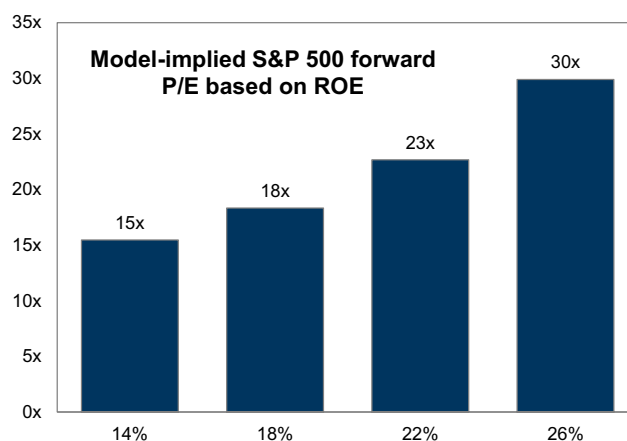
Exhibit 5: Difference in profitability helps explain variation in valuations across equity indices


Source: FactSet, Goldman Sachs Global Investment Research

Whether S&P 500 valuations revert toward their long-term average or prove structurally higher going forward will depend in large part on the trajectory of corporate profitability. Based on our macro model, every 1 percentage point change in S&P 500 ROE is associated with a change of roughly 1x in S&P 500 P/E.

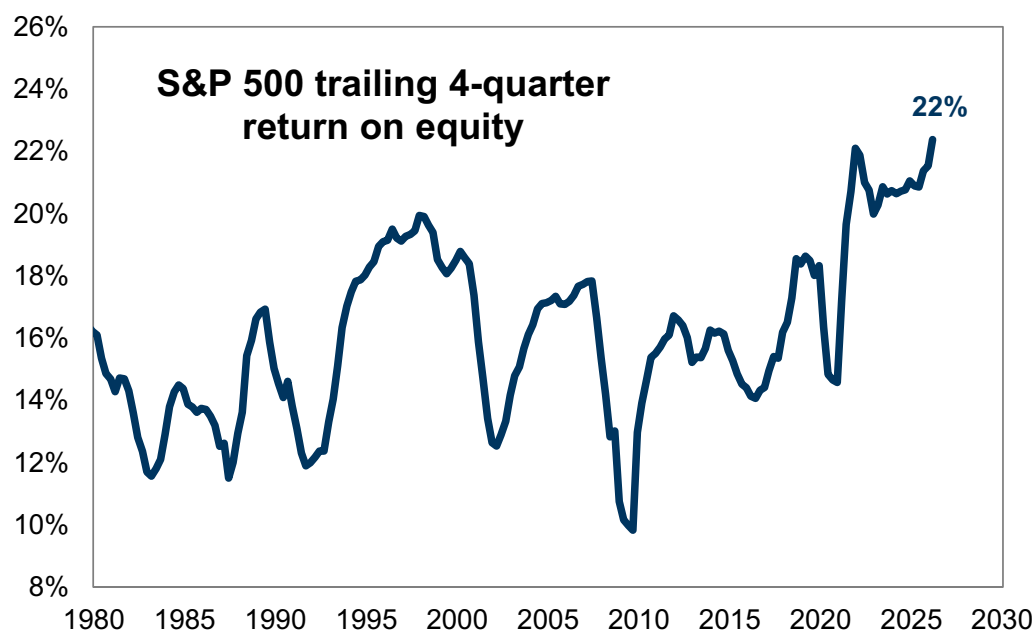
Exhibit 6: Drivers of our macro model of S&P 500 valuation


Source: Goldman Sachs Global Investment Research

Exhibit 7: Sensitivity of S&P 500 valuation to ROE
 assumes constant inflation and interest rates


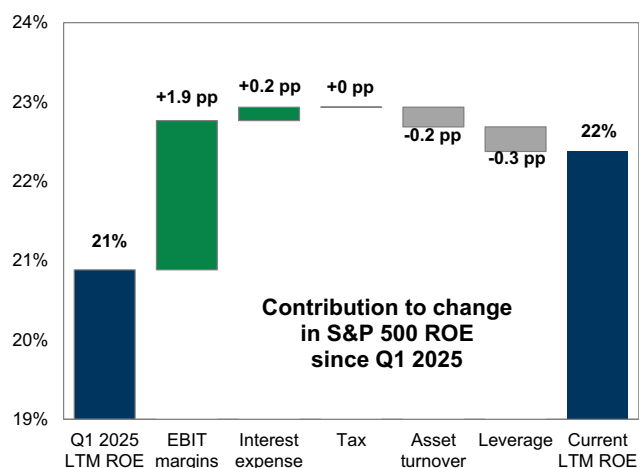
Source: Goldman Sachs Global Investment Research

S&P 500 trailing 4-quarter ROE reached a record high of 22% in Q1 2026, surpassing the previous high reached in 2021. ROE rose by 150 basis points during the past four quarters. During the past 40 years, S&P 500 return on equity has expanded by 800 basis points.

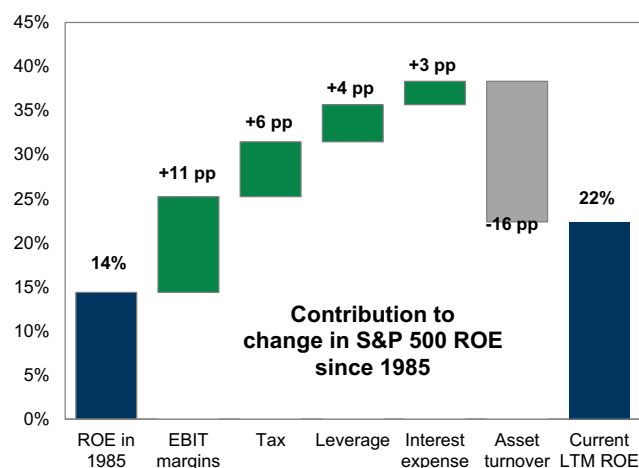
Exhibit 8: S&P 500 ROE has surged to a record high

Source: Compustat, Goldman Sachs Global Investment Research

EBIT margins have been the primary driver of ROE expansion during the past year and during the past few decades. Declining tax rates and falling borrow costs have also been tailwinds to ROE. Rising corporate leverage has been a contributor to long-term S&P 500 ROE expansion but a modest detractor during the past few quarters.

Exhibit 9: Recent ROE growth decomposition

Source: Compustat, Goldman Sachs Global Investment Research

Exhibit 10: Long-term ROE growth decomposition

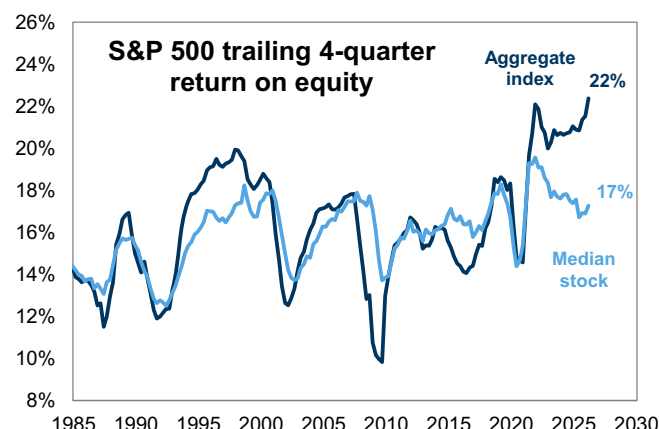
Source: Compustat, Goldman Sachs Global Investment Research

In contrast with surging S&P 500 Index ROE, ROE for the median S&P 500 stock has declined during the past few years and remains below previous highs. Since 1985, ROE for the median S&P 500 company has expanded from 14% to 17%. However, during the past few years, ROE for the median stock has declined due to a combination of higher interest expense, lower leverage, and modestly higher tax expense.

ROE for the aggregate S&P 500 has benefited from the growing weight of mega-cap

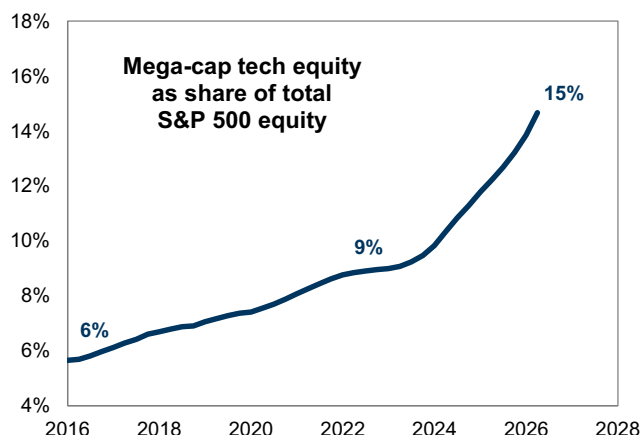
tech stocks, which generate above-average profitability. ROE for the seven largest stocks in the S&P 500 (NVDA, AAPL, GOOGL, MSFT, AMZN, AVGO, and META) currently registers a collective 44% and has expanded by 9 pp during the past three years ([Exhibit 13](#)).

Exhibit 11: Divergence between aggregate and median stock ROE



Source: Compustat, Goldman Sachs Global Investment Research

Exhibit 12: The share of S&P 500 equity attributable to mega-cap tech has more than doubled during the past decade

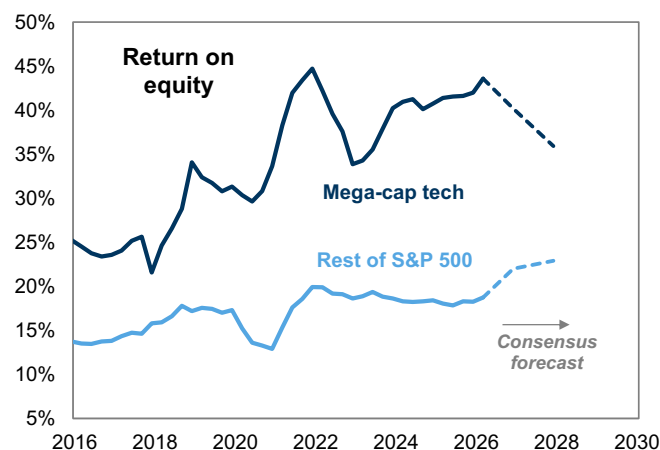


Mega-cap tech = AAPL, AMZN, AVGO, GOOGL, META, MSFT, NVDA

Source: Compustat, Goldman Sachs Global Investment Research

The AI capex boom is likely to weigh on ROE for the mega-cap tech stocks in coming years. While ROE for the mega-cap tech stocks is extremely elevated, the forward outlook faces increasing challenges. Consensus estimates suggest each of the mega-cap tech stocks will experience declining ROEs next year, by an average of 7 pp.

Exhibit 13: ROE for mega-cap tech and the rest of the S&P 500



Mega-cap tech = AAPL, AMZN, AVGO, GOOGL, META, MSFT, NVDA

Source: Compustat, FactSet, Goldman Sachs Global Investment Research

Exhibit 14: Consensus estimates point to declining ROE for mega-cap tech

	ROE (consensus)			2027 vs. 2026
	2025	2026	2027	
AVGO	43 %	51 %	51 %	(0)pp
AMZN	19	18	16	(2)
MSFT	29	26	24	(2)
META	28	28	23	(5)
GOOGL	32	30	24	(6)
NVDA	74	72	61	(12)
AAPL	140	109	83	(26)
Average				(7)pp

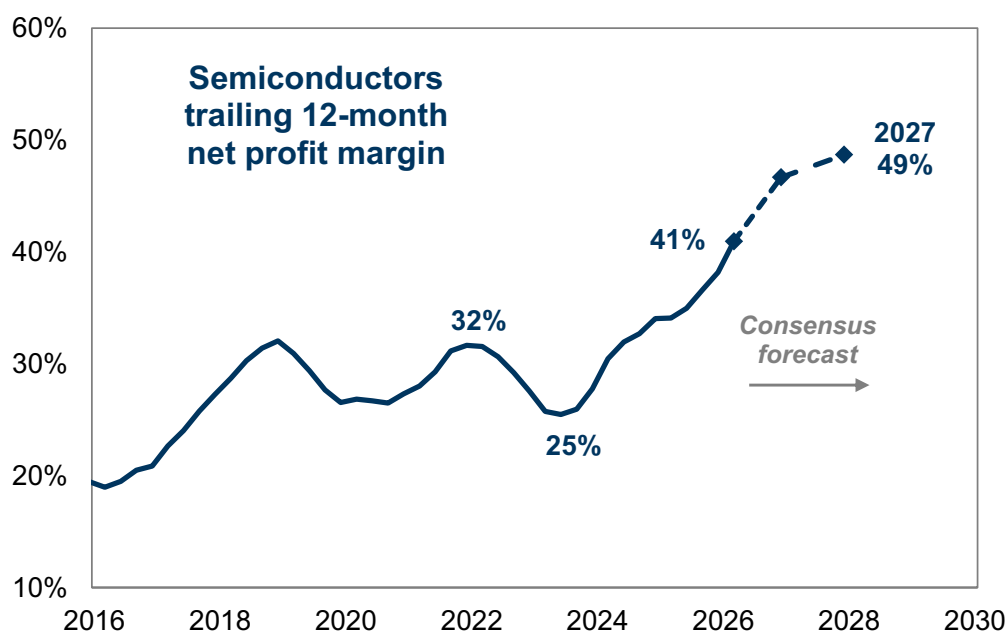
Based on calendar years

Source: FactSet, Goldman Sachs Global Investment Research

The demand for chips has pushed semiconductor profitability to extremely high levels, but investors remain skeptical about the durability of those earnings. The largest direct earnings beneficiaries of the AI capex boom have been semiconductor

firms. Strong moats and a shortfall of supply relative to demand have given these semiconductor firms strong pricing power, boosting their margins and lifting ROE. However, investors fear the combination of slowing demand and increasing supply, including from new competitors, will eventually weigh on semiconductor profitability. Analyst forecasts imply that semiconductor companies will largely be able to defend profit margins during the next year, but their estimates imply a decline in ROE as growth in equity outpaces growth in earnings.

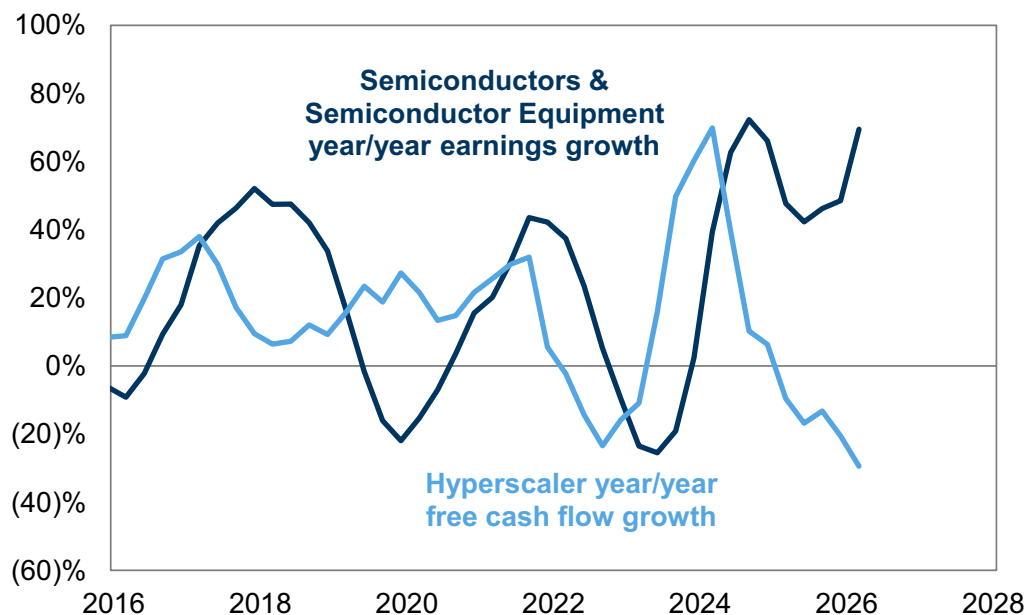
Exhibit 15: Semiconductor margins have climbed to a record high



Source: FactSet, Goldman Sachs Global Investment Research

The capex boom boosting semiconductor earnings today will weigh on hyperscaler ROE going forward as depreciation expenses build. Analyst estimates imply that depreciation and amortization will climb from 7% of hyperscaler revenues in 2022 to 12% in 2027 ([Exhibit 17](#)).

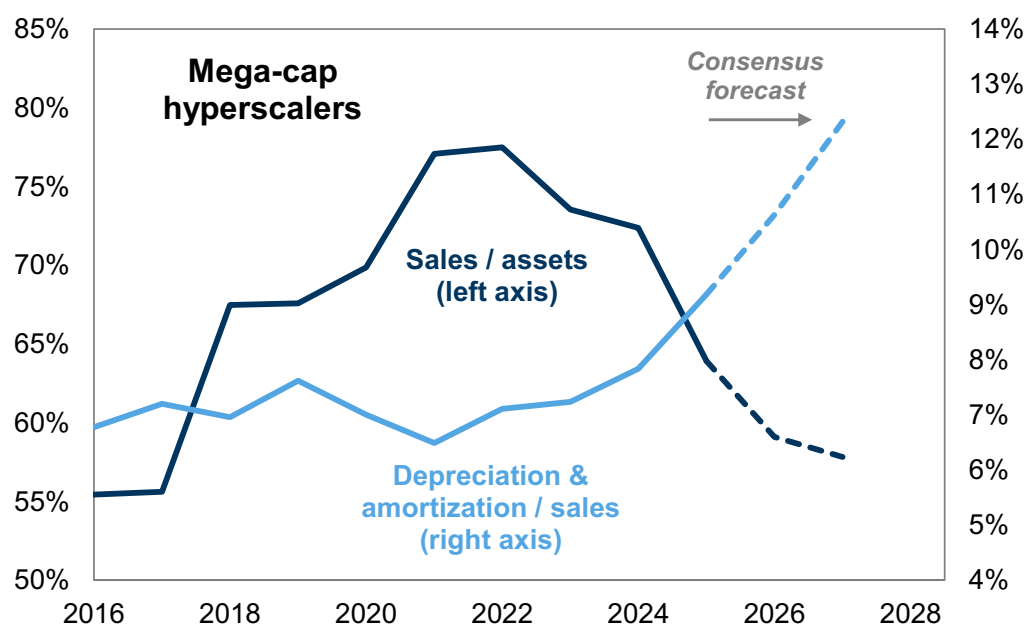
Exhibit 16: Semiconductor earnings have risen at the expense of hyperscaler free cash flows
trailing 4-quarter earnings and free cash flows



Hyperscalers = AMZN, GOOGL, META, MSFT, ORCL

Source: FactSet, Goldman Sachs Global Investment Research

The hyperscalers have also become increasingly asset-heavy businesses. For the decade prior to AI capex build-out, the asset-light business models of the hyperscalers were reflected in rising sales/asset ratios, supporting ROE expansion. However, the physical footprint required to support AI workloads has led to a steady decline in asset turnover, which analysts expect will continue in 2027.

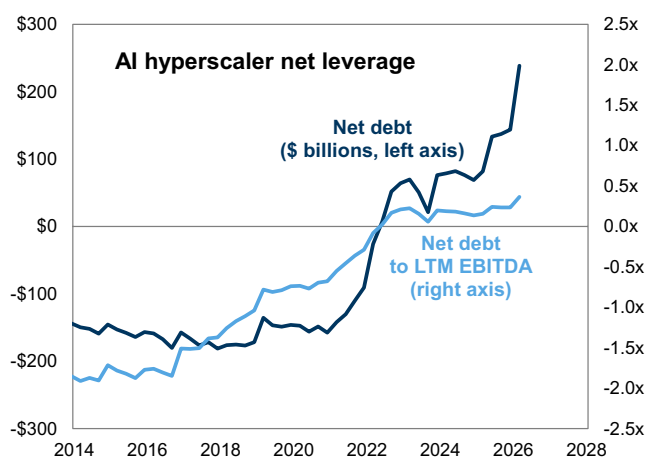
Exhibit 17: AI hyperscaler capex will create headwinds to ROE

Hyperscalers = AMZN, META, MSFT, GOOGL, ORCL

Source: FactSet, Goldman Sachs Global Investment Research

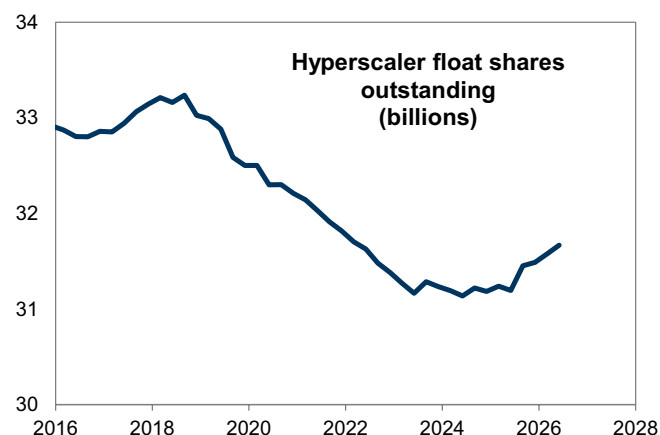
Capital raising to fund the AI build-out signals additional headwinds to ROE.

Consensus estimates suggest the hyperscalers will spend \$770 billion on capex in 2026, equivalent to 100% of cash flows from operations. In order to fund continued capex growth, the companies have increasingly turned to debt and equity issuance and pulled back on buybacks. Net debt for the hyperscalers has increased by \$170 billion since start of 2025 and share counts have begun to rise. While further increases in corporate leverage would support ROE expansion going forward, all else equal, higher leverage would also likely result in increased interest expenses, weighing on ROE.

Exhibit 18: Hyperscaler net leverage has been rising

Hyperscalers = AMZN, GOOGL, META, MSFT, ORCL

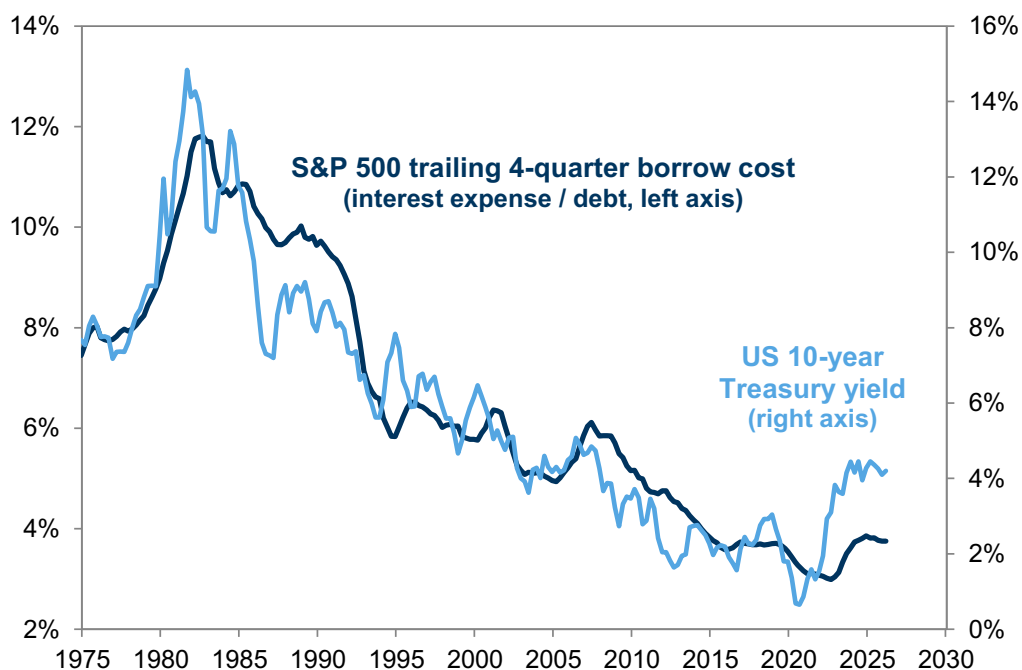
Source: Compustat, Goldman Sachs Global Investment Research

Exhibit 19: Hyperscaler shares outstanding have grown in recent quarters

Hyperscalers = AMZN, GOOGL, META, MSFT, ORCL

Source: FactSet, Goldman Sachs Global Investment Research

S&P 500 borrow costs have risen only modestly during the past few years relative to the increase in Treasury yields. However, the increase in bond yields raises the cost of financing for new issuance, in turn pressuring profitability.

Exhibit 20: S&P 500 borrow costs have increased modestly relative to the increase in bond yields

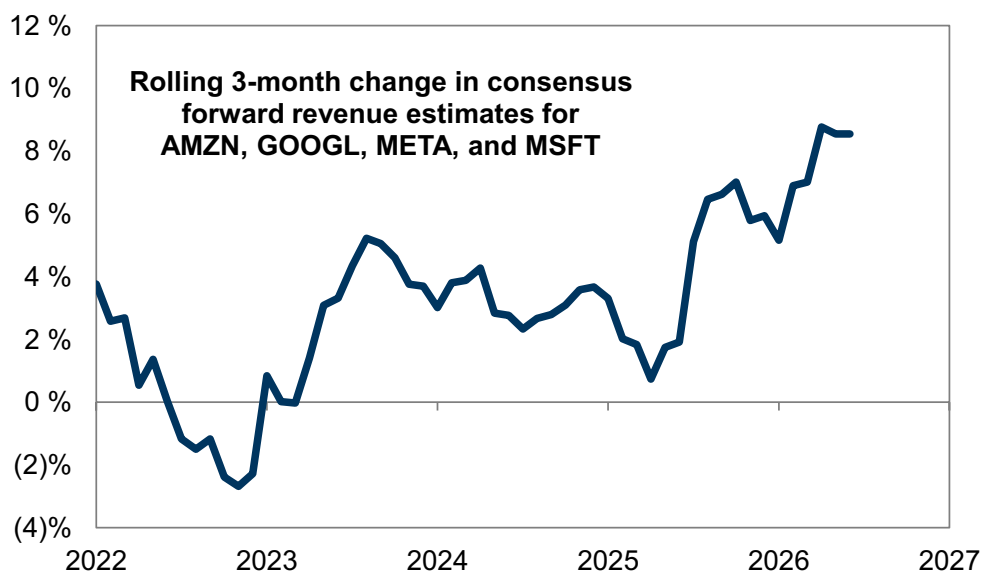
Source: Compustat, Goldman Sachs Global Investment Research

Despite growing headwinds, the net impact of the AI capex boom on the hyperscalers' ROE will depend on the magnitude of returns on their investments.

The Q1 earnings season offered evidence of the hyperscalers' ability to monetize AI, with positive revenue revisions, large backlogs, and rising gross margins. Our equity analysts

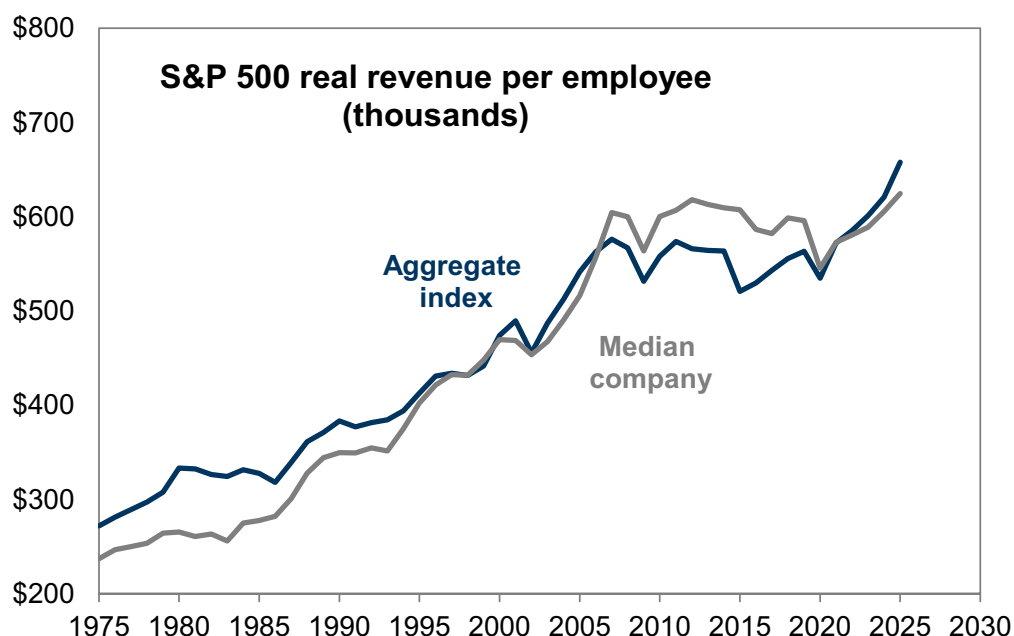
also expect an inflection in token economics ahead for hyperscalers and model providers, enabled by stabilizing token prices but falling compute cost per token.

Exhibit 21: Hyperscaler revenue estimates have been improving recently



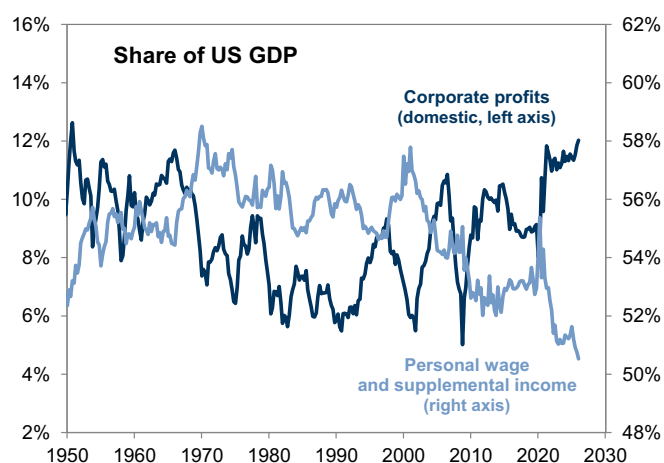
Source: FactSet, Goldman Sachs Global Investment Research

For most of corporate America, AI productivity represents a potential tailwind to profitability. We remain in the early stages of enterprise AI adoption. Last quarter, roughly 54% of companies discussed AI in the context of productivity on their earnings calls, but just 11% of companies quantified the AI productivity gains on a specific use case and only 2% of companies quantified the impact of AI productivity on earnings. The majority of companies focused on cost savings and efficiencies related to AI productivity. For the average company in the index, eventual AI productivity gains should boost sales and earnings per employee, in turn lifting profitability.

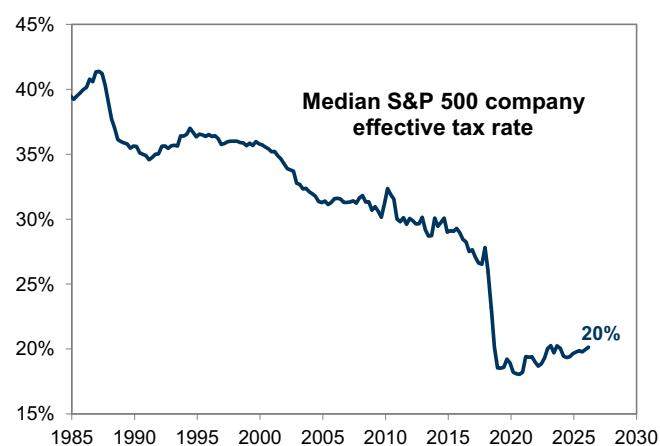
Exhibit 22: We expect AI productivity gains will eventually boost sales per employee

Source: Compustat, Goldman Sachs Global Investment Research

The share of AI-related productivity gains that accrue to corporate profits over the long term will depend in part on the response of government policy. Corporate profits as a share of GDP have risen to the highest level since 1950 alongside a decline in the personal income share and a multi-decade decline in the effective S&P 500 tax rate. Lower corporate tax rates have been one of the largest contributors to ROE expansion during the past 40 years.

Exhibit 23: Corporate profit share has risen

Source: US Bureau of Economic Analysis (BEA), Goldman Sachs Global Investment Research

Exhibit 24: The effective tax rate for S&P 500 companies has declined

Source: Compustat, Goldman Sachs Global Investment Research

We rebalance our ROE Growth basket (GSTHGROE) in this report. Our sector-neutral basket includes the 50 stocks with the highest consensus expected ROE growth during the next 12 months. The median stock in the basket has expected ROE growth of 18%

vs. 0% for the median S&P 500 stock and trades at a similar valuation.

Exhibit 25: Constituents of our ROE Growth basket (GSTHGROE)

bold indicates new constituent

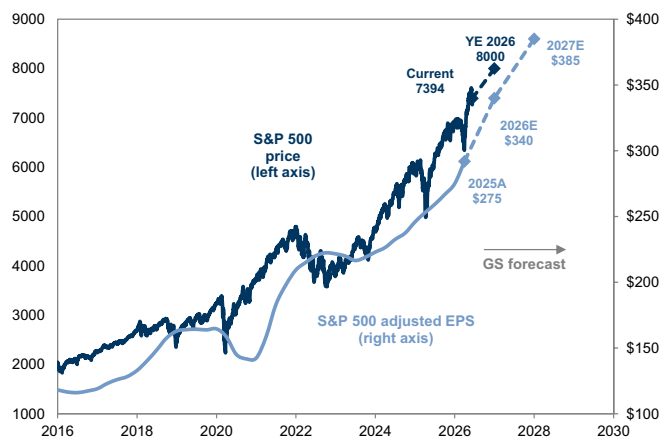
Consensus						Consensus							
		YTD				YTD							
Company Name	Ticker	Total Return	Price to Book	ROE LTM	Forward ROE	ROE Growth	Company Name	Ticker	Total Return	Price to Book	ROE LTM	Forward ROE	ROE Growth
COMMUNICATION SERVICES													
Paramount Skydance	PSKY	(21)%	0.9x	5 %	7 %	46 %	Centene Corp.	CNC	54 %	1.4x	6 %	8 %	29 %
Take-Two Interactive Software	TTWO	(17)	11.3	26	35	34	Viatriis Inc.	VTRS	33	1.2	18	20	11
T-Mobile US Inc.	TMUS	(8)	3.5	19	24	22	Abbott Laboratories	ABT	(28)	3.0	19	20	7
News Corp.	NWSA	1	1.7	7	7	13	UnitedHealth Group	UNH	24	3.8	15	16	6
AT&T Inc.	T	(5)	1.5	13	14	7							
CONSUMER DISCRETIONARY													
NIKE Inc.	NKE	(27)%	4.0x	17 %	20 %	21 %	CSX Corp.	CSX	31 %	6.8x	25 %	29 %	18 %
Ford Motor Co.	F	15	1.4	15	18	20	GE Aerospace	GE	8	18.7	39	45	15
DoorDash Inc.	DASH	(32)	6.5	9	11	19	C.H. Robinson Worldwide	CHRW	20	12.6	36	41	14
Williams-Sonoma	WSM	23	12.8	51	61	19	Westinghouse Air Brake Technologies	WAB	23	4.1	15	17	13
Chipotle Mexican Grill	CMG	(16)	13.6	54	64	18							
CONSUMER STAPLES													
Bunge Global SA	BG	42 %	1.7x	9 %	11 %	27 %							
Archer-Daniels-Midland	ADM	39	1.7	9	10	19							
ENERGY													
Phillips 66	PSX	40 %	2.6x	15 %	21 %	38 %							
Marathon Petroleum	MPC	62	4.5	33	44	33							
FINANCIALS													
Block Inc.	XYZ	6 %	1.7x	8 %	10 %	30 %	Salesforce Inc.	CRM	(37)%	2.9x	24 %	34 %	43 %
Citigroup Inc.	C	20	1.2	8	9	24	Cohesent Corp.	COHR	97	8.9	10	14	38
Citizens Financial Group	CFG	15	1.2	8	9	20	Ciena Corp.	CIEN	90	22.4	24	32	33
Global Payments	GPN	(15)	0.8	14	17	18	ON Semiconductor	ON	114	5.9	14	19	32
Invesco Ltd.	IVZ	9	1.2	8	9	15	Analog Devices Inc.	ADI	53	5.9	15	19	27
KeyCorp.	KEY	10	1.4	10	12	15	Applied Materials	AMAT	116	20.5	31	39	25
MATERIALS													
LyondellBasell Industries	LYB	50 %	1.9x	15 %	22 %	46 %	Teradyne Inc.	TER	97	20.8	28	34	23
HEALTH CARE													
							Texas Instruments	TXN	73	16.3	34	41	18
							Hewlett Packard Enterprise	HPE	96	2.5	14	17	18
							Corning Inc.	GLW	102	13.1	19	22	16
							Synopsys Inc.	SNPS	(3)	3.0	8	9	12
							Monolithic Power Systems	MPWR	76	21.8	25	28	11
							Broadcom Inc.	AVGO	12	22.7	47	53	11
							First Solar Inc.	FSLR	4	3.1	16	18	8
							Qnity Electronics	Q	83	3.6	10	11	7
							Trimble Inc.	TRMB	(36)	2.1	14	14	6
							Roper Technologies	ROP	(25)	1.8	11	12	5
							NXP Semiconductors	NXPI	40	7.5	30	31	5
							Keysight Technologies	KEYS	67	9.7	24	25	4
REAL ESTATE													
							CoStar Group Inc.	CSGP	(51)%	1.7x	6 %	8 %	43 %
UTILITIES													
							Sempra	SRE	4 %	1.9x	9 %	10 %	9 %
ROE Basket Median													
S&P 500 Median													

Source: FactSet, Goldman Sachs Global Investment Research

S&P 500 earnings and return forecasts

Pricing as of June 11, 2026, unless otherwise noted.

Exhibit 26: Goldman Sachs top-down S&P 500 forecasts



Source: FactSet, Goldman Sachs Global Investment Research

Exhibit 27: Goldman Sachs and consensus forecasts for S&P 500 earnings and returns

	Goldman Sachs forecast	Bottom-up analyst consensus	Top-down strategist consensus
S&P 500 EPS			
2025	\$275	\$275	\$275
Y/Y growth	12 %	12 %	12 %
2026	\$340	\$341	\$320
Y/Y growth	24 %	24 %	16 %
2027	\$385	\$398	\$365
Y/Y growth	13 %	17 %	14 %
S&P 500 P/E			
Current on NTM EPS	21x	20x	22x
S&P 500 price target			
YE 2026	8000	NA	7650
Return to target	8 %		3 %
12-month	8300	8900	NA
Return to target	12 %	21 %	

Source: Bloomberg, FactSet, Goldman Sachs Global Investment Research

Biggest stock movers this week

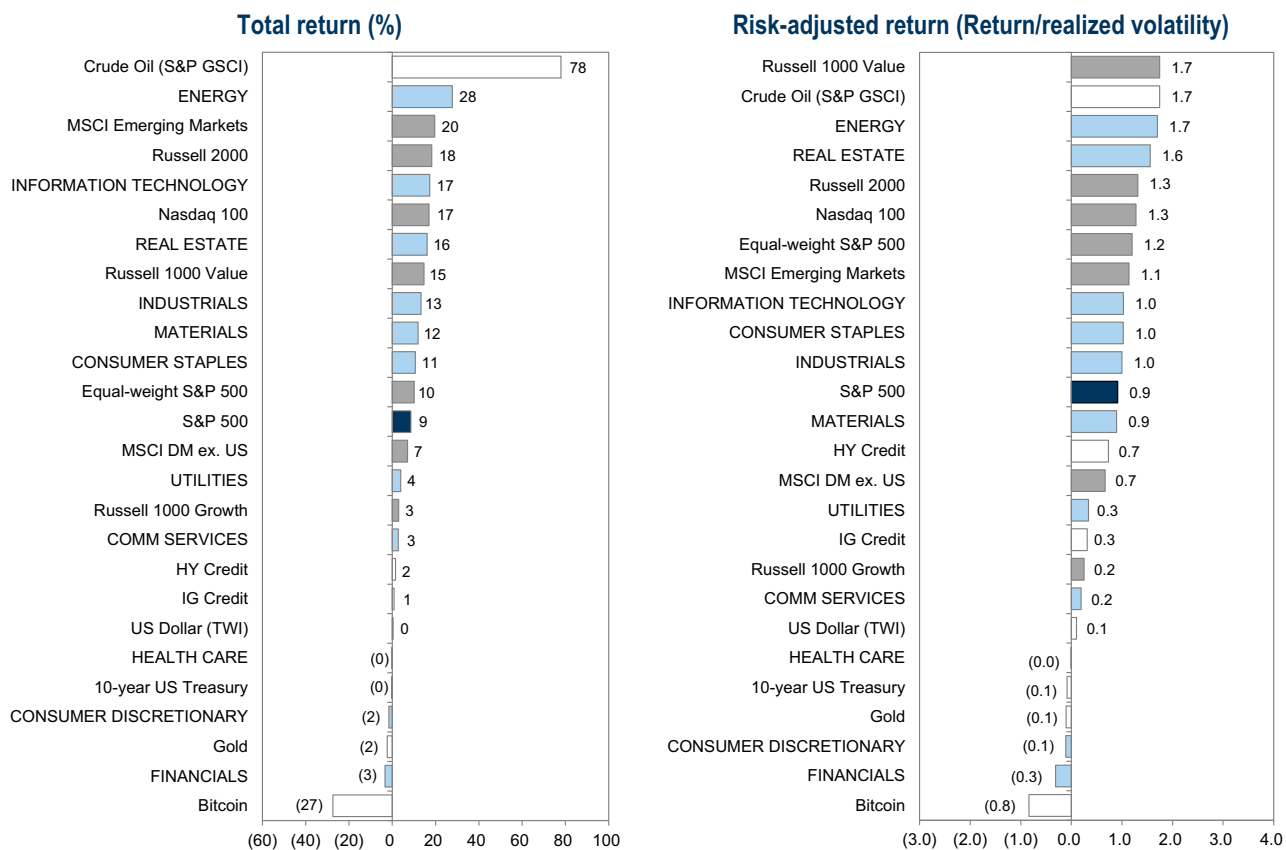
Exhibit 28: S&P 500 stocks with the largest moves this week
as of June 12, 2026

Company	Ticker	Industry group	This week	Total return			Market cap (bn)	Consensus	Fwd 12m
				3m	6m	12m		2026 EPS growth	P/E
Top 10 returns this week									
KLA Corp.	KLAC	Semiconductors & Semi Equip	32 %	65 %	94 %	178 %	\$315	889 %	53x
Sandisk Corp.	SNDK	Tech Hardware & Equipment	27	187	679	4577	279	1871	12
Intel Corp.	INTC	Semiconductors & Semi Equip	26	144	196	466	588	173	117
Applied Materials	AMAT	Semiconductors & Semi Equip	25	58	105	222	439	44	38
Lam Research Corp.	LRCX	Semiconductors & Semi Equip	21	66	115	302	453	39	49
Casey's General Stores	CASY	Consumer Staples Retail	17	34	69	84	34	22	42
Micron Technology	MU	Semiconductors & Semi Equip	14	138	286	760	1,123	638	10
Robinhood Markets	HOOD	Financial Services	13	17	(25)	23	83	(8)	47
Teradyne Inc.	TER	Semiconductors & Semi Equip	13	26	87	341	60	83	57
J.M. Smucker	SJM	Food Beverage & Tobacco	13	10	18	27	12	16	12
Bottom 10 returns this week									
Super Micro Computer	SMCI	Tech Hardware & Equipment	(27)%	1 %	(6)%	(26)%	\$19	67 %	10x
Adobe Inc.	ADBE	Software & Services	(19)	(20)	(38)	(47)	88	11	8
PTC Inc.	PTC	Software & Services	(17)	(26)	(33)	(31)	14	(10)	15
Oracle Corp.	ORCL	Software & Services	(14)	13	(7)	5	529	9	23
Autodesk Inc.	ADSK	Software & Services	(14)	(18)	(32)	(31)	43	20	15
AppLovin Corp.	APP	Software & Services	(11)	4	(33)	25	161	64	29
Salesforce Inc.	CRM	Software & Services	(11)	(14)	(36)	(37)	136	11	12
Akamai Technologies	AKAM	Software & Services	(11)	24	55	69	19	(6)	20
Gartner Inc.	IT	Software & Services	(10)	(7)	(36)	(64)	10	1	10
GoDaddy Inc.	GDDY	Software & Services	(10)	(13)	(40)	(58)	10	14	10
Median S&P 500 stock			1 %	3 %	6 %	12 %	\$42	10 %	18x
S&P 500 Index			1	9	8	24	66,169	24	20

Source: FactSet, Goldman Sachs Global Investment Research

YTD absolute and risk-adjusted returns

Exhibit 29: YTD asset returns and return/volatility ratios

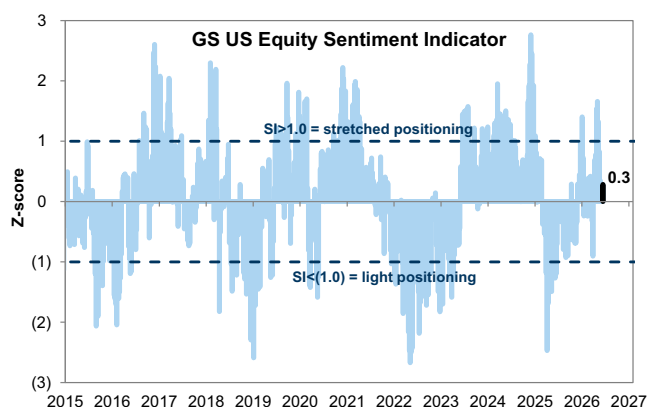


Crude oil represents S&P GSCI Crude Oil Index.

Source: FactSet, Haver Analytics, Goldman Sachs Global Investment Research

Sentiment and flows

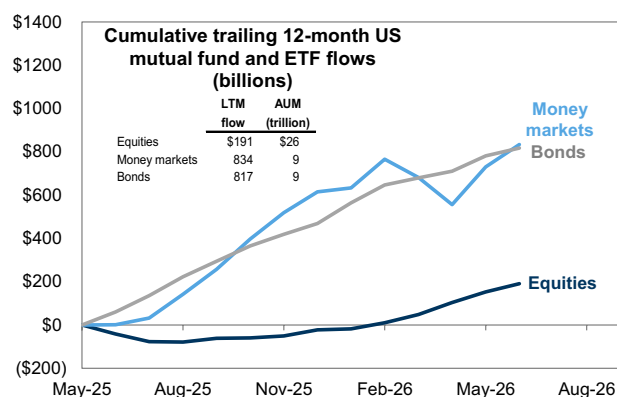
Exhibit 30: GS US Equity Sentiment Indicator of investor positioning
as of June 12, 2026



The Sentiment Indicator combines 9 measures of positioning across institutional, retail, and foreign investors and has historically been a statistically significant signal for near-term S&P 500 returns.

Source: Goldman Sachs Global Investment Research

Exhibit 31: Recent mutual fund and ETF flows



Source: EPFR, Goldman Sachs Global Investment Research

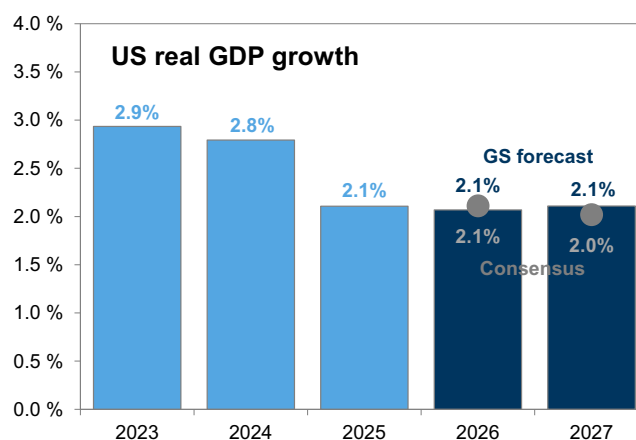
Economic growth

Exhibit 32: US equity market internal pricing of economic growth



Source: Bloomberg, Goldman Sachs Global Investment Research

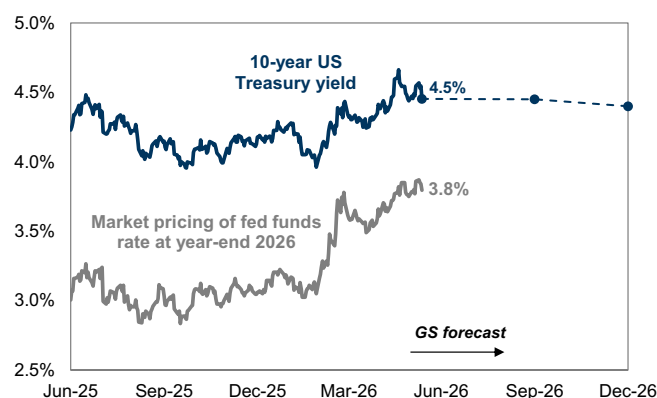
Exhibit 33: Goldman Sachs and consensus forecasts for US real GDP growth



Source: Bloomberg, Goldman Sachs Global Investment Research

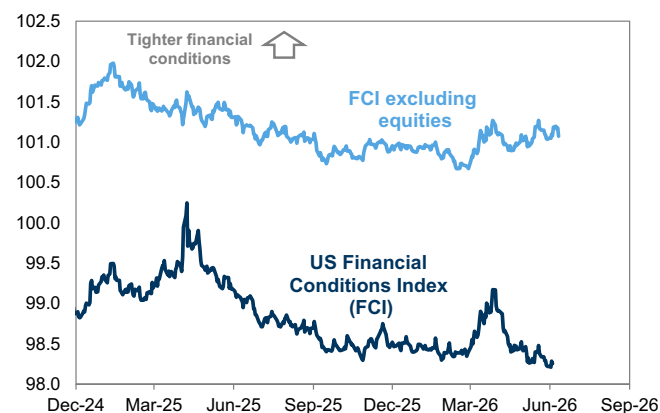
Interest rates and financial conditions

Exhibit 34: Market pricing of near-term and long-term interest rate outlook



Source: Goldman Sachs Global Investment Research

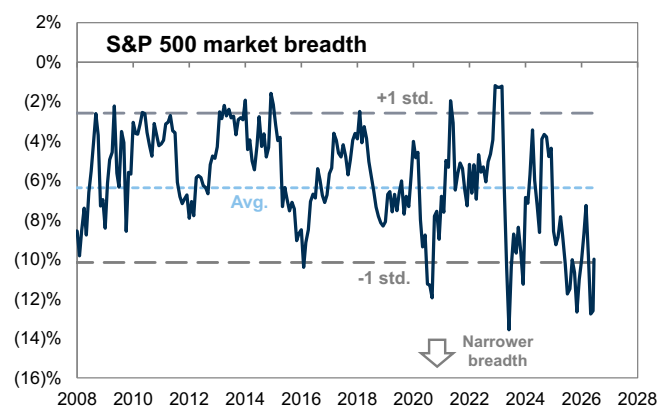
Exhibit 35: Goldman Sachs Financial Conditions Index



Source: Goldman Sachs Global Investment Research

Market breadth and concentration

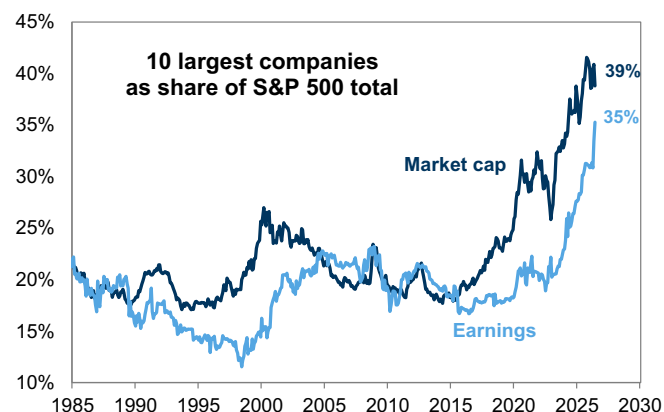
Exhibit 36: S&P 500 52-week market breadth



Market breadth calculated as the difference between the distance of the aggregate S&P 500 Index from its 52-week high and the distance of the median S&P 500 constituent from its 52-week high. When the aggregate index is much closer to its 52-week high than is the median stock, market breadth is narrow.

Source: Goldman Sachs Global Investment Research

Exhibit 37: Concentration of S&P 500 market cap and earnings in the 10 largest index constituents

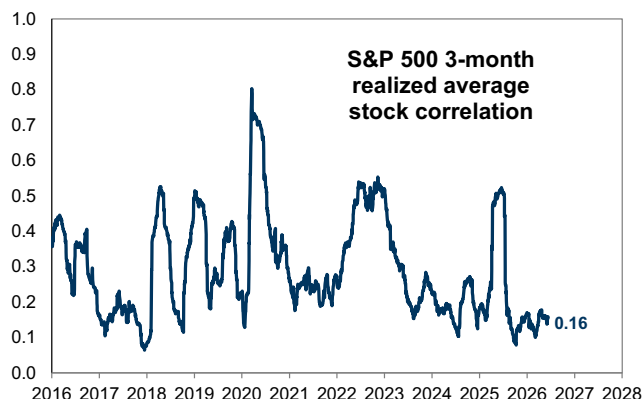


Earnings reflect consensus forward 12-month estimates.

Source: Compustat, IBES, FactSet, Goldman Sachs Global Investment Research

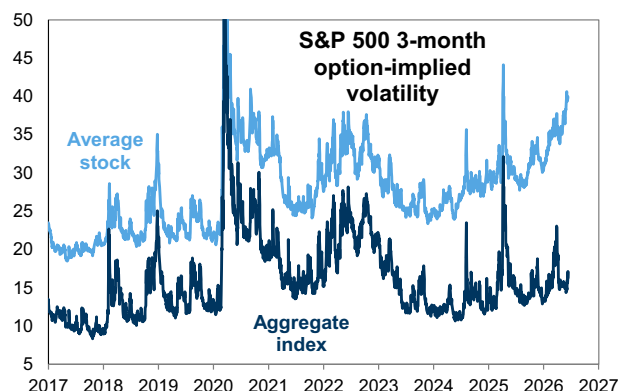
Correlation and volatility

Exhibit 38: S&P 500 realized average stock correlation



Source: Goldman Sachs Global Investment Research

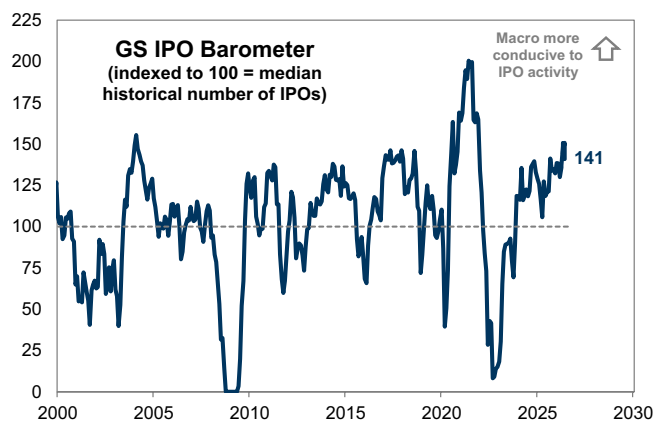
Exhibit 39: S&P 500 implied volatility



Source: Goldman Sachs Global Investment Research

IPO Barometer and mutual fund performance

Exhibit 40: Goldman Sachs IPO Barometer



The Goldman Sachs IPO Barometer combines five indicators that gauge the conduciveness of the macro environment to IPO activity.

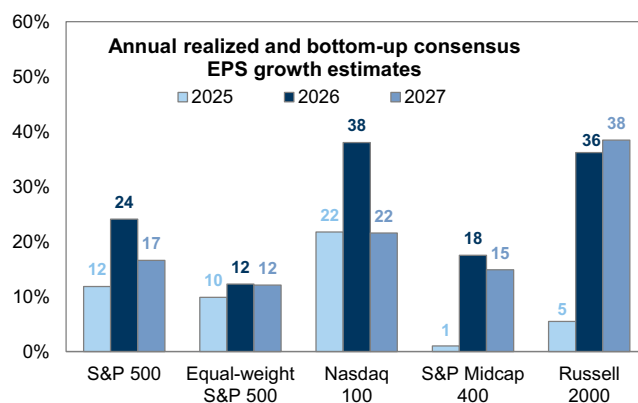
Source: Goldman Sachs Global Investment Research

Exhibit 41: US equity mutual fund returns relative to benchmarks

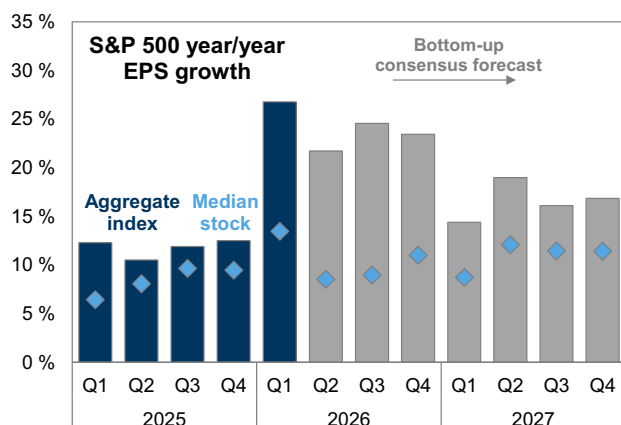
Mutual fund style	Average	% outperforming
	YTD return	benchmarks YTD
Large-cap core	8%	36%
Large-cap growth	5	63
Large-cap value	10	15
Large-cap total	7%	38%
Small-cap core	17%	33%

Source: FactSet, Goldman Sachs Global Investment Research

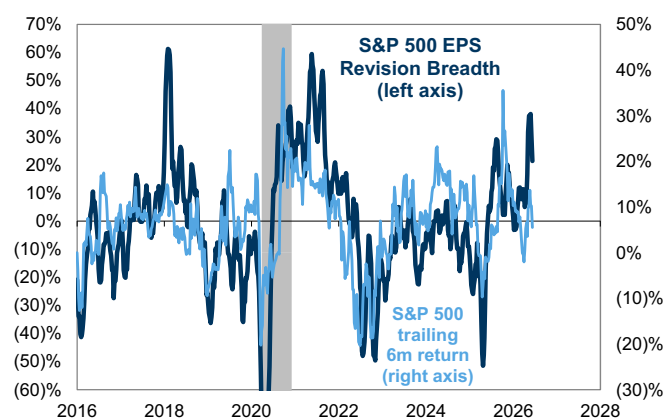
Earnings growth

Exhibit 42: Realized and consensus EPS growth for select US equity indices

Source: FactSet, Goldman Sachs Global Investment Research

Exhibit 43: Realized and consensus year/year S&P 500 EPS growth

Source: FactSet, Goldman Sachs Global Investment Research

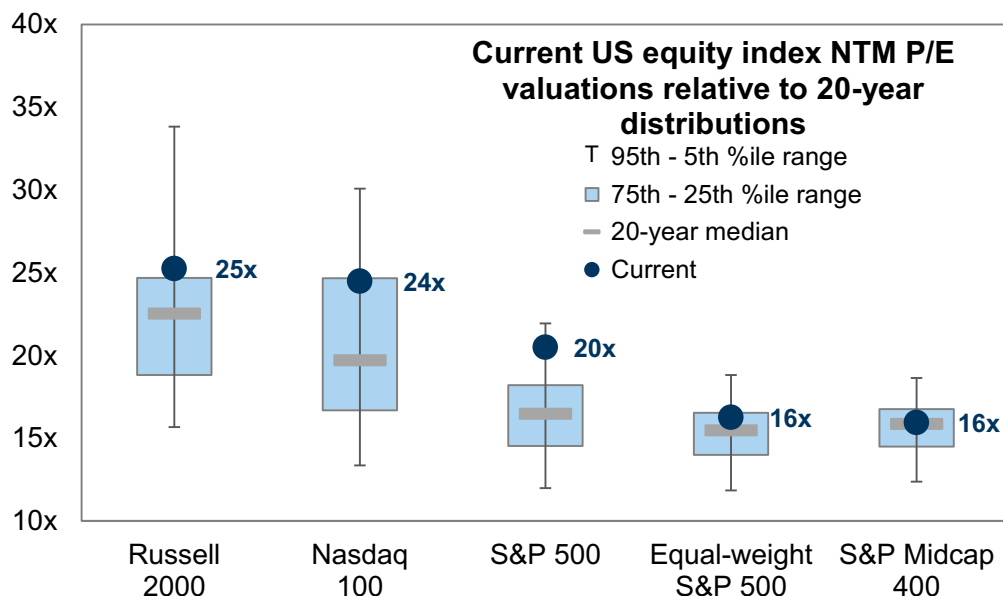
Exhibit 44: S&P 500 FY2 earnings revision breadth

Revision breadth calculated as $\frac{[(\# \text{ of companies with positive EPS revisions}) - (\# \text{ of companies with negative revisions})]}{(\text{total } \# \text{ of companies})}$. Revisions reflect FY2 EPS estimate changes during a rolling 1-month window.

Source: FactSet, Goldman Sachs Global Investment Research

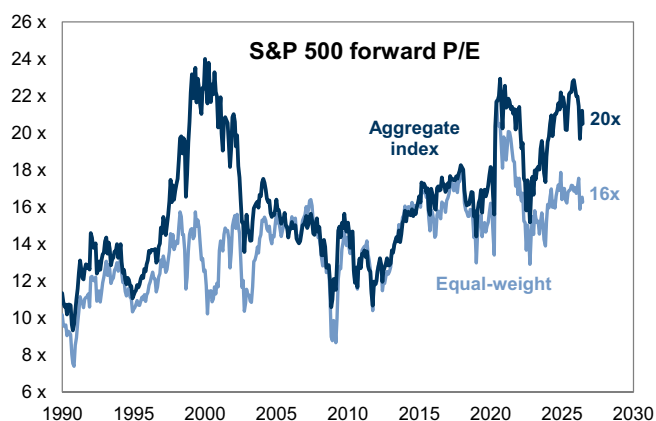
Valuations

Exhibit 45: US equity index P/E valuations vs. history



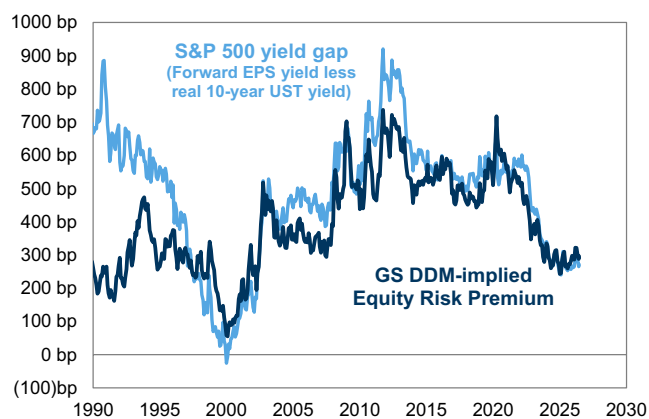
Source: Compustat, FactSet, IBES, Goldman Sachs Global Investment Research

Exhibit 46: S&P 500 consensus forward 12-month P/E



Source: Compustat, FactSet, IBES, Goldman Sachs Global Investment Research

Exhibit 47: S&P 500 valuation relative to US Treasury yields



Equity risk premium calculated by solving for the discount rate that equates the fair value of our multi-stage S&P 500 dividend discount model to the current market price and subtracting the 10-year US Treasury yield.

Source: Goldman Sachs Global Investment Research

Sector returns, earnings, and valuations

Exhibit 48: Sector and industry group returns

			Top quartile						
			Bottom quartile						
S&P 500			Index weight	1 Week	1 Month	3 Months	Last 12 Mo	YTD	GS allocation
			100 %	(2.5)%	(0)%	9 %	24 %	9 %	
S E C T O R	Consumer Staples	5 %	3.7 %	1 %	1 %	8 %	11 %	OW	
	Health Care	9	1.4	8	1	16	(0)		
	Real Estate	2	1.2	1	7	15	16		
	Financials	12	0.8	3	7	5	(3)		
	Utilities	2	0.2	(2)	(4)	12	4	OW	
	Industrials	9	(0.5)	0	4	23	13		
	Materials	2	(0.9)	(2)	2	17	12		
	Consumer Discretionary	9	(1.7)	(5)	4	10	(2)		
	Energy	3	(2.8)	(0)	0	37	28		
	Communication Services	10	(3.8)	(6)	3	27	3		
Information Technology		37	(5.6)	0	22	43	17		
I N D U S T R Y G R O U P	Household & Personal Products	1 %	5.7 %	4 %	(2)%	(6)%	6 %		
	Food Beverage & Tobacco	2	3.8	2	4	10	13		
	Insurance	2	3.7	4	3	(2)	(2)		
	Consumer Durables & Apparel	0	3.5	6	(3)	0	(5)		
	Telecommunication Services	1	3.3	(4)	(12)	(8)	0		
	Consumer Staples Retail	2	2.6	(2)	(1)	14	11		
	Consumer Services	2	2.2	4	(1)	(2)	(4)		
	Banks	3	1.7	7	13	27	2		
	Pharma Biotech & Life Sciences	6	1.4	9	1	29	4		
	Transportation	1	1.4	6	11	26	14		
	Health Care Equipment & Services	3	1.3	6	1	(3)	(6)		
	Real Estate Investment Trusts (REITs)	2	1.2	1	7	15	16		
	Utilities	2	0.2	(2)	(4)	12	4		
	Financial Services	7	(0.2)	1	5	(2)	(6)		
	Commercial & Professional Svcs	1	(0.8)	3	(7)	(23)	(12)		
	Capital Goods	7	(0.9)	(1)	4	32	17		
	Materials	2	(0.9)	(2)	2	17	12		
	Consumer Discretionary Retail	5	(2.2)	(6)	8	10	3		
	Energy	3	(2.8)	(0)	0	37	28		
	Semiconductors & Semi Equipment	18	(4.2)	(1)	34	89	39		
	Media & Entertainment	9	(4.4)	(7)	5	31	3		
	Automobiles & Components	2	(4.5)	(8)	(0)	25	(10)		
	Technology Hardware & Equipment	11	(5.0)	4	25	70	26		
	Software & Services	9	(9.3)	(1)	0	(17)	(17)		

Source: FactSet, Goldman Sachs Global Investment Research

Exhibit 49: Earnings contributions and growth by S&P 500 sector

Sector	GS top-down					Consensus bottom-up			
	Contribution			EPS growth		Contribution		EPS growth	
	2025	2026	2027	2026	2027	2026	2027	2026	2027
Info Tech	\$70	\$110	\$140	56 %	27 %	\$107	\$139	52 %	30 %
Financials	52	56	61	8	9	56	63	8	11
Comm Services	29	36	38	24	5	37	39	26	6
Health Care	33	34	39	2	14	34	40	1	19
Cons Discretionary	23	24	26	7	7	26	29	13	14
Industrials	21	23	25	9	9	23	28	9	19
Energy	12	19	17	60	(10)	19	18	62	(7)
Consumer Staples	15	15	16	4	4	15	17	4	8
Utilities	8	8	9	7	6	9	9	14	8
Materials	5	8	8	39	5	8	8	42	10
Real Estate	6	7	7	5	6	7	8	15	8
S&P 500	\$275	\$340	\$385	24 %	13 %	\$341	\$398	24 %	17 %

Source: FactSet, Goldman Sachs Global Investment Research

Exhibit 50: S&P 500 sector P/E valuations relative to history

	Consensus forward 12-month P/E valuation				
	Current P/E	Absolute P/E		P/E vs. S&P 500	
		%ile rank vs. history		%ile rank vs. history	
		10-year	30-year	10-year	30-year
Consumer Discretionary	26x	46%	81%	25%	71%
Industrials	25	86	95	97	99
Information Technology	23	55	62	53	42
Consumer Staples	22	85	85	56	35
S&P 500	20	56	76		
Communication Services	20	53	53	35	12
Real Estate	19	65	81	39	35
Health Care	18	74	66	39	19
Utilities	17	56	86	69	70
Materials	17	41	61	21	14
Financials	15	38	66	9	8
Energy	13	34	45	37	18

Source: Compustat, FactSet, IBES, Goldman Sachs Global Investment Research

Thematic baskets

Exhibit 51: Recent performance of select Goldman Sachs Research baskets

basket returns expressed relative to the equal-weight S&P 500 except for long/short basket pairs as noted

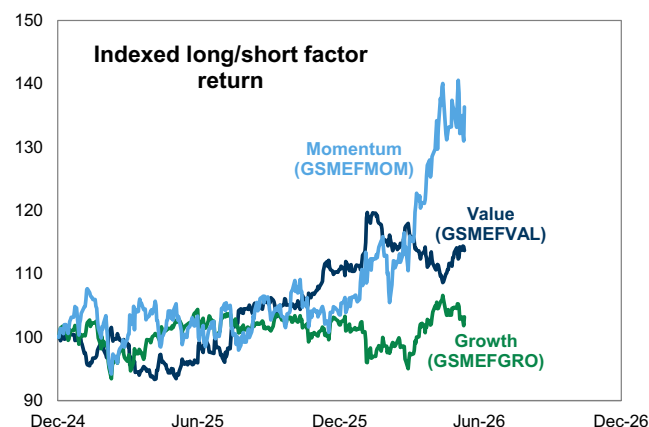
	Basket / Index	Ticker (GSTH_____)	Total returns			
			1 Wk	Top quartile		
				Bottom quartile		
				3 Mo	12 Mo	YTD
	S&P 500	SPX	(2.5)%	9 %	24 %	9 %
	Equal-weight S&P 500	SPW	(0.4)	7	19	10
Macro-economic	High vs. Low Tax Rate	HTAX / LTAX	2.6 %	(3)%	(1)%	2 %
	Interest Rate Sensitive	USTY	0.6	23	31	22
	Dual Beta	BETA	(1.1)	40	89	48
	Low vs. High Labor Cost	LLAB / HLAB	(1.2)	8	21	13
Geographic sales	US Sales	AINT	(0.4)%	5 %	(12)%	(4)%
	International Sales	INTL	(1.8)	27	48	32
	EM Sales	BRIC	(2.3)	39	82	52
	Western Europe Sales	WEUR	(4.6)	5	6	3
Fundamental	Strong vs. Weak Balance Sheet	SBAL / WBAL	2.8 %	(4)%	(20)%	(10)%
	High Quality	QUAL	(0.1)	(6)	(14)	(9)
	Stable Growers	STGR	(2.2)	(7)	(19)	(10)
	Long vs. Short Duration	LDUR / SDUR	(2.4)	(4)	(21)	(11)
	ROE Growth	GROE	(2.9)	21	41	31
	High vs. Low Operating Leverage	OPHI / OPLO	(3.5)	(0)	(9)	(5)
Uses of cash	Debt Reducers vs. Issuers	DRED / DISS	0.7 %	12 %	27 %	13 %
	Dividend Growth	DIVG	(1.9)	9	15	6
	Total Cash Return	CASH	(2.0)	9	13	5
	Buybacks	REPO	(2.2)	7	4	1
	Capex and R&D	CAPX	(4.1)	27	57	34
	High Growth Investment	HGIR	(4.9)	19	37	23
Ownership, risk & liquidity	Mutual Fund Over- vs. Underweights	MFOV / MFUW	3.2 %	(14)%	(31)%	(22)%
	Low vs. High Liquidity	LLIQ / HLIQ	2.6	(8)	(18)	(12)
	High Sharpe Ratio	SHRP	(1.8)	(7)	(19)	(10)
	Hedge Fund VIP	HVIP	(2.8)	9	13	5
Equity research	Oil Input Cost	GSRHOILX	7.6 %	(1)%	(11)%	(1)%
	SMB Exposure	GSRHSMBX	(1.5)	(1)	(21)	(12)
	Gov't Exposure	GSRHGOVT	(1.8)	(5)	15	(3)
	M&A Candidates	GSRHACQN	(2.8)	3	15	(8)

Basket returns and constituents can be accessed via GS Marquee or Bloomberg using the indicated tickers.

Source: Goldman Sachs Global Investment Research

Factors

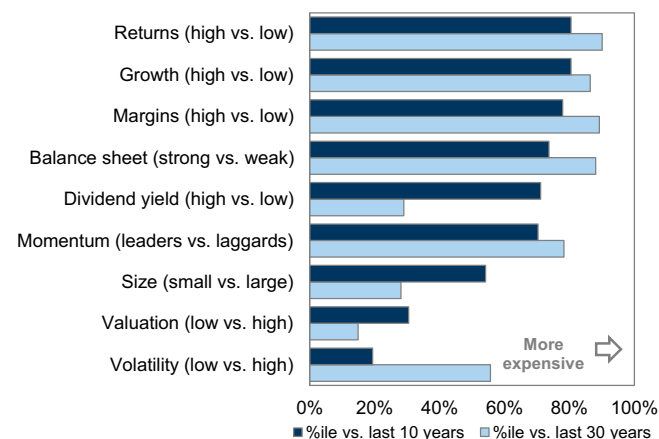
Exhibit 52: Indexed return of select equity factors



Factor performance and constituents can be viewed on GS Marquee or Bloomberg using the tickers indicated on the chart.

Source: Goldman Sachs Global Investment Research

Exhibit 53: Equity factor valuations relative to history



Source: Goldman Sachs Global Investment Research

Goldman Sachs global macro research cross-asset forecasts

Exhibit 54: Goldman Sachs macro research asset forecasts

Goldman Sachs Global Macro Forecasts

	units	Current	3m	6m	12m	Change to 12m target
Equities						
MXAPJ	level	856	980	1030	1080	26 %
TOPIX	level	3830	4100	4200	4400	15
S&P 500	level	7394	7600	8000	8300	12
STOXX Europe 600	level	622	640	645	660	6
10-year rates						
Euro Area (Germany)	%	2.5	3.0	3.0	3.0	55 bp
US	%	4.5	4.5	4.4	4.3	(15)
Japan	%	2.7	2.5	2.5	2.4	(27)
Corporate bonds						
Investment grade	bp	75	85	82	78	3 bp
High yield	bp	271	305	295	280	9
Currencies						
Euro / US dollar	EUR/\$	1.15	1.14	1.18	1.20	4 %
Sterling / US dollar	£/\$	1.33	1.33	1.34	1.33	0
US dollar / Yen	\$/¥	161	160	158	155	(3)
Commodities						
COMEX gold	\$/troy oz	4114	4805	4990	5445	32 %
NYMEX nat. gas	\$/mmBtu	3	3.50	3.50	3.50	13
LME copper	\$/mt	13420	13620	13735	13800	3
Brent crude oil	\$/bbl	90	92	88	80	(11)

Source: FactSet, Goldman Sachs Global Investment Research

Disclosure Appendix

Reg AC

We, Ben Snider, Ryan Hammond, Jenny Ma, Daniel Chavez, Kartik Jayachandran and Christophe Sung, hereby certify that all of the views expressed in this report accurately reflect our personal views, which have not been influenced by considerations of the firm's business or client relationships.

Unless otherwise stated, the individuals listed on the cover page of this report are analysts in Goldman Sachs' Global Investment Research division.

Contributing Authors: Ben Snider Goldman Sachs & Co. LLC, Ryan Hammond Goldman Sachs & Co. LLC, Jenny Ma Goldman Sachs & Co. LLC, Daniel Chavez Goldman Sachs & Co. LLC, Kartik Jayachandran Goldman Sachs & Co. LLC, Christophe Sung Goldman Sachs & Co. LLC.

Unless otherwise stated, the individuals listed in the Contributing Authors disclosure of this report are analysts in Goldman Sachs' Global Investment Research division.

Disclosures

Basket disclosure

The ability to trade the basket(s) in this report will depend upon market conditions, including liquidity and borrow constraints at the time of trade.

Marquee disclosure

Marquee is a product of Goldman Sachs FICC and Equities. Any Marquee content linked in this report is not necessarily representative of GS Research views. Because the Marquee platform and third-party platforms (e.g., Bloomberg) are only permitted to reflect changes to these baskets after publication of this note, there will be a delay between the rebalancing of baskets in this note and when the changes to these baskets are reflected on such platforms. If you need access to Marquee, please contact your GS salesperson or email the Marquee team at gs-marquee-sales@gs.com.

Regulatory disclosures

Disclosures required by United States laws and regulations

See company-specific regulatory disclosures above for any of the following disclosures required as to companies referred to in this report: manager or co-manager in a pending transaction; 1% or other ownership; compensation for certain services; types of client relationships; managed/co-managed public offerings in prior periods; directorships; for equity securities, market making and/or specialist role. Goldman Sachs trades or may trade as a principal in debt securities (or in related derivatives) of issuers discussed in this report.

The following are additional required disclosures: **Ownership and material conflicts of interest:** Goldman Sachs policy prohibits its analysts, professionals reporting to analysts and members of their households from owning securities of any company in the analyst's area of coverage. **Analyst compensation:** Analysts are paid in part based on the profitability of Goldman Sachs, which includes investment banking revenues. **Analyst as officer or director:** Goldman Sachs policy generally prohibits its analysts, persons reporting to analysts or members of their households from serving as an officer, director or advisor of any company in the analyst's area of coverage. **Non-U.S. Analysts:** Non-U.S. analysts may not be associated persons of Goldman Sachs & Co. LLC and therefore may not be subject to FINRA Rule 2241 or FINRA Rule 2242 restrictions on communications with a subject company, public appearances and trading in securities covered by the analysts.

Additional disclosures required under the laws and regulations of jurisdictions other than the United States

The following disclosures are those required by the jurisdiction indicated, except to the extent already made above pursuant to United States laws and regulations. **Australia:** Goldman Sachs Australia Pty Ltd and its affiliates are not authorised deposit-taking institutions (as that term is defined in the Banking Act 1959 (Cth)) in Australia and do not provide banking services, nor carry on a banking business, in Australia. This research, and any access to it, is intended only for "wholesale clients" within the meaning of the Australian Corporations Act, unless otherwise agreed by Goldman Sachs. In producing research reports, members of Global Investment Research of Goldman Sachs Australia may attend site visits and other meetings hosted by the companies and other entities which are the subject of its research reports. In some instances the costs of such site visits or meetings may be met in part or in whole by the issuers concerned if Goldman Sachs Australia considers it is appropriate and reasonable in the specific circumstances relating to the site visit or meeting. To the extent that the contents of this document contains any financial product advice, it is general advice only and has been prepared by Goldman Sachs without taking into account a client's objectives, financial situation or needs. A client should, before acting on any such advice, consider the appropriateness of the advice having regard to the client's own objectives, financial situation and needs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests and a copy of Goldman Sachs' Australian Sell-Side Research Independence Policy Statement are available at: <https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html>. **Brazil:** Disclosure information in relation to CVM Resolution n. 20 is available at <https://www.gs.com/worldwide/brazil/area/gir/index.html>. Where applicable, the Brazil-registered analyst primarily responsible for the content of this research report, as defined in Article 20 of CVM Resolution n. 20, is the first author named at the beginning of this report, unless indicated otherwise at the end of the text. **Canada:** This information is being provided to you for information purposes only and is not, and under no circumstances should be construed as, an advertisement, offering or solicitation by Goldman Sachs & Co. LLC for purchasers of securities in Canada to trade in any Canadian security. Goldman Sachs & Co. LLC is not registered as a dealer in any jurisdiction in Canada under applicable Canadian securities laws and generally is not permitted to trade in Canadian securities and may be prohibited from selling certain securities and products in certain jurisdictions in Canada. If you wish to trade in any Canadian securities or other products in Canada please contact Goldman Sachs Canada Inc., an affiliate of The Goldman Sachs Group Inc., or another registered Canadian dealer. **Hong Kong:** Further information on the securities of covered companies referred to in this research may be obtained on request from Goldman Sachs (Asia) L.L.C. **India:** Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (India) Securities Private Limited, Research Analyst - SEBI Registration Number INH000001493, 10th Floor, Ascent-Worli, Sudam Kalu Ahire Marg, Worli, Mumbai-400 025, India, Corporate Identity Number U74140MH2006FTC160634, Phone +91 22 6616 9000, Fax +91 22 6616 9001. Goldman Sachs may beneficially own 1% or more of the securities (as such term is defined in clause 2 (h) the Indian Securities Contracts (Regulation) Act, 1956) of the subject company or companies referred to in this research report. Investment in securities market are subject to market risks. Read all the related documents carefully before investing. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Goldman Sachs (India) Securities Private Limited compliance officer and investor grievance contact details can be found at: <https://www.goldmansachs.com/worldwide/india/documents/Grievance-Redressal-and-Escalation-Matrix.pdf>, and a copy of the annual audit compliance report can be found at this link: <https://publishing.gs.com/content/site/india-annual-compliance-report.html>. **Japan:** See below. **Korea:** This research, and any access to it, is intended only for "professional investors" within the meaning of the Financial Services and Capital Markets Act, unless otherwise agreed by Goldman Sachs. Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (Asia) L.L.C., Seoul Branch. **New Zealand:** Goldman Sachs New Zealand Limited and its affiliates are neither "registered banks" nor "deposit takers" (as defined in the Reserve Bank of New Zealand Act 1989) in New Zealand. This research, and any access to it, is intended for "wholesale clients" (as defined in the Financial Advisers Act 2008) unless otherwise agreed by Goldman Sachs. A copy of certain Goldman Sachs

Australia and New Zealand disclosure of interests is available at: <https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html>.

Russia: Research reports distributed in the Russian Federation are not advertising as defined in the Russian legislation, but are information and analysis not having product promotion as their main purpose and do not provide appraisal within the meaning of the Russian legislation on appraisal activity. Research reports do not constitute a personalized investment recommendation as defined in Russian laws and regulations, are not addressed to a specific client, and are prepared without analyzing the financial circumstances, investment profiles or risk profiles of clients. Goldman Sachs assumes no responsibility for any investment decisions that may be taken by a client or any other person based on this research report. **Singapore:** Goldman Sachs (Singapore) Pte. (Company Number: 198602165W), which is regulated by the Monetary Authority of Singapore, accepts legal responsibility for this research, and should be contacted with respect to any matters arising from, or in connection with, this research. **Taiwan:** This material is for reference only and must not be reprinted without permission. Investors should carefully consider their own investment risk. Investment results are the responsibility of the individual investor. **United Kingdom:** Persons who would be categorized as retail clients in the United Kingdom, as such term is defined in the rules of the Financial Conduct Authority, should read this research in conjunction with prior Goldman Sachs research on the covered companies referred to herein and should refer to the risk warnings that have been sent to them by Goldman Sachs International. A copy of these risks warnings, and a glossary of certain financial terms used in this report, are available from Goldman Sachs International on request.

European Union and United Kingdom: Disclosure information in relation to Article 6 (2) of the European Commission Delegated Regulation (EU) (2016/958) supplementing Regulation (EU) No 596/2014 of the European Parliament and of the Council (including as that Delegated Regulation is implemented into United Kingdom domestic law and regulation following the United Kingdom's departure from the European Union and the European Economic Area) with regard to regulatory technical standards for the technical arrangements for objective presentation of investment recommendations or other information recommending or suggesting an investment strategy and for disclosure of particular interests or indications of conflicts of interest is available at <https://www.gs.com/disclosures/europeanpolicy.html> which states the European Policy for Managing Conflicts of Interest in Connection with Investment Research.

Japan: Goldman Sachs Japan Co., Ltd. is a Financial Instrument Dealer registered with the Kanto Financial Bureau under registration number Kinsho 69, and a member of Japan Securities Dealers Association, Financial Futures Association of Japan Type II Financial Instruments Firms Association, and Investment Management Association of Japan. Sales and purchase of equities are subject to commission pre-determined with clients plus consumption tax. See company-specific disclosures as to any applicable disclosures required by Japanese stock exchanges, the Japanese Securities Dealers Association or the Japanese Securities Finance Company.

Global product; distributing entities

Goldman Sachs Global Investment Research produces and distributes research products for clients of Goldman Sachs on a global basis. Analysts based in Goldman Sachs offices around the world produce research on industries and companies, and research on macroeconomics, currencies, commodities and portfolio strategy. This research is disseminated in Australia by Goldman Sachs Australia Pty Ltd (ABN 21 006 797 897); in Brazil by Goldman Sachs do Brasil Corretora de Títulos e Valores Mobiliários S.A.; Public Communication Channel Goldman Sachs Brazil: 0800 727 5764 and / or contatogoldmanbrasil@gs.com. Available Weekdays (except holidays), from 9am to 6pm. Canal de Comunicação com o Público Goldman Sachs Brasil: 0800 727 5764 e/ou contatogoldmanbrasil@gs.com. Horário de funcionamento: segunda-feira à sexta-feira (exceto feriados), das 9h às 18h; in Canada by Goldman Sachs & Co. LLC; in Hong Kong by Goldman Sachs (Asia) L.L.C.; in India by Goldman Sachs (India) Securities Private Ltd.; in Japan by Goldman Sachs Japan Co., Ltd.; in the Republic of Korea by Goldman Sachs (Asia) L.L.C., Seoul Branch; in New Zealand by Goldman Sachs New Zealand Limited; in Russia by OOO Goldman Sachs; in Singapore by Goldman Sachs (Singapore) Pte. (Company Number: 198602165W); and in the United States of America by Goldman Sachs & Co. LLC. Goldman Sachs International has approved this research in connection with its distribution in the United Kingdom.

Goldman Sachs International ("GSI"), authorised by the Prudential Regulation Authority ("PRA") and regulated by the Financial Conduct Authority ("FCA") and the PRA, has approved this research in connection with its distribution in the United Kingdom.

European Economic Area: Goldman Sachs Bank Europe SE ("GSBE") is a credit institution incorporated in Germany and, within the Single Supervisory Mechanism, subject to direct prudential supervision by the European Central Bank and in other respects supervised by German Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin) and Deutsche Bundesbank and disseminates research within the European Economic Area.

General disclosures

This research is for our clients only. Other than disclosures relating to Goldman Sachs, this research is based on current public information that we consider reliable, but we do not represent it is accurate or complete, and it should not be relied on as such. The information, opinions, estimates and forecasts contained herein are as of the date hereof and are subject to change without prior notification. We seek to update our research as appropriate, but various regulations may prevent us from doing so. Other than certain industry reports published on a periodic basis, the large majority of reports are published at irregular intervals as appropriate in the analyst's judgment.

Goldman Sachs conducts a global full-service, integrated investment banking, investment management, and brokerage business. We have investment banking and other business relationships with a substantial percentage of the companies covered by Global Investment Research. Goldman Sachs & Co. LLC, the United States broker dealer, is a member of SIPC (<https://www.sipc.org>).

Our salespeople, traders, and other professionals may provide oral or written market commentary or trading strategies to our clients and principal trading desks that reflect opinions that are contrary to the opinions expressed in this research. Our asset management area, principal trading desks and investing businesses may make investment decisions that are inconsistent with the recommendations or views expressed in this research.

We and our affiliates, officers, directors, and employees will from time to time have long or short positions in, act as principal in, and buy or sell, the securities or derivatives, if any, referred to in this research, unless otherwise prohibited by regulation or Goldman Sachs policy.

The views attributed to third party presenters at Goldman Sachs arranged conferences, including individuals from other parts of Goldman Sachs, do not necessarily reflect those of Global Investment Research and are not an official view of Goldman Sachs.

Any third party referenced herein, including any salespeople, traders and other professionals or members of their household, may have positions in the products mentioned that are inconsistent with the views expressed by analysts named in this report.

This research is focused on investment themes across markets, industries and sectors. It does not attempt to distinguish between the prospects or performance of, or provide analysis of, individual companies within any industry or sector we describe.

Any trading recommendation in this research relating to an equity or credit security or securities within an industry or sector is reflective of the investment theme being discussed and is not a recommendation of any such security in isolation.

This research is not an offer to sell or the solicitation of an offer to buy any security in any jurisdiction where such an offer or solicitation would be illegal. It does not constitute a personal recommendation or take into account the particular investment objectives, financial situations, or needs of individual clients. Clients should consider whether any advice or recommendation in this research is suitable for their particular circumstances and, if appropriate, seek professional advice, including tax advice. The price and value of investments referred to in this research and the income from them may fluctuate.

Past performance is not a guide to future performance, future returns are not guaranteed, and a loss of original capital may occur. Fluctuations in exchange rates could have adverse effects on the value or price of, or income derived from, certain investments.

Certain transactions, including those involving futures, options, and other derivatives, give rise to substantial risk and are not suitable for all investors. Investors should review current options and futures disclosure documents which are available from Goldman Sachs sales representatives or at <https://www.theocc.com/about/publications/character-risks.jsp> and https://www.goldmansachs.com/disclosures/cftc_fcm_disclosures. Transaction costs may be significant in option strategies calling for multiple purchase and sales of options such as spreads. Supporting documentation will be supplied upon request.

Differing Levels of Service provided by Global Investment Research: The level and types of services provided to you by Goldman Sachs Global Investment Research may vary as compared to that provided to internal and other external clients of GS, depending on various factors including your individual preferences as to the frequency and manner of receiving communication, your risk profile and investment focus and perspective (e.g., marketwide, sector specific, long term, short term), the size and scope of your overall client relationship with GS, and legal and regulatory constraints. As an example, certain clients may request to receive notifications when research on specific securities is published, and certain clients may request that specific data underlying analysts' fundamental analysis available on our internal client websites be delivered to them electronically through data feeds or otherwise. No change to an analyst's fundamental research views (e.g., ratings, price targets, or material changes to earnings estimates for equity securities), will be communicated to any client prior to inclusion of such information in a research report broadly disseminated through electronic publication to our internal client websites or through other means, as necessary, to all clients who are entitled to receive such reports.

All research reports are disseminated and available to all clients simultaneously through electronic publication to our internal client websites. Not all research content is redistributed to our clients or available to third-party aggregators, nor is Goldman Sachs responsible for the redistribution of our research by third party aggregators. For research, models or other data related to one or more securities, markets or asset classes (including related services) that may be available to you, please contact your GS representative or go to <https://research.gs.com>.

Disclosure information is also available at <https://www.gs.com/research/hedge.html> or from Research Compliance, 200 West Street, New York, NY 10282.

© 2026 Goldman Sachs.

You are permitted to store, display, analyze, modify, reformat, and print the information made available to you via this service only for your own use. You may not resell or reverse engineer this information to calculate or develop any index for disclosure and/or marketing or create any other derivative works or commercial product(s), data or offering(s) without the express written consent of Goldman Sachs. You are not permitted to publish, transmit, or otherwise reproduce this information, in whole or in part, in any format to any third party without the express written consent of Goldman Sachs. This foregoing restriction includes, without limitation, using, extracting, downloading or retrieving this information, in whole or in part, to train or finetune a machine learning or artificial intelligence system, or to provide or reproduce this information, in whole or in part, as a prompt or input to any such system.

GLOBAL ECONOMICS ANALYST

The Path to 2075 — Is Convergence Really Over?

- Do poor economies grow faster than rich economies? Many economists argue that the evidence for convergence between low- and high-income economies is weak, suggesting that generalized convergence was largely confined to the ‘hyper-globalization’ decade of 2000–10. If true, this would be bad news for global well-being, and it would fundamentally weaken a key justification for long-term investment in emerging markets.
- We argue that previous studies that reported only limited evidence of convergence are incomplete, because they fail to weight countries by population size – effectively treating China and Cyprus’s experiences as equally significant. With this simple adjustment, we find that economic convergence has been a consistent feature of the global economy since the 1980s.
- The key drivers of this convergence have been Asia’s most populous nations: China (since the early 1980s), and India, Indonesia and Bangladesh (from the late 1990s onwards). Together, these four countries account for more than 40% of the global population.
- Among developed economies, the US has been an important positive outlier. Despite possessing one of the world’s highest levels of GDP per capita, its GDP per capita growth has continued to outpace most other developed markets.
- The contrast that we find with unweighted convergence studies suggests an intuitive point: periods of rapid globalization, such as 2000–10, are especially beneficial for the development of small economies with limited domestic markets. However, for large emerging market economies – especially those in East Asia – economic convergence appears relatively robust. This ongoing convergence implies that the share of global GDP represented by emerging markets (and East Asian EMs in particular) will continue to rise over time.
- Looking ahead, risks exist in both directions. The development of AI seems likely to bolster the strong performance of the US and parts of East Asia (in particular, China, South Korea and Taiwan). On the downside, increased protectionism could result in a reversal of globalization, while climate change poses a particular risk to Southern Asia and parts of Africa.

Kevin Daly
+44(20)7774-5908 | kevin.daly@gs.com
Goldman Sachs International

The Path to 2075 — Is Convergence Really Over?

Economic Convergence, Led by Asia's Powerhouses

In this *Global Economics Analyst*, we review the performance of GDP per capita across 121 developed (DM) and emerging market (EM) economies since 1980, with a particular emphasis on the process of long-term economic convergence.

The principle of convergence – that poorer economies generally grow faster than richer economies over time – is a central assumption underpinning our *Path to 2075 projections* for the global economy (see Box 1).¹ The logic is that the marginal return on investment should be higher in poorer economies where capital is relatively scarce, which attracts investment, increases capital stock, and raises living standards.

We begin by outlining some key stylized facts about GDP per capita levels and growth rates across the world. *Exhibit 1* (left-hand side) depicts GDP per capita across various regions since 1980, while the right-hand panel presents the same data indexed to 1980. Several features stand out:

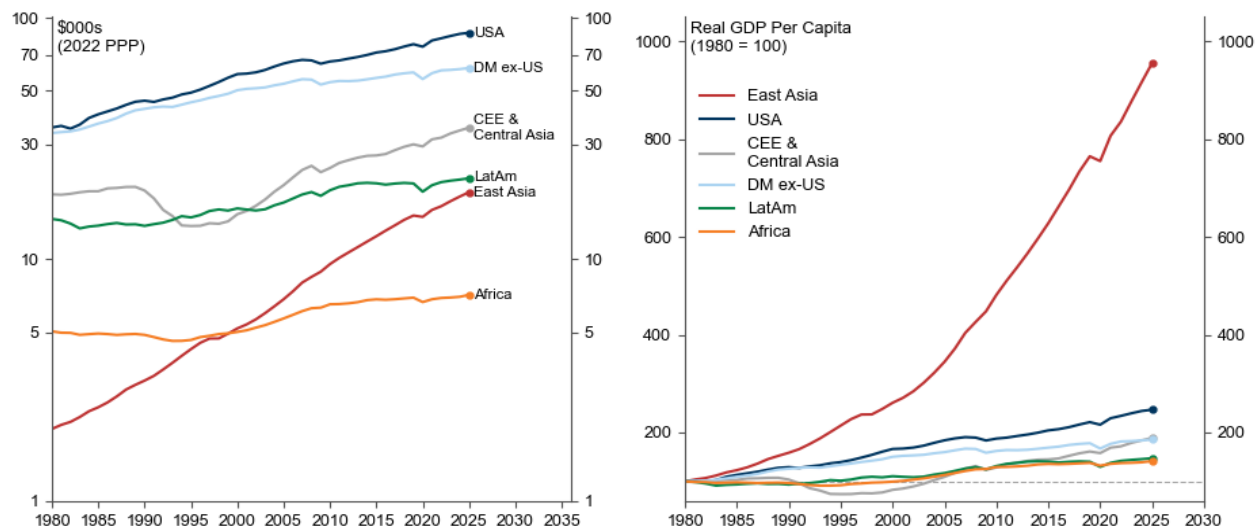
1. Cross-country differences in GDP per capita levels are large, even after adjusting for differences in purchasing power parity.² To take one striking example, GDP per person in the US is more than 12 times higher than the average across Africa.
2. East Asia's emerging markets have seen rapid GDP per capita growth, with average output per person now nearly 10 times higher than in 1980. However, GDP per capita in the region remains well below US and Western European levels, suggesting further potential for convergence.
3. US GDP per capita growth since 1980 has either matched or exceeded GDP per capita growth in the majority of other regions (with East Asia the only exception), despite having a much higher starting level of GDP per capita.
4. GDP per capita growth in Latin American and African economies has disappointed, averaging +0.9% and +0.8% per year respectively since 1980.
5. The break-up of the Soviet Union and the dissolution of the Eastern Bloc was initially very negative for GDP per capita in Eastern Europe and Central Asia in the early 1990s. However, GDP per capita levels have since rebounded in this region.

¹ “*The Path to 2075 — Slower Global Growth, But Convergence Remains Intact*”, Goldman Sachs Global Economics Paper, 6 December 2022.

² The choice between PPP weights and market weights depends on the comparison's purpose. PPP weights account for lower prices in poorer economies and better reflect welfare gains, while market weights show the Dollar value of each economy's GDP without adjusting for spending power. Since we are focused on income convergence, we use PPP weights. See “*How Fast Does the World Grow?*”, *GS Global Economics Analyst*, 27 October 2021.

Exhibit 1: GDP per Capita in East Asian EMs Has Risen Nearly Tenfold Since 1980, But Remains Well Below DM Levels

LHS: Real GDP Per Capita (\$000s log scale; 2022 PPP); RHS: Real GDP Per Capita (2022 PPP, 1980=100)



Regions are weighted by population size

Source: The Conference Board Total Economy Database™ (April 2026), Goldman Sachs Global Investment Research

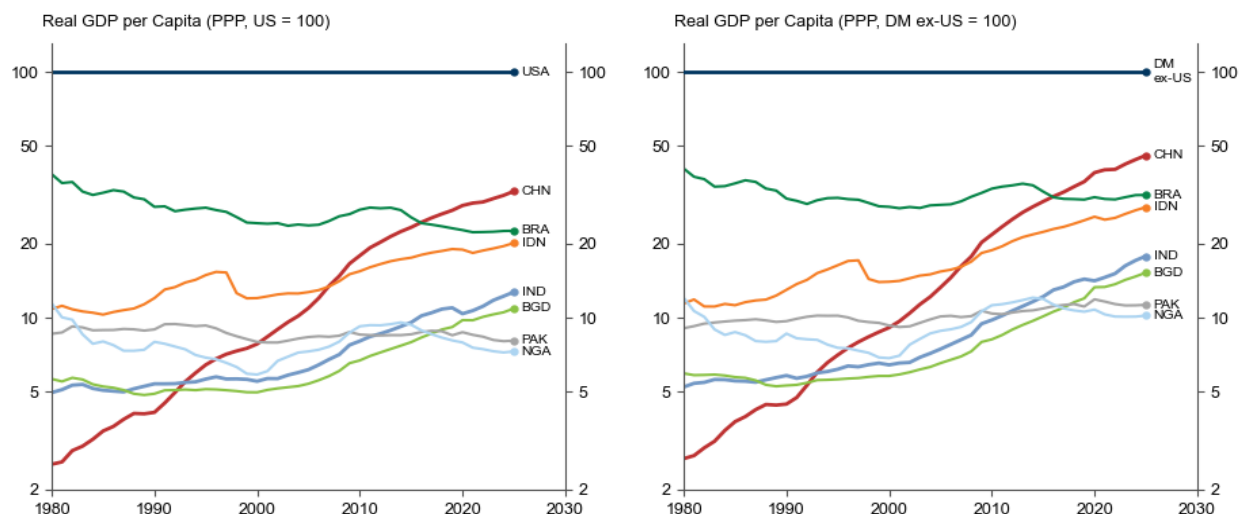
Exhibit 2 compares the GDP per capita of the eight most populous countries. The US is fixed as the 'frontier economy' in the left panel, while developed economies (excluding the US) are set as the benchmark in the right panel. China has seen significant income convergence, rising from just over 2% of US levels in 1980 to around 33% today. Since the late 1990s, India, Indonesia, and Bangladesh have also shown clear progress.

When compared to US GDP per capita, Brazil, Pakistan, and Nigeria have underperformed, with Brazil slipping further behind the US over time, while Pakistan and Nigeria have remained largely unchanged. However, as the US has shown particularly strong performance among developed markets, the relative picture shifts if we compare these economies to developed markets more broadly – Pakistan and Nigeria now show some convergence, whereas Brazil has remained more or less flat.³

³ The aim of this piece is to record, rather than explain, the convergence that has occurred. However, the question remains as to why the GDP per capita performance of Brazil, Pakistan and Nigeria has been disappointing. In Brazil's case, macroeconomic instability, high government spending crowding out private investment, significant trade barriers and inadequate infrastructure are all cited as contributing factors. For Pakistan and Nigeria, macroeconomic instability, elite capture and incomplete property rights are frequently mentioned.

Exhibit 2: China Has Converged Fastest Among the World's Most Populous States, But India, Indonesia and Bangladesh Are Also Converging

Real GDP Per Capita in 2022 PPP, log scale (LHS: US = 100, RHS: DM ex-US=100)



Source: The Conference Board Total Economy Database™ (April 2026), Goldman Sachs Global Investment Research

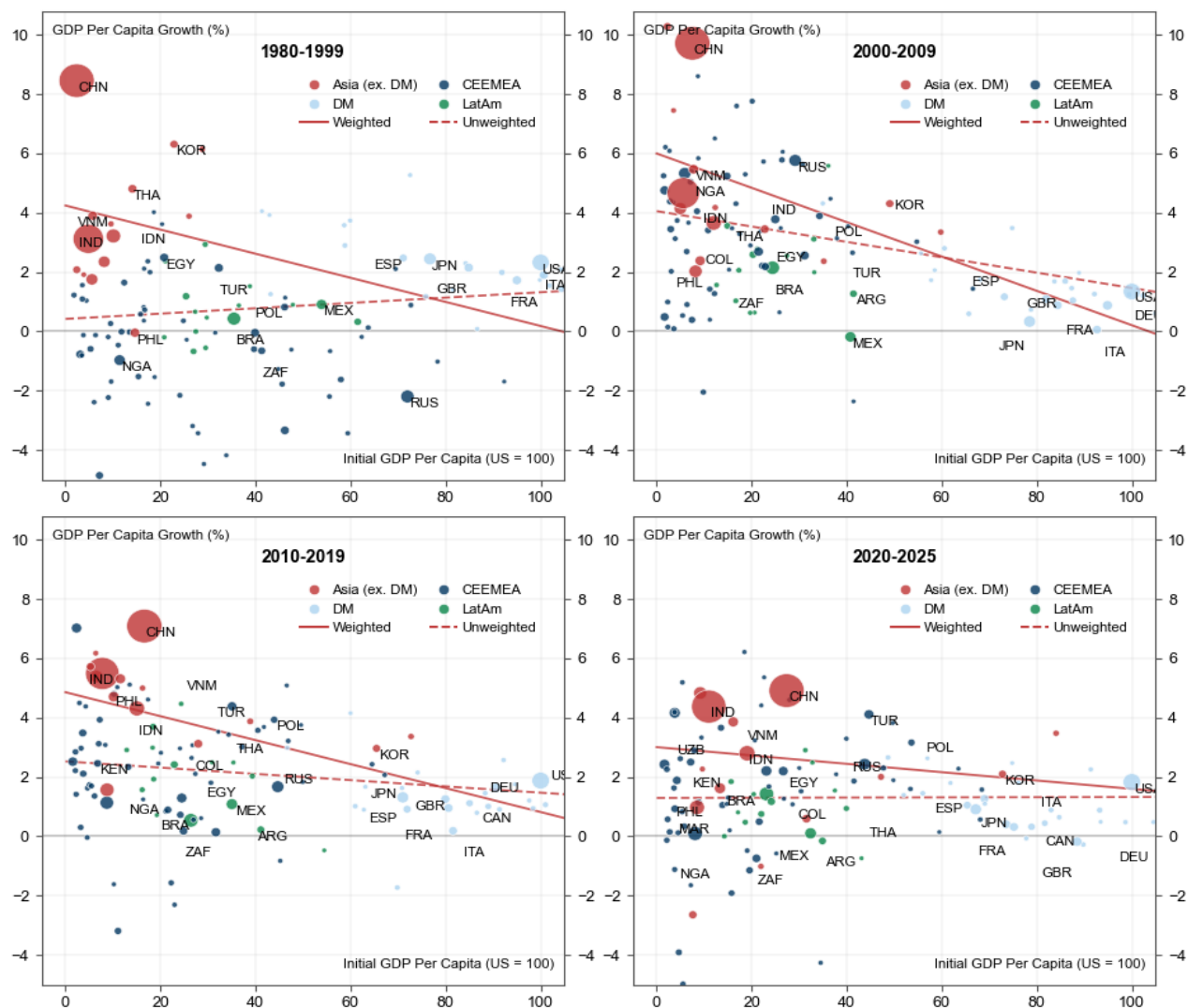
Population-Weighted Convergence is Reasonably Stable Across Decades

Exhibit 3 illustrates average GDP per capita growth in relation to initial GDP per capita levels for EM and DM economies across four distinct periods: 1980-1999, 2000-2009, 2010-2019, and 2020-2025. The level of GDP per capita is shown relative to the US (US = 100). We also show lines of best fit for all 121 economies on both a population-weighted and an unweighted basis.

Across all four periods, the population-weighted regression line slopes downwards, indicating that GDP per capita growth has been higher in low-income economies than in high-income economies (i.e., that there has been convergence). In contrast, the regression lines for unweighted countries are mixed, showing convergence in some periods but not in others. Moreover, the slope of the regression line is steeper for population-weighted countries in all four periods. Note that, although the performance of China and India is central to the narrative of population-weighted convergence – unsurprising given that together they comprise more than a third of the world's population – the regression lines continue to slope downwards from 2000 even when these two countries are excluded.

Exhibit 3: GDP per Capita Has Grown Faster in Low-Income Economies than in High-Income Economies

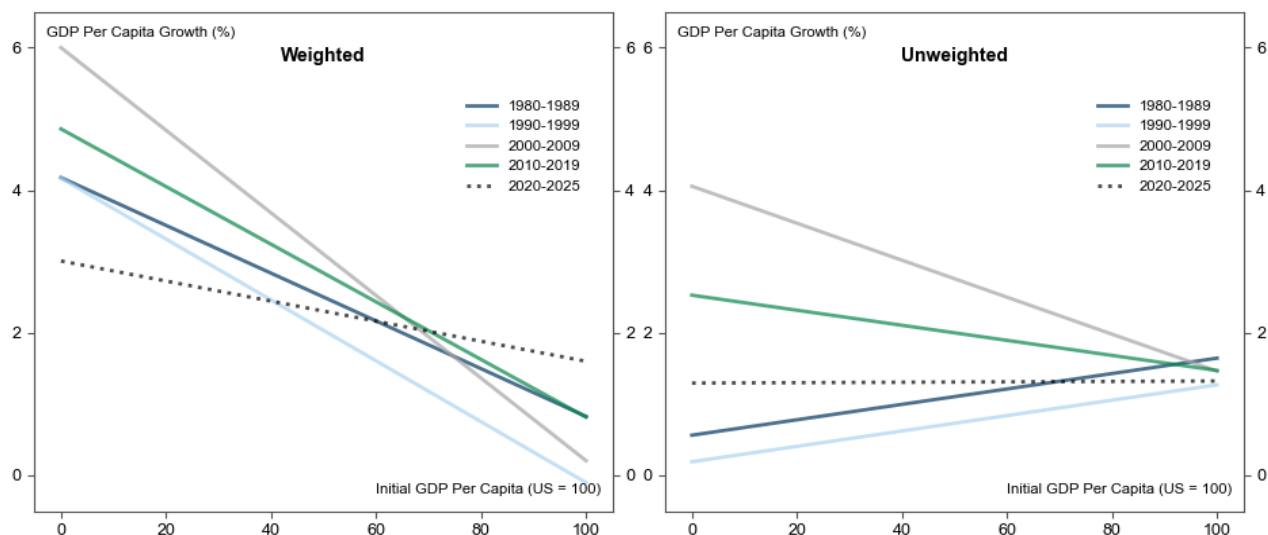
Real GDP Per Capita (PPP, %) vs Initial GDP Per Capita (US=100); Size of bubble = average population size



Source: The Conference Board Total Economy Database™ (April 2026), Goldman Sachs Global Investment Research

Exhibit 4 examines decade-by-decade changes in regression slopes. The population-weighted regressions show broadly similar slopes, except for 2020-2025, where the slope is still downward but less marked. The past half decade has seen a series of major shocks – the pandemic, the 2022 energy crisis, and 2025 US tariffs – and it remains to be seen how the slope will look for the decade as a whole. In comparison with the weighted lines, the regression slopes for countries on an unweighted basis display much greater variability.

Exhibit 4: GDP per Capita Growth Has Been Faster in Low-Income Economies in Every Decade on a Weighted Basis, But the Pattern is More Mixed on an Unweighted Basis
Population weighted vs unweighted fitted lines



Source: The Conference Board Total Economy Database™ (April 2026), Goldman Sachs Global Investment Research

How rapid is this convergence? Since 2000, individuals in countries with a GDP per capita at 50% of the US level have, on average, experienced GDP per capita growth approximately 2.3pp higher than those in developed economies with productivity levels similar to the US.⁴ Over a 10-year period, GDP per capita levels for such individuals would have risen from 50% of US levels to around 61% of US levels.

How Our Results Compare with the Existing Growth Literature

On one level, our finding that the process of economic convergence is relatively robust and has been a consistent feature since the early 1980s should not be particularly surprising, as it is a central prediction of many economic models.⁵ However, the majority of cross-country studies have found little evidence that poor countries grow faster than rich ones, irrespective of other factors – in other words, there is little evidence of absolute or generalized convergence in the GDP per capita levels of countries over time.⁶

By making the simple adjustment of weighting each country by population size, our results suggest that convergence between poor and rich economies is a more robust feature than implied by previous studies of this type. **If the unit of aggregation is taken to be people, rather than countries – which is what population-weighting achieves – then convergence is alive and well.**

⁴ The slope of the regression line for 2000-2025 is around -0.05.

⁵ Standard (neo-classical) growth theory suggests that relatively poor, low-productivity economies should grow at a faster rate than relatively rich, high-productivity countries. This result follows directly from the assumption of diminishing marginal returns on capital: in developing economies where capital is limited, the returns on capital should be greater than in richer economies where capital is abundant. In turn, the relatively high returns on capital in less developed economies should attract investment, increasing the capital stock and raising living standards.

⁶ A key paper in this literature is Barro and Sala-i-Martin (1992), 'Convergence', *Journal of Political Economy*. Other studies found evidence of 'conditional' convergence – meaning that convergence occurs only when specific factors, such as institutional quality, are present. See Mankiw, Romer and Weil (1992) 'A Contribution to the Empirics of Economic Growth', *Quarterly Journal of Economics*.

We are not the first to adjust for population size in this way. Previous studies have also weighted by population size and, consistent with our findings, have found stronger evidence of convergence on this basis.⁷ Nonetheless, those studies maintain that, when assessing which economic policies drive growth and convergence, it is still reasonable to treat each country as an equal unit, since such policies are largely determined at the national level. We disagree for two reasons: First, economic welfare is a study of people, not nations, and if more people reside in some nations than others, they should be weighted accordingly. Second, a country's size is itself a potentially significant factor influencing growth and convergence. If larger countries tend to grow faster than smaller ones, this may partially be attributable to their size. Overlooking this aspect risks missing a potentially important determinant of growth.⁸

The contrast that we find with unweighted convergence studies suggests an important and intuitive point: periods of rapid globalization, such as 2000–10, tend to be particularly beneficial for the development of small economies with limited domestic markets. However, for large emerging market economies, economic convergence appears to be more resilient to the pace of globalization.

EM Growth Continues to Outstrip DM Growth by a Wide Margin

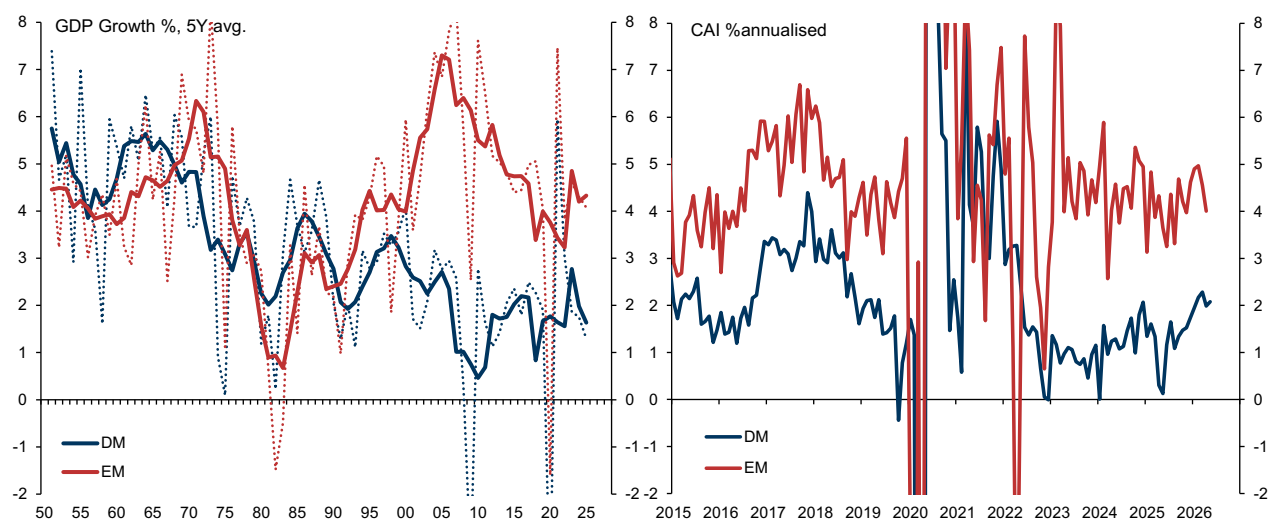
Our findings also explain why EMs consistently outgrow DM countries in our coverage – if convergence were not a fairly consistent feature of the global economy, this would not occur. While population growth is also higher in EMs, it does not fully explain the gap in potential growth rates (4.2% vs 1.6%). Moreover, in the four years since we published our *Path to 2075 long-term economic projections*, the gap between EM and DM growth has so far been larger than we expected (2.6pp vs. 2.0pp).

⁷ In a follow-up paper to Barro and Sala-i-Martin (1992)), Sala-i-Martin (2006) acknowledges that there is stronger evidence of convergence in income distribution at the level of the individual rather than the country but maintains that a country framework makes sense when evaluating which economic policies drive growth and convergence. See Sala-i-Martin (2006), 'The World Distribution of Income: Falling Poverty and... Convergence, Period', *Quarterly Journal of Economics*.

⁸ A large domestic market makes it easier to achieve economies of scale while still focused primarily on domestic demand, especially if industries are protected during their early stages of development. This strategy – protecting fledgling industries until they reach sufficient size – is less likely to succeed in countries with a small domestic market. There is also a related but distinct point concerning economic geography: large economies are able to support bigger cities and benefit from agglomeration effects, and economic convergence is often associated with rising urbanization and increased specialization. This is more difficult to achieve in small economies with small populations.

Exhibit 5: After a Brief Hiatus During the Pandemic, EM Economies Are Outgrowing DM Economies

LHS: GDP Growth (% 5Y avg); RHS: Current Activity Indicator (% change, annualized)



Source: Goldman Sachs Global Investment Research

The ongoing convergence between (population-weighted) poor and rich economies implies that the incomes of EM economies will gradually move closer to those of developed economies – resulting in a further reduction in global income inequality – and that the share of global GDP represented by emerging markets will continue to increase over time.

Looking forward, there are both positive and negative risks to long-term economic growth and convergence. The development of AI seems likely to bolster the strong performance of the US and parts of South-East Asia and lift global GDP productivity growth more generally.⁹ In terms of AI-related capital expenditure, South Korea and Taiwan are already the main beneficiaries, while from a development and adoption standpoint, China seems well-placed to gain.

On the downside, we consider two risks especially significant: First, although global growth and trade have so far been surprisingly resilient to rising tariffs, increased protectionism could still trigger a reversal of globalization. Second, climate change represents a particular threat to Southern Asia and parts of Africa.

Kevin Daly and Alessa Abraham*

**Alessa is an intern in the CEEMEA economics team.*

⁹ For a discussion of the potential boost to EM productivity from AI and technology adoption more generally, see “EM Technology Adoption Cycles and Implications for Artificial Intelligence”, *Global Economics Comment*, 21 May 2026.

Box 1: The Path to 2075 — Long-Term Projections for the Global Economy

The Path to 2075 projections builds on our original BRICs analysis, offering long-term forecasts for the global economy, covering 104 countries out to the year 2075.¹⁰ Exhibit 6 provides a high-level summary of our projections for the major regional aggregates, broken down by decade.

Our projections imply that global growth in 2024-29 (2.8%) will be faster than in 2020-24, but slower than in the 2010-19 decade (3.2%, based on market FX weights).¹¹ We project that EM growth will continue to outstrip DM growth (3.8% vs. 1.8%), with around half of this difference due to demographic factors and the other half due to productivity growth differentials. We expect Asia (ex-DM) to remain the fastest-growing region.

Exhibit 6: A Gradual Slowdown in Global Economic Growth, With EM Growth Continuing to Outstrip DM

	Market FX Weighted								
	2000-2009	2010-2019	2020-2029	2024-2029	2030-2039	2040-2049	2050-2059	2060-2069	2070-2079
World	2.7	3.2	2.4	2.8	2.5	2.1	2.0	1.8	1.7
DM	1.6	1.9	1.5	1.8	1.6	1.4	1.3	1.2	1.1
EM	5.7	5.1	3.6	3.8	3.2	2.6	2.3	2.1	1.9
Asia (ex. DM)	7.6	6.7	4.1	4.2	3.1	2.4	2.1	1.8	1.5
CEEMEA	4.8	3.5	2.6	3.2	3.3	3.1	3.0	2.9	2.7
LatAm	2.8	2.4	2.3	3.0	3.1	2.7	2.3	1.9	1.6

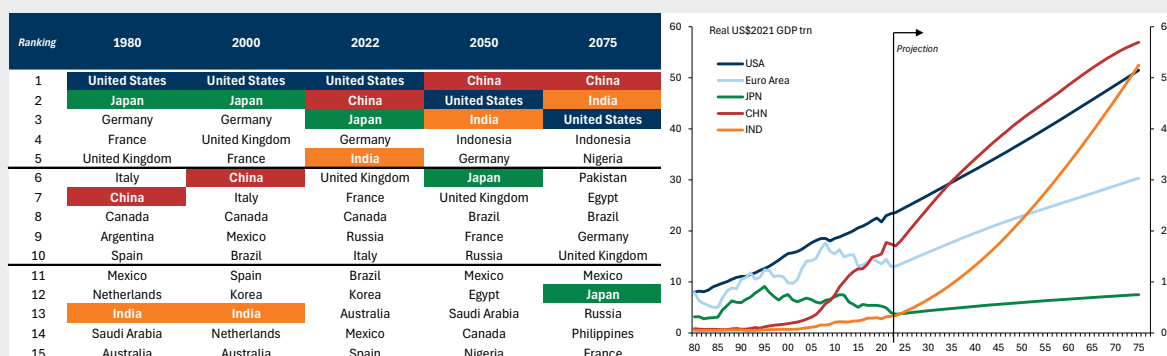
Source: Goldman Sachs Global Investment Research

Exhibit 7 combines our GDP estimates with our long-term real exchange rate projections, to project the real US Dollar value of major economies over time.

In 2050, we project that the world's five largest economies will be China, the US, India, Indonesia, and Germany (with Indonesia displacing Brazil and Russia among the list of largest EMs over this horizon). We estimate that China will overtake the US as the world's largest economy around 2035.¹² If we extend the projection horizon to 2075, the world's three largest economies are China, India, and the US, with India (just) overtaking the US. Interestingly, US potential GDP growth is expected to be materially faster than China's at that horizon because of its better demographic outlook. Given the right policies and institutions, we project that current EMs will make up seven of the world's top ten economies in 2075.

Exhibit 7: Our Projections Imply that China, the US, India, Indonesia, and Germany Will be the World's Five Largest Economies in 2050

World's largest economies (measured in USD)



Source: Goldman Sachs Global Investment Research

¹⁰ "The Path to 2075 — Slower Global Growth, But Convergence Remains Intact", Goldman Sachs Global

Making economic projections over a 50-year horizon for 104 countries inevitably involves a considerable degree of risk and uncertainty. Indeed, given the difficulties in forecasting even one or two years ahead, some readers may be skeptical of the value of forecasts that extend so far into the future. However, one advantage of making long-term projections is that cyclical risk – an important source of short-term forecasting error – tends to mean-revert over time, leaving GDP mostly determined by slower-moving trends in population, capital, and technology.

Economics Paper, 6 December 2022.

¹¹ Because of the large swings in GDP growth generated by the Covid pandemic, the 2024-29 column provides the cleanest indication of what our model indicates for near-term potential growth.

¹² Given the recent pessimism around China's long-term growth prospects, some readers may be surprised that we expect China to overtake the US at this horizon. However, three points are worth bearing in mind in this regard: First, China has already closed most of the gap with US GDP (China's GDP has risen from 12% of the US in 2000 to around 80% currently, measured in market exchange rates). Second, despite significant downward revisions, potential growth in China remains significantly higher than in the US on our revised estimates (4.0% vs. 1.9% for 2024-29). Third, in addition to differences in potential growth, we expect some of the US Dollar's real overvaluation vs. the Chinese Yuan to be unwound over the next 10-15 years.

Box 2: Comparing Our Results Across Different Growth Databases

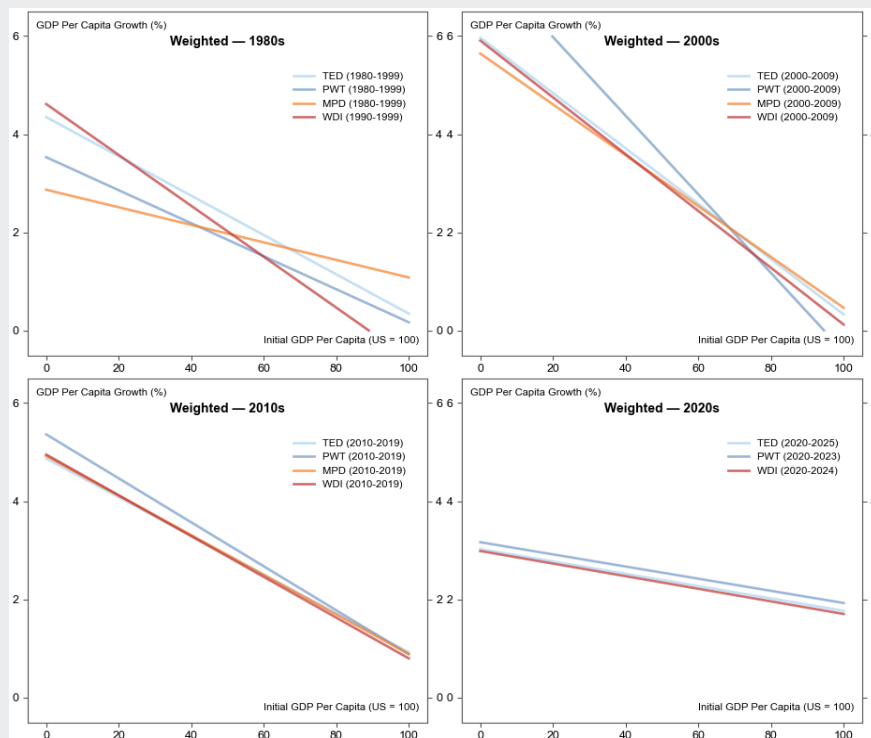
Growth and convergence studies typically use one of four different databases when analyzing long-term economic performance: World Development Indicators (WDI) by the World Bank, the Penn World Tables (PWT), the Maddison Project Database (MPD), and the Total Economy Database (TED) developed by the Conference Board. We generally use the Total Economy Database, partly because it provides the most up-to-date GDP figures. However, to cross-check our findings, we have replicated our analysis using each of these databases to ensure the robustness of our results.

Each of these databases possesses distinct advantages and limitations, making some more suitable than others depending on the research objectives. Among them, the Maddison database offers the most extensive historical coverage, in some cases reaching back centuries, making it ideal for studies in economic history. The World Bank's database has the shortest historical span, beginning only in 1990. Another key distinction is how they adjust GDP figures for differences in Purchasing Power Parity (PPP), which in turn impacts growth indicators.

The results of replicating our convergence analysis across each of the databases are presented in Exhibit 8. Although there are minor variations in the slopes of the fitted lines, the principal result—that weighted convergence persists—remains broadly consistent across all datasets.

Exhibit 8: Our Results Are Similar Across Each of the Main Growth Databases

Fitted lines - weighted by population, across the decades



Source: The Conference Board Total Economy Database™ (April 2026), Maddison Project Database (2020), Penn World Table 11.0, World Bank World Development Indicators, Goldman Sachs Global Investment Research

The Global Economics Team

Jan Hatzius
+1(212)902-0394
jan.hatzius@gs.com
Goldman Sachs & Co. LLC

Joseph Briggs
+1(212)902-2163
joseph.briggs@gs.com
Goldman Sachs & Co. LLC

Sarah Dong
+1(212)357-9741
sarah.dong@gs.com
Goldman Sachs & Co. LLC

Megan Peters
+44(20)7051-2058
megan.l.peters@gs.com
Goldman Sachs International

Disclosure Appendix

Reg AC

I, Kevin Daly, hereby certify that all of the views expressed in this report accurately reflect my personal views, which have not been influenced by considerations of the firm's business or client relationships.

Unless otherwise stated, the individuals listed on the cover page of this report are analysts in Goldman Sachs' Global Investment Research division.

Contributing Authors: Kevin Daly Goldman Sachs International.

Unless otherwise stated, the individuals listed in the Contributing Authors disclosure of this report are analysts in Goldman Sachs' Global Investment Research division.

Disclosures

Regulatory disclosures

Disclosures required by United States laws and regulations

See company-specific regulatory disclosures above for any of the following disclosures required as to companies referred to in this report: manager or co-manager in a pending transaction; 1% or other ownership; compensation for certain services; types of client relationships; managed/co-managed public offerings in prior periods; directorships; for equity securities, market making and/or specialist role. Goldman Sachs trades or may trade as a principal in debt securities (or in related derivatives) of issuers discussed in this report.

The following are additional required disclosures: **Ownership and material conflicts of interest:** Goldman Sachs policy prohibits its analysts, professionals reporting to analysts and members of their households from owning securities of any company in the analyst's area of coverage. **Analyst compensation:** Analysts are paid in part based on the profitability of Goldman Sachs, which includes investment banking revenues. **Analyst as officer or director:** Goldman Sachs policy generally prohibits its analysts, persons reporting to analysts or members of their households from serving as an officer, director or advisor of any company in the analyst's area of coverage. **Non-U.S. Analysts:** Non-U.S. analysts may not be associated persons of Goldman Sachs & Co. LLC and therefore may not be subject to FINRA Rule 2241 or FINRA Rule 2242 restrictions on communications with a subject company, public appearances and trading in securities covered by the analysts.

Additional disclosures required under the laws and regulations of jurisdictions other than the United States

The following disclosures are those required by the jurisdiction indicated, except to the extent already made above pursuant to United States laws and regulations. **Australia:** Goldman Sachs Australia Pty Ltd and its affiliates are not authorised deposit-taking institutions (as that term is defined in the Banking Act 1959 (Cth)) in Australia and do not provide banking services, nor carry on a banking business, in Australia. This research, and any access to it, is intended only for "wholesale clients" within the meaning of the Australian Corporations Act, unless otherwise agreed by Goldman Sachs. In producing research reports, members of Global Investment Research of Goldman Sachs Australia may attend site visits and other meetings hosted by the companies and other entities which are the subject of its research reports. In some instances the costs of such site visits or meetings may be met in part or in whole by the issuers concerned if Goldman Sachs Australia considers it is appropriate and reasonable in the specific circumstances relating to the site visit or meeting. To the extent that the contents of this document contains any financial product advice, it is general advice only and has been prepared by Goldman Sachs without taking into account a client's objectives, financial situation or needs. A client should, before acting on any such advice, consider the appropriateness of the advice having regard to the client's own objectives, financial situation and needs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests and a copy of Goldman Sachs' Australian Sell-Side Research Independence Policy Statement are available at: <https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html>. **Brazil:** Disclosure information in relation to CVM Resolution n. 20 is available at <https://www.gs.com/worldwide/brazil/area/gir/index.html>. Where applicable, the Brazil-registered analyst primarily responsible for the content of this research report, as defined in Article 20 of CVM Resolution n. 20, is the first author named at the beginning of this report, unless indicated otherwise at the end of the text. **Canada:** This information is being provided to you for information purposes only and is not, and under no circumstances should be construed as, an advertisement, offering or solicitation by Goldman Sachs & Co. LLC for purchasers of securities in Canada to trade in any Canadian security. Goldman Sachs & Co. LLC is not registered as a dealer in any jurisdiction in Canada under applicable Canadian securities laws and generally is not permitted to trade in Canadian securities and may be prohibited from selling certain securities and products in certain jurisdictions in Canada. If you wish to trade in any Canadian securities or other products in Canada please contact Goldman Sachs Canada Inc., an affiliate of The Goldman Sachs Group Inc., or another registered Canadian dealer. **Hong Kong:** Further information on the securities of covered companies referred to in this research may be obtained on request from Goldman Sachs (Asia) L.L.C. **India:** Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (India) Securities Private Limited, Research Analyst - SEBI Registration Number INH000001493, 10th Floor, Ascent-Worli, Sudam Kalu Ahire Marg, Worli, Mumbai-400 025, India, Corporate Identity Number U74140MH2006FTC160634, Phone +91 22 6616 9000, Fax +91 22 6616 9001. Goldman Sachs may beneficially own 1% or more of the securities (as such term is defined in clause 2 (h) the Indian Securities Contracts (Regulation) Act, 1956) of the subject company or companies referred to in this research report. Investment in securities market are subject to market risks. Read all the related documents carefully before investing. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Goldman Sachs (India) Securities Private Limited compliance officer and investor grievance contact details can be found at: <https://www.goldmansachs.com/worldwide/india/documents/Grievance-Redressal-and-Escalation-Matrix.pdf>, and a copy of the annual audit compliance report can be found at this link: <https://publishing.gs.com/content/site/india-annual-compliance-report.html>. **Japan:** See below. **Korea:** This research, and any access to it, is intended only for "professional investors" within the meaning of the Financial Services and Capital Markets Act, unless otherwise agreed by Goldman Sachs. Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (Asia) L.L.C., Seoul Branch. **New Zealand:** Goldman Sachs New Zealand Limited and its affiliates are neither "registered banks" nor "deposit takers" (as defined in the Reserve Bank of New Zealand Act 1989) in New Zealand. This research, and any access to it, is intended for "wholesale clients" (as defined in the Financial Advisers Act 2008) unless otherwise agreed by Goldman Sachs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests is available at: <https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html>. **Russia:** Research reports distributed in the Russian Federation are not advertising as defined in the Russian legislation, but are information and analysis not having product promotion as their main purpose and do not provide appraisal within the meaning of the Russian legislation on appraisal activity. Research reports do not constitute a personalized investment recommendation as defined in Russian laws and regulations, are not addressed to a specific client, and are prepared without analyzing the financial circumstances, investment profiles or risk profiles of clients. Goldman Sachs assumes no responsibility for any investment decisions that may be taken by a client or any other person based on this research report. **Singapore:** Goldman Sachs (Singapore) Pte. (Company Number: 198602165W), which is regulated by the Monetary Authority of Singapore, accepts legal responsibility for this research, and should be contacted with respect to any matters arising from, or in connection with, this research. **Taiwan:** This material is for reference only and must not be reprinted without permission. Investors should carefully consider their own investment risk. Investment results are the responsibility of the individual investor. **United Kingdom:** Persons who would be categorized as retail clients in the United Kingdom, as such term is defined in the rules of the Financial Conduct Authority, should read this research in conjunction with prior Goldman Sachs research on the covered

companies referred to herein and should refer to the risk warnings that have been sent to them by Goldman Sachs International. A copy of these risks warnings, and a glossary of certain financial terms used in this report, are available from Goldman Sachs International on request.

European Union and United Kingdom: Disclosure information in relation to Article 6 (2) of the European Commission Delegated Regulation (EU) (2016/958) supplementing Regulation (EU) No 596/2014 of the European Parliament and of the Council (including as that Delegated Regulation is implemented into United Kingdom domestic law and regulation following the United Kingdom's departure from the European Union and the European Economic Area) with regard to regulatory technical standards for the technical arrangements for objective presentation of investment recommendations or other information recommending or suggesting an investment strategy and for disclosure of particular interests or indications of conflicts of interest is available at <https://www.gs.com/disclosures/europeanpolicy.html> which states the European Policy for Managing Conflicts of Interest in Connection with Investment Research.

Japan: Goldman Sachs Japan Co., Ltd. is a Financial Instrument Dealer registered with the Kanto Financial Bureau under registration number Kinsho 69, and a member of Japan Securities Dealers Association, Financial Futures Association of Japan Type II Financial Instruments Firms Association, and Investment Management Association of Japan. Sales and purchase of equities are subject to commission pre-determined with clients plus consumption tax. See company-specific disclosures as to any applicable disclosures required by Japanese stock exchanges, the Japanese Securities Dealers Association or the Japanese Securities Finance Company.

Global product; distributing entities

Goldman Sachs Global Investment Research produces and distributes research products for clients of Goldman Sachs on a global basis. Analysts based in Goldman Sachs offices around the world produce research on industries and companies, and research on macroeconomics, currencies, commodities and portfolio strategy. This research is disseminated in Australia by Goldman Sachs Australia Pty Ltd (ABN 21 006 797 897); in Brazil by Goldman Sachs do Brasil Corretora de Títulos e Valores Mobiliários S.A.; Public Communication Channel Goldman Sachs Brazil: 0800 727 5764 and / or contatogoldmanbrasil@gs.com. Available Weekdays (except holidays), from 9am to 6pm. Canal de Comunicação com o Público Goldman Sachs Brasil: 0800 727 5764 e/ou contatogoldmanbrasil@gs.com. Horário de funcionamento: segunda-feira à sexta-feira (exceto feriados), das 9h às 18h; in Canada by Goldman Sachs & Co. LLC; in Hong Kong by Goldman Sachs (Asia) L.L.C.; in India by Goldman Sachs (India) Securities Private Ltd.; in Japan by Goldman Sachs Japan Co., Ltd.; in the Republic of Korea by Goldman Sachs (Asia) L.L.C., Seoul Branch; in New Zealand by Goldman Sachs New Zealand Limited; in Russia by OOO Goldman Sachs; in Singapore by Goldman Sachs (Singapore) Pte. (Company Number: 198602165W); and in the United States of America by Goldman Sachs & Co. LLC. Goldman Sachs International has approved this research in connection with its distribution in the United Kingdom.

Goldman Sachs International ("GSI"), authorised by the Prudential Regulation Authority ("PRA") and regulated by the Financial Conduct Authority ("FCA") and the PRA, has approved this research in connection with its distribution in the United Kingdom.

European Economic Area: Goldman Sachs Bank Europe SE ("GSBE") is a credit institution incorporated in Germany and, within the Single Supervisory Mechanism, subject to direct prudential supervision by the European Central Bank and in other respects supervised by German Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin) and Deutsche Bundesbank and disseminates research within the European Economic Area.

General disclosures

This research is for our clients only. Other than disclosures relating to Goldman Sachs, this research is based on current public information that we consider reliable, but we do not represent it is accurate or complete, and it should not be relied on as such. The information, opinions, estimates and forecasts contained herein are as of the date hereof and are subject to change without prior notification. We seek to update our research as appropriate, but various regulations may prevent us from doing so. Other than certain industry reports published on a periodic basis, the large majority of reports are published at irregular intervals as appropriate in the analyst's judgment.

Goldman Sachs conducts a global full-service, integrated investment banking, investment management, and brokerage business. We have investment banking and other business relationships with a substantial percentage of the companies covered by Global Investment Research. Goldman Sachs & Co. LLC, the United States broker dealer, is a member of SIPC (<https://www.sipc.org>).

Our salespeople, traders, and other professionals may provide oral or written market commentary or trading strategies to our clients and principal trading desks that reflect opinions that are contrary to the opinions expressed in this research. Our asset management area, principal trading desks and investing businesses may make investment decisions that are inconsistent with the recommendations or views expressed in this research.

We and our affiliates, officers, directors, and employees will from time to time have long or short positions in, act as principal in, and buy or sell, the securities or derivatives, if any, referred to in this research, unless otherwise prohibited by regulation or Goldman Sachs policy.

The views attributed to third party presenters at Goldman Sachs arranged conferences, including individuals from other parts of Goldman Sachs, do not necessarily reflect those of Global Investment Research and are not an official view of Goldman Sachs.

Any third party referenced herein, including any salespeople, traders and other professionals or members of their household, may have positions in the products mentioned that are inconsistent with the views expressed by analysts named in this report.

This research is focused on investment themes across markets, industries and sectors. It does not attempt to distinguish between the prospects or performance of, or provide analysis of, individual companies within any industry or sector we describe.

Any trading recommendation in this research relating to an equity or credit security or securities within an industry or sector is reflective of the investment theme being discussed and is not a recommendation of any such security in isolation.

This research is not an offer to sell or the solicitation of an offer to buy any security in any jurisdiction where such an offer or solicitation would be illegal. It does not constitute a personal recommendation or take into account the particular investment objectives, financial situations, or needs of individual clients. Clients should consider whether any advice or recommendation in this research is suitable for their particular circumstances and, if appropriate, seek professional advice, including tax advice. The price and value of investments referred to in this research and the income from them may fluctuate. Past performance is not a guide to future performance, future returns are not guaranteed, and a loss of original capital may occur. Fluctuations in exchange rates could have adverse effects on the value or price of, or income derived from, certain investments.

Certain transactions, including those involving futures, options, and other derivatives, give rise to substantial risk and are not suitable for all investors. Investors should review current options and futures disclosure documents which are available from Goldman Sachs sales representatives or at <https://www.theocc.com/about/publications/character-risks.jsp> and https://www.goldmansachs.com/disclosures/cftc_fcm_disclosures. Transaction costs may be significant in option strategies calling for multiple purchase and sales of options such as spreads. Supporting documentation will be supplied upon request.

Differing Levels of Service provided by Global Investment Research: The level and types of services provided to you by Goldman Sachs Global Investment Research may vary as compared to that provided to internal and other external clients of GS, depending on various factors including your individual preferences as to the frequency and manner of receiving communication, your risk profile and investment focus and perspective (e.g.,

marketwide, sector specific, long term, short term), the size and scope of your overall client relationship with GS, and legal and regulatory constraints. As an example, certain clients may request to receive notifications when research on specific securities is published, and certain clients may request that specific data underlying analysts' fundamental analysis available on our internal client websites be delivered to them electronically through data feeds or otherwise. No change to an analyst's fundamental research views (e.g., ratings, price targets, or material changes to earnings estimates for equity securities), will be communicated to any client prior to inclusion of such information in a research report broadly disseminated through electronic publication to our internal client websites or through other means, as necessary, to all clients who are entitled to receive such reports.

All research reports are disseminated and available to all clients simultaneously through electronic publication to our internal client websites. Not all research content is redistributed to our clients or available to third-party aggregators, nor is Goldman Sachs responsible for the redistribution of our research by third party aggregators. For research, models or other data related to one or more securities, markets or asset classes (including related services) that may be available to you, please contact your GS representative or go to <https://research.gs.com>.

Disclosure information is also available at <https://www.gs.com/research/hedge.html> or from Research Compliance, 200 West Street, New York, NY 10282.

© 2026 Goldman Sachs.

You are permitted to store, display, analyze, modify, reformat, and print the information made available to you via this service only for your own use. You may not resell or reverse engineer this information to calculate or develop any index for disclosure and/or marketing or create any other derivative works or commercial product(s), data or offering(s) without the express written consent of Goldman Sachs. You are not permitted to publish, transmit, or otherwise reproduce this information, in whole or in part, in any format to any third party without the express written consent of Goldman Sachs. This foregoing restriction includes, without limitation, using, extracting, downloading or retrieving this information, in whole or in part, to train or finetune a machine learning or artificial intelligence system, or to provide or reproduce this information, in whole or in part, as a prompt or input to any such system.